

IDENTIFICATION

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1. ABSTRACT

THE RK8E DRIVE CONTROL TEST IS DESIGNED FOR THE PURPOSE OF CHECKOUT OF THE RK8E DISK CONTROL LOGIC REQUIRING THE USE OF THE DISK DRIVE(S).

IN GENERAL, THE TEST IS AN INSTRUCTION TEST TO VERIFY BASIC OPERATION OF THE SEEK ONLY, RESTORE, WRITE DATA, READ DATA, WRITE ALL, AND READ ALL FUNCTIONS WITH ALL DRIVES ON THE CONTROL. SIMPLE COMPLEMENT DATA PATTERNS OF 2525 + 5252, 5252 + 2525, AND 0000 + 7777 ARE USED TO VERIFY ADDRESSING AND DATA TRANSFERS TO AND FROM EACH INDIVIDUAL DRIVE.

A MANUAL INTERVENTION TEST IS ALSO INCLUDED (SEE SECTION 5.7), TO ALLOW THE OPERATOR TO SELECT DATA PATTERNS AND COMMAND FUNCTIONS VIA THE SWITCH REGISTER.

CONSIDERING NO ERROR CONDITIONS, THE DRIVES THAT HAVE RUN THIS TEST ARE FORMATTED, IF THE PROGRAM WAS STOPPED AT END OF PROGRAM PASS COMPLETION BY SWR9=1.

2. REQUIREMENTS

2.1 HARDWARE

- A. PDP-8/A, 8/E, 8/F, OR 8/M COMPUTER OR OTHER FAMILY OF 8 COMPATIBLE COMPUTER WITH NECESSARY DW8E BUS ADAPTER.
- B. AT LEAST 4K OF READ/WRITE MEMORY. AT LEAST 8K OF MEMORY IS NEEDED FOR OPERATION OF THE CONSOLE PACKAGE.
- C. ASR-33 TELETYPE OR EQUIVALENT
- D. RK8E DISK CONTROL
- E. RK05J OR RK05F DISK DRIVE(S)
- F. UNFORMATTED OR FORMATTED 2200 BPI-1600 SECTOR PACK(S)

2.2 STORAGE

THE PROGRAM OCCUPIES OR UTILIZES LOCATIONS 0000 TO LOCATION 7577 OF FIELD 0 AND LOCATIONS 0 TO 1377 OF FIELD 1.

3. PRELIMINARY PROGRAMS

ALL BASIC AND EXTENDED MEMORY DIAGNOSTICS AND THE RK8E DISKLESS CONTROL TEST SHOULD BE RUN PRIOR TO THIS TEST.

4. SWITCH REGISTER SETTINGS

SWR0=1 SCOPE LOOP ON ERROR. AFTER AN ERROR HALT AT LOCATION "ERHLT9" RAISING THIS SWITCH AND PRESSING KEY CONTINUE WILL RESULT IN A SCOPE LOOP ON THE CURRENT FAILING TEST IF THE TEST CONTINUES TO FAIL. THE ERROR TYPEOUT AND THE ERROR HALT AT LOCATION "ERHLT9" WILL BE INHIBITED. THE TTY BELL WILL RING INDICATING AN ERROR IF SWR2=0.

SWR1=1 SCOPE LOOP ON CURRENT NON-FAILING TEST. RAISING THIS SWITCH CAUSES THE PROGRAM TO LOOP ON THE CURRENT TEST IF THE TEST IS WORKING CORRECTLY. MAY BE USED IN CONJUNCTION WITH SWR0=1 FOR INTERMITTENT PROBLEMS.

SWR2=1 INHIBIT BELL ON SCOPE LOOP. WHEN IN A SCOPE LOOP DUE TO SWR0=1, RAISING THIS SWITCH INHIBITS THE SCOPE LOOP ERROR BELL.

SWR4=1 STOP PROGRAM OR HALT SWITCH. RAISING THIS SWITCH WILL RESULT IN A PROGRAM STOP UPON COMPLETION OF THE NEXT NON-FAILING TEST. IF POSSIBLE, THIS SWITCH SHOULD ALWAYS BE USED TO STOP THE PROGRAM.

SWR5=1 INHIBIT THE RECOVERABLE ERROR HALT AFTER A RECOVERABLE ERROR TYPEOUT. AFTER AN ERROR HALT AT LOCATION "ERHLT9", RAISING THIS SWITCH AND PRESSING KEY CONTINUE WILL INHIBIT ALL FUTURE RECOVERABLE ERROR HALTS. IF SWR1=0 THE PROGRAM WILL PROCEED TO NEXT TEST AFTER EACH ERROR TYPEOUT. IF SWR1=1 THE PROGRAM WILL PROCEED BACK TO THE SAME OR CURRENT FAILING TEST.

SWR6=1 RECALIBRATE IN SCOPE LOOPS. RAISING THIS SWITCH WILL RESULT IN A DISK RECALIBRATION WHEN IN A SCOPE LOOP DUE TO SWR0=1, SWR1=1, OR WHEN SWR5=1.

SWR7=1 PROGRAM WAIT LOOP FOR DISK IN SCOPE LOOPS. RAISING THIS SWITCH WILL RESULT IN A PROGRAM WAIT LOOP FOR APROX. 500 MS WHEN IN A SCOPE LOOP DUE TO SWR0=1, SWR1=1, OR WHEN SWR5=1. IN SOME CASES, THIS MAY BE USEFUL FOR WAITING FOR THE DISK MOVEMENT TO COMPLETE IF CONTROL OR DRIVE ERRORS OCCUR, BEFORE REPEATING THE TEST AGAIN. IN SOME CASES, FAILURE TO WAIT, MAY CAUSE ADDITIONAL ERRORS.

SWR8=1 GET ALL REGISTERS AFTER THE RECOVERABLE ERROR HALT "ERHLT9". AFTER AN ERROR HALT AT LOCATION "ERHLT9", RAISING THIS SWITCH AND PRESSING KEY CONTINUE RESULTS IN AN ERROR TYPEOUT OF THE ACTUAL CONTENTS OF

- SWR9=1 PROGRAM HALT OR STOP AT END OF PROGRAM
PASS COMPLETION.
- SWR10=11 DISK DRIVE(S) TO TEST. WHEN RUNNING
THE CHECK WRITE PROTECT TEST SECTION 5.4,
THE CHECK WRITE PROTECT TEST SECTION 5.5,
THE MANUAL FUNCTIONS SECTION 5.6, AND THE
THE SEEK FROM SWITCHES SECTION 5.8, THESE
SWITCHES INDICATE THE DRIVE NUMBER TO SELECT.

5. OPERATOR AND/OR PROGRAM ACTION

5.1 STANDARD TEST PROCEDURE

- A. START AS SPECIFIED THROUGH OUT THIS DOCUMENTATION
IS KEY CLEAR AND THEN KEY CONTINUE ON PDP8/E, PDP8/M,
AND PDP8/F COMPUTERS.
- B. LOAD THE PROGRAM INTO FIELD 0 USING
THE STANDARD BINARY LOADER TECHNIQUE.
- C. IF IT IS DESIRED TO CHANGE THE IOT CODES WITHIN THE
PROGRAM, FOLLOW THE PROCEDURE IN SECTION 5.8.
- D. RUN THE DRIVE CONTROL TEST WITH ALL DRIVES ON THE DISK
SYSTEM (SEE SECTION 5.3).
- E. THE PROGRAM EXECUTION TIME IS APROX. 30 MINUTES PER
DISK DRIVE.
- F. RUN THE WRITE PROTECT CHECK TESTS ON ALL DRIVES ON THE
DISK SYSTEM BY FOLLOWING THE PROCEDURES IN SECTIONS 5.5
AND 5.6.
- G. MANUAL FUNCTIONS, SECTION 5.7, MAY BE USED FOR TROUBLE
SHOOTING, IF DESIRED.
- H. SEEK FROM SWITCHES, SECTION 5.9, MAY BE USED FOR
TROUBLE SHOOTING, IF DESIRED.
- I. IF THE PROGRAM WAS STOPPED BY SWR4=1 OR BY "ERHLT9",
ADDRESS 0210 CAN BE USED TO RESTART THE PROGRAM AT THE
LAST SUBTEST EXECUTED. (NOTE: WATCH YOUR SWITCH SETTINGS.)

5.2 RK05 DRIVE CARTRIDGE MOUNTING PROCEDURE

THE FOLLOWING IS THE CORRECT CARTRIDGE MOUNTING PROCEDURE
FOR THE RK05 DISK DRIVE. ANY DEVIATION ENCOUNTERED DURING
THIS PROCEDURE WILL BE CONSIDERED AS AN ERROR CONDITION.

- A. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION.
- B. TURN AC POWER TO DISK DRIVE ON.

- C. VERIFY THAT LIGHT LABELED "PWR" IS ON.
- D. WAIT FOR LIGHT LABELED "LOAD" TO COME ON.
- E. VERIFY THAT LIGHTS LABELED "RDY", "ON CYL", "FAULT", "WT", AND "RD" ARE OFF.
- F. OPEN ACCESS DOOR.
- G. INSERT CARTRIDGE.
- H. CLOSE ACCESS DOOR.
- I. SET SWITCH LABELED "RUN/LOAD" TO THE "RUN" POSITION.
- J. WAIT FOR THE LIGHTS LABELED "RDY" AND "ON CYL" TO COME ON.
- K. TOGGLE SWITCH LABELED "WT PROT" AND VERIFY THAT THE LIGHT LABELED "WT PROT" GOES ON AND OFF.
- L. TOGGLE SWITCH LABELED "WT PROT" UNTIL THE LIGHT LABELED "WT PROT" GOES OFF.
- M. VERIFY THAT LIGHTS LABELED "FAULT", "WT", "RD", AND "LOAD" ARE OFF.

5.3 DRIVE CONTROL TEST

- A. MAKE READY THE DISK DRIVE TO BE TESTED USING THE RK05 DRIVE CARTRIDGE MOUNTING PROCEDURE SECTION 5.2.
- B. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON ALL DRIVES NOT BEING TESTED.
- C. VERIFY THAT AC POWER TO ALL DRIVES IS ON.
- D. SET THE SWITCH REGISTER TO 0200 AND PRESS LOAD ADDRESS.
- E. SET THE SWITCH REGISTER TO 0000.
- F. PRESS CLEAR AND THEN CONTINUE.
- G. THE TTY WILL RESPOND WITH THE FOLLOWING MESSAGE QUESTIONING THE OPERATOR ON THE DISK DRIVES TO TEST. THE RESPONSE SHOULD BE Y FOR YES OR N FOR NO:

RK05 DRIVE CONTROL TEST
TEST (Y=YES OR N=NO):
DISK0? DISK1? DISK2? DISK3?

AFTER THE QUESTIONS ARE ANSWERED THE PROGRAM WILL BEGIN TESTING THE DRIVES SPECIFIED.

- H. THE PROGRAM SHOULD PRINT THE FOLLOWING MESSAGE AT THE COMPLETION OF EACH PASS.

"RK05 DRIVE CONTROL TEST PASS COMPLETE"

- I. ALWAYS USE SWR4=1 FOR STOPPING THE TEST.

- J. IF IT IS DESIRED TO HAVE THE PROGRAM HALT OR STOP AT
END OF PROGRAM PASS COMPLETION SET SWR9=1.
- K. ANY HALTS OR TIMEOUTS OTHER THAN THE PASS COMPLETE
TIMEOUT OR END OF TEST HALT MENTIONED ABOVE WILL BE
CONSIDERED AN ERROR CONDITION. IN ALL CASES ACCESS
"ERRORS" SECTION 6 IN THIS DOCUMENTATION.
- L. FOR THE ABSOLUE LOCATIONS OF ALL KNOWN HALTS ACCESS PAGE
1-22 OF THE PROGRAM LISTING.

5.4 CHECK WRITE PROTECT (MANUAL)

- A. RUN THE DRIVE CONTROL TEST WITH ALL DRIVES ON THE
CONTROL BEFORE RUNNING THIS "WRITE PROTECT" PORTION.
- B. MAKE READY A DRIVE TO TEST USING THE RK05 DRIVE CARTRIDGE
MOUNTING PROCEDURE SECTION 5.2.
- C. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION
ON ALL OTHER DRIVES.
- D. VERIFY THAT AC POWER TO ALL DRIVES IS ON.
- E. VERIFY THAT THE LIGHT LABELED "WT PROT" IS "OFF" ON THE
CURRENT DRIVE UNDER TEST.
- F. SET THE SWITCH REGISTER TO 0206 AND PRESS LOAD ADDRESS.
- G. SET THE SWITCH REGISTER TO 0000.
- H. SET SWR10-11 TO THE CURRENT DRIVE NUMBER UNDER TEST.
- I. PRESS START AND THE COMPUTER SHOULD HALT AT LOCATION "MPHLT1".
- J. PRESS SWITCH LABELED "WT PROT" TO TURN "WRITE PROTECT" AND
THE LIGHT LABELED "WT PROT" ON.
- K. PRESS KEY CONTINUE AND THE COMPUTER SHOULD HALT AT LOCATION
"MPHLT2" INDICATING A SUCCESSFUL TEST.
- M. IF ANY ERRORS ARE ENCOUNTERED OR IF IT IS DESIRED TO TRY
THE TEST AGAIN, REPEAT STEPS A-K.
- N. FOR POSSIBLE ERROR TIMEOUTS ACCESS SECTION 6 IN THIS DOC-
UMENTATION. (NOTE: NO SCOPE LOOPS ARE AVAILABLE FOR THIS
TEST.)
- O. THE "CHECK WRITE PROTECT PROCEDURE" AS DESCRIBED ABOVE
SHOULD BE RUN TWICE WITH ALL DRIVES ON THE CONTROL.

5.5 CHECK WRITE PROTECT (PROGRAM CONTROL)

- A. RUN THE DRIVE CONTROL TEST WITH ALL DRIVES ON THE
CONTROL BEFORE RUNNING THIS "WRITE PROTECT" PORTION.

- B. MAKE READY A DRIVE TO TEST USING THE RK05 DRIVE CARTRIDGE MOUNTING PROCEDURE SECTION 5.2.
- C. SET SWITCH LABELED "RUN/LOAD" TO THE "LOAD" POSITION ON ALL OTHER DRIVES.
- D. VERIFY THAT AC POWER TO ALL DRIVES IS ON.
- E. VERIFY THAT THE LIGHT LABELED "WT PROT" IS "OFF" ON THE CURRENT DRIVE UNDER TEST.
- F. SET THE SWITCH REGISTER TO 0207 AND PRESS LOAD ADDRESS.
- G. SET THE SWITCH REGISTER TO 0000.
- H. SET SW10-11 TO THE CURRENT DRIVE NUMBER UNDER TEST.
- I. PRESS START AND THE COMPUTER SHOULD HALT AT LOCATION "APHLT1" INDICATING A SUCCESSFUL TEST.
- J. VERIFY THAT THE WRITE PROTECT LIGHT LABELED "WT PROT" IS ON, ON THE CURRENT DRIVE.
- L. IF ANY ERRORS ARE ENCOUNTERED OR IF IT IS DESIRED TO TRY THE TEST AGAIN, REPEAT STEPS A-J.
- M. FOR POSSIBLE ERROR TYPEDOUTS ACCESS SECTION 6 IN THIS DOCUMENTATION. (NOTE: NO SCOPE LOOPS ARE AVAILABLE FOR THIS TEST.)
- N. THE "CHECK WRITE PROTECT PROCEDURE" AS DESCRIBED ABOVE SHOULD BE RUN TWICE WITH ALL DRIVES ON THE CONTROL.

5.6 MANUAL FUNCTIONS (FOR TROUBLE SHOOTING ONLY)

THE MANUAL FUNCTIONS ENABLES THE OPERATOR TO SELECT FUNCTIONS, DISK ADDRESS, AND DATA PATTERNS VIA THE SWITCH REGISTER. THIS IS NOT PART OF THE STANDARD TEST PROCEDURE AND SHOULD ONLY BE USED FOR TROUBLE SHOOTING IF DESIRED.

- A. SET THE SWITCH REGISTER TO 0204 AND PRESS LOAD ADDRESS.
- B. SET THE SWITCH REGISTER TO THE DESIRED FUNCTION TO BE LOADED INTO THE COMMAND REGISTER. (SEE SECTION 8.)
(NOTE: THE EXTENDED MEMORY BITS 6-8, THE ENABLE INTERRUPT BIT 3, AND THE ENABLE SET DONE BIT ON SEEK COMPLETE BIT 4, ARE NOT RECOGNIZED. THIS MANUAL PORTION IS ONLY FLAG DRIVEN AND ALL DATA TRANSFERS ARE TO THE CURRENT FIELD.)
- C. PRESS START AND THE COMPUTER SHOULD HALT.
- D. SET THE SWITCH REGISTER TO THE DESIRED DISK ADDRESS TO BE LOADED INTO THE CYLINDER, SURFACE, AND SECTOR REGISTER. (SEE SECTION 8.)
- E. PRESS START AND THE COMPUTER SHOULD HALT.

F. SET THE SWITCH REGISTER TO THE COMPLEMENT TYPE DATA PATTERN TO BE WRITTEN ON OR READ FROM THE DISK DEPENDING ON THE FUNCTION PREVIOUSLY LOADED INTO THE COMMAND REGISTER. (NOTE: A SETTING OF 0000 WILL RESULTS IN A COMPLEMENT DATA PATTERN OF 0000 + 7777. A SETTING OF 2525 WILL RESULT IN A COMPLEMENT DATA PATTERN OF 2525 + 5252.)

G. PRESS START AND THE COMPUTER SHOULD HALT.

H. SET THE SWITCH REGISTER TO 0000, PRESS START, AND THE FUNCTION SELECTED WILL BE EXECUTED.

I. IF POSSIBLE, ALWAYS USE SWR4=1 FOR STOPPING PROGRAM.

J. IN CASE OF ERRORS OR DESIRED LOOPS, USE THE REGULAR SWITCH REGISTER SETTINGS (SECTION 4.)

K. IF A WRITE ALL OR THE WRITE DATA FUNCTION WAS SELECTED, THE DATA PATTERN SELECTED WILL BE WRITTEN ON THE DISK ADDRESS SELECTED.

L. IF A READ ALL OR READ DATA FUNCTION WAS SELECTED, THE DATA WILL BE READ OFF THE DISK ADDRESS SELECTED AND COMPARED AGAINST THE DATA PATTERN SELECTED.

M. IF A SEEK ONLY FUNCTION WAS SELECTED, A SEEK ONLY WILL BE EXECUTED TO THE DISK ADDRESS SELECTED.

N. IF A WRITE LOCK FUNCTION WAS THE SELECTED THE DISK DRIVE SELECTED WILL BE WRITE LOCKED.

5.7 CHANGE PROGRAM DEVICE IOT CODES

THE PROGRAM NORMALLY RECOGNIZES DEVICE IOT CODE X74X. TO CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM:

A. SET THE SWITCH REGISTER TO 0205 AND PRESS LOAD ADDRESS.

B. SET THE SWITCH REGISTER TO 0000, SET SWITCH REGISTER BITS 3-8 TO THE DESIRED DEVICE IOT CODE, AND PRESS START.

C. THE PROGRAM WILL CHANGE THE DEVICE IOT CODES WITHIN THE PROGRAM AND THEN HALT.

D. PRESSING KEY CONTINUE AT THIS TIME WILL RESULT IN A START OF THE PROGRAM AT LOCATION 0200 (SEE SECTIONS 5.3 OR 5.4 FOR OPERATION INSTRUCTIONS).

5.8 SEEK FROM SWITCHES (FOR RK05 ALIGNMENT)

THE FOLLOWING SURTEST WAS REQUESTED BY FIELD SERVICE TO AID IN RK05 ALIGNMENT. THE PROGRAM WILL SEEK ONLY BETWEEN ADDRESSES FROM SWITCH REGISTER.

A. SET THE SWITCH REGISTER TO 4000 AND PRESS LOAD ADDRESS.

B. SET THE SWITCH REGISTER TO 0000.

- C. SET SWR9-11 TO THE DRIVE NUMBER AND EXTENDED CYLINDER BIT OF THE FIRST SEEK ADDRESS (BITS 9-10 TO DRIVE NUMBER AND BIT 11 TO EXTENDED CYLINDER).
- D. SET SWR0-7 TO THE REMAINDER OF THE CYLINDER BITS AND THE SURFACE OF THE FIRST SEEK ADDRESS.
- E. PRESS START AND THE COMPUTER SHOULD HALT.
- F. SET THE SWITCH REGISTER TO 0000.
- G. SET SWR9-11 TO THE DRIVE NUMBER AND EXTENDED CYLINDER BIT OF THE SECOND SEEK ADDRESS (BITS 9-10 TO THE DRIVE NUMBER AND BIT 11 TO THE EXTENDED CYLINDER).
- H. SET SWR0-7 TO THE CYLINDER BITS AND SURFACE OF THE SECOND SEEK ADDRESS.
- I. PRESS START AND THE DRIVE SHOULD SEEK BETWEEN THE ADDRESSES SPECIFIED BY THE SWITCH REGISTER.
- J. THE SECOND SEEK ADDRESS CAN BE CHANGED AT ANY TIME BY SIMPLY CHANGING THE SWITCH REGISTER TO SELECT A NEW ADDRESS.
- K. CARE SHOULD BE TAKEN TO NOT SELECT A NON-EXISTENT DISK DRIVE OR NON-EXISTENT CYLINDER.
- L. NO ERROR CHECKING IS DONE DURING THIS SURTEST.
- M. IT IS POSSIBLE TO SEEK TO A CONSTANT ADDRESS BY MAKING THE FIRST AND SECOND ADDRESS EQUAL.

6. ERRORS

6.1 USEFUL ERROR INFORMATION

IN THE DRIVE CONTROL TEST, THE DISK SKIP IOT IS FIRST CHECKED AND TIMED-OUT USING AN "ISZ" TIME LOOP. IF THE SKIP IOT FAILS, AN ERROR TYPEOUT AND ERROR HALT SHOULD OCCUR. ONCE PROVEN TO WORK, THE IOT IS NOT TIMED-OUT. THE PROGRAM MAY HANG-UP IF THE SKIP IOT FAILS INTERMITTENTLY. (NOTE: THE MANUAL FUNCTIONS, SECTION 5.7, ALWAYS TIMES OUT THE SKIP IOT TO PREVENT HANGING UP.

ALL ERRORS FOUND WHEN RUNNING THIS TEST SHOULD BE CORRECTED BEFORE PROCEEDING ON IN THE TEST.

WHEN AN OPERATOR ENCOUNTERS AN ERROR WHEN RUNNING THIS TEST HE SHOULD, IN ALL CASES, READ THE ERROR TYPEOUT INFORMATION, NOTE THE LOCATION OF THE FAILURE, READ ALL THE INFORMATION UNDER ERRORS IN THIS DOCUMENTATION, AND THEN ACCESS THE PROGRAM LISTING FOR FURTHER INFORMATION.

THE ABSOLUTE LOCATION OF ALL KNOWN HALTS CAN BE FOUND A COMPLEMENT TYPE DATA PATTERN (I.E. 2525 + 5252, 5252 + 2525, OR 0000 + 7777) IS ALWAYS USED IN THIS TEST WHEN DATA IS WRITTEN AND THEN CHECKED. IN SOME CASES, ALL 0'S IS USED IN CHECKING CRC AND STATUS REGISTERS, HOWEVER, THE DATA IS NOT CHECKED.

THE PROGRAM USES THE SAME PROGRAM BUFFER FOR WRITING AND READING DATA. THE BUFFER IS SETUP BEFORE A WRITE FUNCTION AND CLEARED BEFORE THE DATA IS READ AND CHECKED. THE BUFFER OCCUPIES THE CURRENT FIELD FROM THE END OF THE PROGRAM +400 LOCATIONS.

BEFORE DATA IS WRITTEN ON THE DISK, THE FIRST TWO WORDS OF THE BUFFER ARE SET TO THE ABSOLUTE DISK ADDRESS. THE FIRST WORD OF THE BUFFER (BITS 9-11) IS SET TO THE DRIVE NUMBER AND THE EXTENDED CYLINDER BIT, THE SECOND WORD TO THE 12 REMAINDER CYLINDER, SURFACE, AND SECTOR BITS, ALSO THE BUFFER +1 IS SET TO THE DATA WORD OF "1234". AFTER THE WRITE THEN READ, THE WORDS ARE CHECKED FOR CORRECT VALUES, INDICATING THAT THE INFORMATION WAS WRITTEN ON AND READ FROM THE SAME PLACE ON THE DISK AND THAT THE DATA BREAK STOPPED CORRECTLY. WHEN AN ERROR EXISTS WITH THE WORDS AS STATED PREVIOUS, THE OPERATOR SHOULD REALIZE THAT THE PROBLEM IS MOST LIKELY ADDRESSING AND SOMETIMES DATA ERRORS.

WHEN DATA IS BEING READ OFF THE DISK AND A CRC ERROR OCCURES THE PROGRAM WILL THEN CHECK THE DATA READ FOR DATA ERRORS. IF NO DATA ERRORS EXIST THE CRC ERROR FOUND WILL BE REPORTED AS A STATUS REGISTER ERROR. IF DATA ERRORS ARE FOUND THE DATA ERRORS WILL BE REPORTED AS DISK DATA ERRORS AND THE CRC STATUS ERROR INDICATED IN THE "ST:". (SEE SECTION 6.4 FOR ERROR HEADERS AND TYPEOUTS).

THE ABSOLUTE ADDRESS LOCATIONS OF THE DATA BUFFER

6.2 NON-RECOVERABLE ERROR HALTS

NON-RECOVERABLE ERROR HALTS FOR WHICH THERE ARE NO TYPEOUTS OR SCOPE LOOPS ARE LISTED AND DEFINED AS FOLLOWS.

ERHLT1	UNDEFINED INTERRUPT
ERHLT2	SKIP TRAP FOR IOT "DCLR"
ERHLT3	SKIP TRAP FOR IOT "DLAG"
ERHLT4	SKIP TRAP FOR IOT "DLCA"
ERHLT5	SKIP TRAP FOR IOT "DRST"
ERHLT6	SKIP TRAP FOR IOT "DLDC"
ERHLT7	SKIP TRAP FOR IOT "DMAN"

6.3 RECOVERABLE ERROR HALT

ALL RECOVERABLE ERRORS, FOR WHICH THERE ARE SCOPE LOOPS AND ERROR TYPEOUTS, SHOULD RESULT IN AN ERROR HALT AT LOCATION "ERHLT9".

ERHLT9 RECOVERABLE ERROR HALT. READ INFORMATION TYPEOUT ON TTY AND ACCESS PROGRAM LISTING AND DOCUMENTATION.

6.4 ERROR TYPEOUTS -----

WHEN A RECOVERABLE ERROR OCCURS THE PROGRAM WILL PRINT AN "ERROR HEADER" WHICH WILL SPECIFY THE PARTICULAR REGISTER OR TYPE OF ERROR FOUND AT THE TIME OF THE FAILURE.

POSSIBLE "ERROR HEADERS" ARE AS FOLLOWS.

STATUS REGISTER ERROR
COMMAND REGISTER ERROR
DISK ADDRESS REGISTER ERROR
DISK DATA ERROR
CRC REGISTER ERROR
DATA REGISTER ERROR
DISK SKIP ERROR
DISK INTERRUPT ERROR

AFTER THE "ERROR HEADER" MENTIONED ABOVE IS TYPED, THE PROGRAM WILL PRINT THE FOLLOWING ERROR INFORMATION FOUND AT THE TIME OF THE FAILURE, PERTAINING TO THE FAILURE. POSSIBLE TYPEOUTS ARE AS FOLLOWS.

PC: PROGRAM LOCATION OF THE ACTUAL FAILURE.
GD: REFERS TO THE DATA EXPECTED IN THE REGISTER OR TYPE OF TEST SPECIFIED IN THE "ERROR HEADER".
CR: CONTENTS OF THE CRC REGISTER.
ST: CONTENTS OF THE STATUS REGISTER.
DB: CONTENTS OF THE LOWER DATA REGISTER.
CM: CONTENTS OF THE COMMAND REGISTER.
DA: CONTENTS OF THE DISK ADDRESS REGISTER OR THE CYLINDER, SURFACE, AND SECTOR BITS.
CA: CONTENTS OF THE INITIAL CURRENT ADDRESS
AD: BREAK ADDRESS OF DATA BREAK IN COMPUTER.
DT: DATA FOUND DURING DATA BREAK.

THE "GD:" INFORMATION TYPED OUT POINTS TO THE DATA EXPECTED IN THE REGISTER OR TYPE OF ERROR TYPED OUT IN THE "ERROR HEADER".

THE ERROR INFORMATION INDICATOR SUGGESTED BY THE "ERROR HEADER" (I.E. DA: FOR DISK ADDRESS ERROR, CM: FOR COMMAND REGISTER ERROR, CR: FOR CRC REGISTER ERROR, ETC.), IS THE ACTUAL CONTENTS OF THAT PARTICULAR REGISTER. ERROR INFORMATION OTHER THAN THAT SUGGESTED BY THE ERROR HEADER IS THE SOFTWARE VALUE LOADED INTO THAT REGISTER PRIOR TO THE FAILURE.

TO TYPE THE ACTUAL CONTENTS OF THE REGISTERS, SET SWR0=1 AFTER AN ERROR HALT AT LOCATION "ERHLT9", AND PRESS KEY CONTINUE. THE CONTENTS OF THE CRC, STATUS, LOWER DATA, COMMAND, AND SURFACE AND SECTOR REGISTERS WILL THEN BE TYPED.

6.5 SCOPE LOOPS

THERE ARE SCOPE LOOPS AVAILABLE FOR ALL ERRORS RESULTING IN AN ERROR HALT AT LOCATION "ERHLT9".

TO ENTER SCOPE LOOP, INHIBIT ERROR TYPEOUT, AND INHIBIT ERROR HALT, AFTER AN ERROR HALT AT "ERHLT9", SET SWR0=1 TO INDICATE SCOPE LOOP AND PRESS KEY CONTINUE.

IF THE SCOPE LOOP IS WORKING CORRECTLY AND THE TEST IS STILL FAILING, THE TTY BELL SHOULD RING INDICATING AN ERROR. THEN SET SWR2=1 TO INHIBIT THE TTY ERROR BELL.

SWR1=1 MAY HAVE TO BE USED IN SCOPE LOOPS IN CONJUNCTION WITH SWR2=1, IF THE CURRENT TEST IS WORKING INTERMITTENTLY.

6.6 TYPICAL ERROR TYPEOUTS

THE FOLLOWING IS AN EXAMPLE OF AN "ERROR HEADER" AND TYPEOUT THAT COULD HAVE OCCURRED IF THE DISK SKIP IOT FAILED TO SKIP.

DISK SKIP ERROR
PC:0267

THE FOLLOWING IS AN EXAMPLE OF AN "ERROR HEADER" AND ERROR TYPEOUT THAT COULD HAVE OCCURRED ON A DATA BREAK ERROR. (NOTE CRC IN THE STATUS INDICATOR "STI")

DISK DATA ERROR
PC:1161 GD:5252 ST:4010 CM:1000 DA:0001 CA:7000 AD:7010 DT:5250

THE FOLLOWING IS A TYPICAL ERROR THAT COULD HAVE OCCURED WHILE READING THE CRC REGISTER.

CRC REGISTER ERROR
PC:2246 GD:116047 CR:116046 CM:1000 DA:7777

THE FOLLOWING IS AN EXAMPLE OF AN ERROR TYPEOUT THAT COULD HAVE OCCURRED IF THE STATUS REGISTER FAILED. (NOTE: IN THIS CASE THE OPERATOR INDICATED TO THE PROGRAM TO TYPE THE ACTUAL CONTENTS OF THE REGISTERS BY SETTING SWR0=1

AFTER THE ERROR HALT AT LOCATION "ERHLT9" AND PRESSING
KEY CONTINUE).

SEQ 0014

STATUS REGISTER ERROR

PC:1100 GD:4000 ST:2000 CM:5002 DA:0000

CR:000000 ST:2000 DB:0000 CM:5002 DA:0000

7. RESTRICTIONS

ALL DISK DRIVES SHOULD BE SET TO THE LOAD POSITION
THAT ARE NOT BEING TESTED.

ALL ERRORS SHOULD BE CORRECTED BEFORE PROCEEDING ON IN
THE PROGRAM.

8. TROUBLE SHOOTING INFORMATION

IOT	FUNCTION
---	-----
6741 DSKP	"SKIP" SKIP IF TRANSFER DONE FLAG OR ERROR FLAG IS SET.
6742 DCLR	"CLEAR" FUNCTION IS REGULATED BY AC BITS 10 AND 11. THE AC IS THEN CLEARED.
AC10 AC11	
---- ----	
0 0	CLEAR THE AC AND STATUS REGISTER.
0 1	CLEAR THE AC, CONTROL, AND MAJOR REGISTERS. THIS INSTRUCTION WILL STOP THE CONTROL EVEN IF IT IS WRITING A HEADER. THIS IS THE ONLY INSTRUCTION THAT CLEARS MAINTENANCE MODE.
1 0	CLEAR AC, RECALIBRATE DISK DRIVE, AND CLEAR STATUS REGISTER.
6743 DLAG	"LOAD DISK ADDRESS AND GO" LOAD THE DISK CYLINDER, SURFACE, AND SECTOR FROM THE AC, CLEAR THE AC, AND DO THE COMMAND IN THE COMMAND REGISTER.
AC	
--	
0-6	CYLINDER
7	SURFACE (1=UPPER) (0=LOWER)
8-11	SECTOR

6744 DLCA

"LOAD CURRENT ADDRESS" LOAD THE
CURRENT ADDRESS FROM AC, THE AC
IS THEN CLEARED.

SEQ 0015

AC

--

0-11

CURRENT ADDRESS

6745 DRST

"READ STATUS" CLEAR THE AC AND
READ THE CONTENTS OF THE STATUS
REGISTER INTO THE AC.

AC

--

0

TRANSFER DONE

1

READY TO SEEK, READ, OR WRITE.

2

NOT USED

3

SEEK FAIL

4

DISK FILE READY

5

CONTROL BUSY ERROR

6

TIME OUT ERROR

7

WRITE LOCK ERROR

8

CRC ERROR

9

DATA RATE ERROR

10

DRIVE STATUS ERROR

11

CYLINDER ADDRESS ERROR

6746 DLDC

"LOAD COMMAND" LOAD THE COMMAND
REGISTER FROM AC, CLEAR THE AC,
AND CLEAR THE STATUS REGISTER.

AC

--

0-2#0

READ DATA

0-2#1

READ ALL

0-2#2

WRITE LOCK

0-2#3

SEEK ONLY

0-2#4

WRITE DATA

0-2#5

WRITE ALL

0-2#6

NOT USED

0-2#7

NOT USED

3

ENABLE INTERRUPT

4

ENABLE SET TRANSFER DONE ON SEEK DONE

5

HALF BLOCK 128 WORDS

6

EXTENDED MEMORY ADDRESS

7

EXTENDED MEMORY ADDRESS

8

EXTENDED MEMORY ADDRESS

9

UNIT SELECT

10

UNIT SELECT

11

EXTENDED CYLINDER ADDRESS

6747 DMAN

"MAINTENANCE IOT" LOAD THE
MAINTENANCE REGISTER FROM THE AC. THE
FUNCTION IS REGULATED BY THE AC BITS.
MAINTENANCE MODE CAN ONLY BE CLEARED
BY DCLR "CLEAR CONTROL".

AC

--

0	ENTER MAINTENANCE MODE
1	ENABLE SHIFT TO LOWER BUFFER
2	AC BIT 10, CRC REGISTER, AND THE LOWER DATA BUFFER ARE CONNECTED AS A SHIFT REGISTER. AC BIT 10 DATA SHIFTS TO THE CRC, THE CRC SHIFTS TO THE LOWER DATA BUFFER.
3	SHIFT COMMAND REGISTER TO THE LOWER DATA BUFFER.
4	SHIFT THE SURFACE AND SECTOR REGISTER TO THE LOWER DATA BUFFER.
5	SHIFT AC 10 DATA TO THE UPPER DATA BUFFER. THE UPPER BUFFER SHOULD SINK IN THE SILO WHEN FULL.
6	ONE SINGLE CYCLE BREAK REQUEST. DIRECTION IS REGULATED BY FUNCTION IN THE COMMAND REGISTER.
7	CLEAR AC THEN READ THE LOWER DATA BUFFER TO THE AC.
8	NOT USED.
9	NOT USED.
10	USED AS DATA WITH OTHER BITS IN THE MAINTENANCE MODE.
11	NOT USED

9. PROGRAM DESCRIPTION

THE RK8E DRIVE CONTROL TEST VERIFIES BASIC FUNCTIONAL OPERATION OF THE RK8E CONTROL LOGIC WITH THE RK05 DISK DRIVE(S). THE PROGRAM IS COMPRISED OF MANY INDIVIDUAL SUBTESTS WHICH ARE AUTOMATICALLY RUN IN A SEQUENTIAL FLOW. ABOVE EACH SUBTEST, IN THE LISTING, IS A BRIEF DESCRIPTION OF EACH SUBTEST.

WHEN SINGLE DRIVE TESTING, ONE PASS THROUGH ALL SUBTESTS (TST0-TST45) RESULTS IN A PASS COMPLETION. WHEN MULTI-DRIVE TESTING, ONE PASS THROUGH ALL SUBTESTS (TST0-TST45) ON ALL DRIVES AND THE RUNNING OF THE OVERLAP SEEK TESTS (OVR LAP, GRONK, AND OVRRED) RESULTS IN A PASS COMPLETION.

CONSIDERING NO ERROR CONDITIONS, THE DRIVES THAT HAVE RUN THIS TEST ARE FORMATTED, IF THE PROGRAM WAS STOPPED AT END OF PROGRAM PASS COMPLETION BY SWR9=1.

10. CONSOLE PACKAGE ADDENDUM

10.1. DESCRIPTION

THE CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE

PACKAGE . 1) RUNNING WITH THE CONSOLE PACKAGE ACTIVE -
THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH
THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO
SWITCH REGISTER, BEFORE CONTINUING WITH EXECUTION OF THE
DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT
THE TERMINAL. NO HALTS WILL BE EXECUTED.
2) CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE
NORMAL STANDALONE OPERATION OF THE PROGRAM AS DISCRIBED IN
SECTIONS 1 THROUGH 9 OF THIS DOCUMENT.

10.2 RESTRICTIONS

-
- 1) RUNNING THE CONSOLE PACKAGE REQUIRES THAT THE
PSEUDO SWITCH REGISTER BE USED.
 - 2) ONCE RUNNING THE CONSOLE PACKAGE NONACTIVE AND
NOW DESIRE TO RUN IT ACTIVE, ONE MUST RELOAD THE DIAGNOSTIC
AND INITILIZE FOR A ACTIVE CONSOLE PACKAGE.

10.3 INITIALIZATION

FOR A ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=0 TO INDICATE USE PSEUDO SWITCH
REGISTER.
- 2.) SET LOCATION 22 BIT3=1 TO INDICATE CONSOLE PACKAGE ACTIVE.

FOR A NON ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 21 BIT0=1 TO INDICATE NOT TO USE PSEUDO
SWITCH REGISTER, BUT TO USE HARDWARE SWITCHES.
- 2.) SET LOCATION 22 BIT3=0 TO INDICATE CONSOLE PACKAGE NOT
ACTIVE.

10.4 CONTROL CHARACTERS

CONTROL CHARACTERS ARE USED TO GIVE THE OPERATOR THE
ABILITY TO PERFORM THE FOLLOWING FUNCTIONS.
NOTE: THE PROGRAM WILL RESPOND TO THE CONTROL
CHARACTER IN FIVE (5) SECONDS OR LESS.

CONTROL C

THIS WILL START THE LOADER THAT IS
IN LOCATION 7600.

CONTROL R

THIS WILL RESTART THE PROGRAM AND
REASK THE SWITCH REGISTER QUESTION AS
DESCRIBED IN SETION 10.6.

CONTROL E

THIS WILL CONTINUE THE PROGRAM FROM AN ERROR IF ALLOWED BY THE DIAGNOSTIC OR FROM A WAITING STATEMENT.

CONTROL L

THIS WILL SWITCH THE TERMINAL MESSAGES FROM THE DISPLAY TO A LINE PRINTER. TO RESTORE THE MESSAGES ON THE TERMINAL CONTROL L MUST BE TYPED AGAIN. IF NO PRINTER IS AVAILABLE AND CONTROL L IS TYPED THE RESULT WILL BE THAT THE CONSOLE PACKAGE WILL WAIT FOR CONTROL C OR R. THE CONTROL L WILL OUTPUT TO THE LINE PRINTER AND THE PROGRAM WILL ATTEMPT TO CONTINUE AS IF A CONTROL E WAS TYPED IN.

CONTROL D

THIS WILL ALLOW THE ABILITY TO CHANGE THE SWITCH REGISTER DURING PROGRAM OPERATION. TYPING THIS CHARACTER WILL RESULT IN AN INTERIGATION OF THE SWITCH REGISTER QUESTION AS DESCRIBED IN SECTION 10.6.

CONTROL S

THIS WILL STOP PROGRAM EXECUTION AND WAIT IN A LOOP FOR A CONTINUE. THE ONLY WAY TO CONTINUE WILL BE TO TYPE A CONTROL D, R OR C. THIS IS A NONPRINTING CHARACTER.

CONTROL Q

THIS IS TO CONTINUE A PROGRAM AFTER A CONTROL S IS TYPED. THIS IS A NONPRINTING CHARACTER.

10.5 WAITING MESSAGE

THE WAITING MESSAGE IS USED TO ALLOW THE OPERATOR TIME TO MAKE A DECISION AS TO WHAT CONTROL CHARACTER TO TYPE. THIS MESSAGE MAY APPEAR AT THE END OF PASS MESSAGE IF THE HALT ON PASS BIT IS SET. THE CONTROL CHARACTERS MAY NOW BE USED TO PERFORM THE NEEDED FUNCTION.

THE WAITING MESSAGE MAY BE PRINTED AFTER A ERROR MESSAGE IF THE HALT ON ERROR BIT IS SET. HERE AGAIN THE CONTROL CHARACTERS MAY BE USED. THE WAITING MESSAGE MAY BE PRINTED IF OPERATOR INTERVENTION IS REQUIRED.

10.6 SWITCH REGISTER MESSAGE

THIS MESSAGE IS USED TO SETUP THE PSEUDO SWITCH REGISTER BEFORE PROGRAM EXECUTION TAKES PLACE. THE SWITCH REGISTER IS SETUP WHEN THE FOURTH CHARACTER IS ENTERED OR A CARRIAGE RETURN IS TYPED

 SR=0000 4000

UNDER SCORING INDICATES OPERATOR RESPONSE

10.7 END OF PASS

AN INDICATION WILL BE GIVEN WHEN THE DIAGNOSTIC HAS MADE A SUCCESSFUL PASS. THE PRINT OUT WILL INDICATE THE DIAGNOSTIC MAINDEC NUMBER THE WORD PASS AND A FOUR DIGIT PASS NUMBER. A PASS WILL BE A TIME PERIOD RATHER THAN A PROGRAM PASS OF THE DIAGNOSTIC. THE TIME PERIOD WILL BE IN THE RANGE OF ONE (1) TO FIVE (5) MINUTES. IF THE DIAGNOSTIC MAKES A PROGRAM PASS IN THE 1 TO 5 MINUTE RANGE THEN THE PASS COUNT WILL BE THE SAME AS THE NUMBER OF PROGRAM PASSES. IF THE PROGRAM MAKES A PROGRAM PASS IN LESS THAN ONE MINUTE THEN THE PASS COUNT WILL NOT BE THE SAME AS THE PASS COUNTER THE PASS COUNTER WILL REFLECT MORE THAN ON PROGRAM PASS. THE NUMBER OF PROGRAM PASSES REQUIRED FOR A PASS MESSAGE CAN BE FOUND IN FIELD 1 LOCATION 0246.

IF HALT AT END OF PASS IS SET THEN THE PASS MESSAGE WILL BE PRINTED AND A WAITING STATEMENT WILL ALSO BE PRINTED. A CONTROL CHARACTER IS NEEDED TO CONTINUE FROM THIS MESSAGE. THE FORMAT OF THE END OF PASS MESSAGE IS

NAME PASS 0001

10.8 ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 6 OF THIS DOCUMENT WILL BE USED.

10.9 SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 4 OF THIS DOCUMENT WILL BE USED.

10.10 PARAMETER CONTROL WORDS

THE CONSOLE PACKAGE USES THE LOCATIONS 20 21 22 FOR THE FOLLOWING PURPOSES.

LOCATION 20

LOCATION 21
HARDWARE IDENTIFIER 1

LOCATION 22
HARDWARE IDENTIFIER 2

LOCATION 0021

BIT	OCTAL VALUE	FUNCTION WHEN 0	FUNCTION WHEN 1
---	-----	-----	
11.	APT-8 HOOKS		

11.1 DESCRIPTION

TWO INTERFACES HAVE BEEN PROVIDED WHICH ALLOW THIS DIAGNOSTIC TO RUN UNDER THE STANDARD APT-8 SYSTEM. THESE INTERFACES ARE:

1. TIMING INTERFACE
2. ERROR INTERFACE

EACH WILL BE EXPLAINED IN DETAIL.

11.2 SETUP

ONLY HARDWARE CONFIGURATION WORD 2, ADDRESS 22, NEED BE ESTABLISHED. THE FOLLOWING INFORMATION MUST BE INDICATED:

1. SINGLE OR MULTIPLE DRIVE TESTING.
2. DRIVE OR DRIVES TO BE TESTED.
3. DIAGNOSTIC RUNNING UNDER APT-8.

IF SINGLE DRIVE TESTING BIT 5 OF ADDRESS 22 MUST BE SET TO A ONE (1) WITH BITS 6-11 CONTAINING THE DRIVE TO BE TESTED. IF MULTIPLE DRIVES ARE TO BE DONE BIT MUST BE SET TO A ZERO (0) AND BIT 6-11 CONTAINING THE HIGHEST NUMBER DRIVE TO BE TESTED. WHEN MULTIPLE DRIVE TESTING ONLY A SPECIFIC NUMBER OF DRIVES CAN BE INDICATED. THE PROGRAM ASSUMES THE DRIVES ARE TO BE DONE BEGINNING WITH DRIVE ZERO (0) AND FINISHING WITH THE HIGHEST DRIVE INDICATED. IF MULTIPLE DRIVES OTHER THAN CONSECUTIVELY NUMBERED DRIVES BEGINNING WITH DRIVE ZERO (0) ARE TO BE DONE, THEY MUST BE DONE AS SINGLE DRIVES AND TESTED INDEPENDANTLY.

THE PROGRAM ALLOWS ONLY DRIVES ZERO (0) THROUGH THREE (3) TO BE TESTED AT THIS TIME.

BIT ZERO OF ADDRESS 22 MUST BE SET TO A ONE TO INDICATE THAT THE PROGRAM WILL RUN UNDER APT-8.

NOTE: IT SHOULD BE NOTED AT THIS TIME THAT WHILE RUNNING UNDER APT-8 THE HARDWARE SWITCH REGISTER IS INOPERATIVE. ONLY THE HALT AND SINGLE STEP SWITCH WILL EFFECT THE PROGRAM RUN.

11.3.1. TIMING

APT-8 IS NOTIFIED OF PROGRAM RUN BETWEEN .2 SEC
AND 2.0 SEC ON A 1.2 MICROSECOND MEMORY CYCLE. THIS WILL ALLOW
THE DIAGNOSTIC TO RUN WITHOUT CAUSING AN APT-8 TIMEOUT ERROR
IF THE DIAGNOSTIC IS TO BE RUN ON ON THE SLOWER MOS MEMORY.

11.3.2. ERRORS

ONLY THE ERROR PC IS REPORTED TO APT-8
SYSTEM. ERRORS WHICH CAUSE A PROGRAMMED HALT CAUSE A TIMEOUT
ERROR. IF A PROGRAMMED HALT SHOULD OCCUR, THE ERROR PC WILL
APPEAR IN THE AC ON THE DEVICE UNDER TEST. PROGRAMMED HALTS
ARE EXPLAINED EARLIER IN THIS DOCUMENT.

11.4. LOADING PRECAUTIONS

THIS PROGRAM SHOULD BE LOADED IN SCRIPT MODE INDICATING
TO APT THAT CROR CHECK SUMS ARE TO BE IGNORED.

SEP 0023

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111 /
112 /*****
113 /CAPASS
114 /THIS IS CALLED AT THE END OF EACH PROGRAM COMPLETION
115 /THE VALUE OF **CNTVAL** WILL BE DETERMINED BY THE TIME IT TAKES
116 /THE PROGRAM TO COMPLETE THIS MANY CAPASS TO BE IN THE 1 TO 4 MINUTE
117 /RANGE
118 /
119 /      CAPASS=JMS      XCOPAS
120 /EX. OF CALL          /
121 /                      JMP      HLT
122 /                      START1
123 /
124 /RETURN TO LOCATION CALL PLUS ONE WITH THE AC=0 IF NON CONSL PACKAGE AND HLT
125 /IF CONTINUE TO RUN THEN RETURN TO CALL PLUS2 AC=0
126 /THE LOCATION SETUP2 IS THE MASK BIT FOR THE HALT AT END OF PASS
127 /CHECK THAT IT IS CORRECT FOR THE CURRENT PROGRAM
128
129 /CALLS USED BY XCOPAS ARE: CHKCLA=XCRCRLF=XCRCCTA=XCBSW=XCOPNT=XC0ING=
130
131
132 XCOPAS, 0
133   CLA      CHKCLA
134   JMP      DOPACK
135   DOPACK   5212
136   DOPACK   4776
137   DOPACK   4262
138   DOPACK   3755
139   DOPACK   3247
140   DOPACK   2740
141   DOPACK   2232
142   DOPACK   1724
143   DOPACK   1216
144   DOPACK   0708
145   DOPACK   0200
146   DOPACK   0000
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166 0241 2251 15-APR-76 13:24 PAGE 1-3
167 0242 2247 15-APR-76 13:24 PAGE 1-3
168 0243 5230 15-APR-76 13:24 PAGE 1-3
169 0244 1251 15-APR-76 13:24 PAGE 1-3
170 0245 2532 15-APR-76 13:24 PAGE 1-3
171 0246 5532 15-APR-76 13:24 PAGE 1-3
172 0247 0000 15-APR-76 13:24 PAGE 1-3
173 0248 0000 15-APR-76 13:24 PAGE 1-3
174 0249 0000 15-APR-76 13:24 PAGE 1-3
175 0250 0000 15-APR-76 13:24 PAGE 1-3
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205 0280 0000 15-APR-76 13:24 PAGE 1-3
206 0281 0000 15-APR-76 13:24 PAGE 1-3
207 0282 0000 15-APR-76 13:24 PAGE 1-3
208 0283 0000 15-APR-76 13:24 PAGE 1-3
209 0284 0000 15-APR-76 13:24 PAGE 1-3
210 0285 0000 15-APR-76 13:24 PAGE 1-3
211 0286 0000 15-APR-76 13:24 PAGE 1-3
212 0287 0000 15-APR-76 13:24 PAGE 1-3
213 0288 0000 15-APR-76 13:24 PAGE 1-3
214 0289 0000 15-APR-76 13:24 PAGE 1-3

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3E0 0027

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270      0321      7012      RTR
271      0322      4764*     JMS
272      0323      1736      TAD I
273      0324      0363      AND
274      0325      7450      SNA
275      0326      5703      JMP I
276      0327      1362      TAD
277      0328      7500      SNA
278      0329      1361      TAD
279      0330      1360      TAD
280      0331      4764*     JMS
281      0332      2336      ISZ
282      0333      5310      PSTOR
283      0334      0000      PSTOR, 0
284      0335      0000      /*****
285
286
287
288      /CAPAUS
289      /THIS ROUTINE WILL CHECK IF THE CONSOL PACKAGE IS ACTIVE, IF ACTIVE
290      /IT WILL RETURN TO CALL PLUS ONE AND 0, AND NO THAT INSTRUCTION.
291      /IF THE CONSOL PACKAGE IS NOT ACTIVE THE CALL WILL BE REPLACED
292      /WITH A 7402 HALT AND THEN RETURN TO THE HALT.
293
294      /
295      CAPAUS= JMS XCBAU
296
297      /
298      /
299      /
300      /
301      /CALLS USED ARE -CHKCLA-
302
303
304
305      0337      0000      XCBAU, 0
306      0338      7380      CLA CLL
307      0341      4777*     JMS
308      0342      5350      JMP CADN3
309      0343      7040      CHA
310      0344      1337      TAD
311      0345      3337      DCA XCBAU
312      0346      1357      TAD
313      0347      3737      DCA I
314      0350      5737      JMP I XCBAU
315
316
317      0357      7402      /CHECK LNC 22 BIT 3 CONSOL BIT
318      0360      0240      /GN ON CONSOL PART RETURN CALL+1
319      0361      0100      /DEACTIVE CONSOL PACKAGE PUT MLY IN CALL
320      0362      3740      /GET CORRECT RETURN ADORS
321      0363      0077      /SET UP RETURN
322      0364      1077      /GET CODE FOR MLY
323      0365      7700      /PUT HALT IN CALL LOCATION
324      1075      /GN TO HALT OR RETURN TO NEXT LOCATION

```

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335 /C6CNTP
336 /THIS ROUTINE WILL CHECK FOR THE PRESENCE OF CONTROL CHARACTERS
337 /IT WILL CHECK FOR THE FOLLOWING CHAR C-R-Q-L-S
338 /C6CNTP: JMS XC6CNT
339 /
340 /FX. JMS XC6CNT
341 /
342 / JMS XC6CNT
343 / JMS XC6CNT
344 / JMS XC6CNT
345 /
346 /RETURN IS TO CALL PLUS ONE IF CONTINUE
347 /RETURN IS TO CALL PLUS TWO IF INMODE SET AND NOT CONTROL CHAR
348 /RETURN IS TO CALL PLUS TWO IF INMODE IS NOT SET AND NO
349 /CONTROL CHAR. THIS WILL POINT THE CHARACTER AND A ?
350 /CLEAR THE AC AND RETURN CALL+2.
351 /
352 /
353 /CALLS USED ARE:CHKCLA-XC6TYPE-XC6CRLF-C6GET-UPAROW-XC6VI-XC6PSX-
354 /
355 /
356 /
357 /
358 /C6CNT, 0
359 /DCA
360 /ACSAVE
361 /CHKCLA
362 /+3
363 /JMS
364 /JMS
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319	0426	1772"	NONEA,	TAD	INMODE	/CHECK IF PROGRAM EXPECTS CHAR
320	0427	7540	NONEA,	JMP CLA	1=CHAR EXPECTED N=NO CHAR EXPECTED	
321	0428	5540	JMP	EXITA	/CHAR EXPECTED	
322	0431	1773"	TAD	CACHAR	/GET CHAR -NOT CONTROL+NOT EXPECTED	
323	0432	4771"	JMS	XCATYPE	/CARBNT CHAR	
324	0433	1372"	TAD	(217	/GET CODE FOR "7"	
325	0434	4771"	JMS	XCATYPE		
326	0435	4767"	JMS	XCCRLF		
327	0436	2500	JMS	XCCRLF	/RUMP RETURN	
328	0437	5500	JMP I	XCCNT	/EXIT CALL+2	
329	0440	2500	132	XCCNT	/RUMP RETURN FOR MAIN PROGRAM CHECK OF CHAR	
330	0441	1773"	TAD	CACHAR	/PUT CHAR IN AC.	
331	0442	5500	JMP I	XCCNT	/EXIT	
332	0443	1773"	TAD	CACHAR	/GET THE CONTENTS OF CHAR	
333	0444	1366	TAD	(102	/AND 100 TO FORM A GOOD ASCII CHARACTER	
334	0445	3773"	DCA	CACHAR	/RESTORE CORRECT CHAR	
335	0446	1260	TAD	XTRALP	/GET START OF TABLE R	
336	0447	1255	TAD	INDEXA	/GET NOW FAR INTO TABLE	
337	0448	1254	DCA	GOTCA	/STORE IT	
338	0449	1654	TAD I	GOTCA	/GET THE ROUTINE STARTING ADDRESS	
339	0452	1254	JMP I	GOTCA	/STORE IT IN HERE	
340	0453	5504	JMP I	GOTCA	/GETO CONTROL CHAR ROUTINE	
341	0454	5504	JMP I	GOTCA	/AND OF CNTRL ROUTINE TO EXECUTE	
342	0455	0000	INDEXA	GOTCA	/DISPLACEMENT INTO CNTRL TABLE	
343	0456	0000	GETDAT	GOTCA	/LOCATION OF ANDRS OF CONTROL CHAR.	
344	0457	0461	XTRALA	TARLA	/ANDRS OF TABLE	
345	0460	0471	XTRALA	TARLA	/ANDRS OF TABLE	
346	0461	7575	TARLA	TARLA	/CNTRL C BACK TO MONITOR 203	
347	0462	7564	TARLA	TARLA	/CNTRL L SWITCH ERROR PRINTING DEVICE 214	
348	0463	7557	TARLA	TARLA	/CNTRL R START DISPLAYING CHAR. AGAIN 221	
349	0464	7556	TARLA	TARLA	/CNTRL R BACK TO REPLYING OF PROGRAM 222	
350	0465	7555	TARLA	TARLA	/CNTRL S STOP SENDING CHAR TO DISPLAY WAIT FOR CNTRL Q 223	
351	0466	7573	TARLA	TARLA	/CNTRL E CONTINUE WITH PROGRAM 225	
352	0467	7574	TARLA	TARLA	/CNTRL R CHANGE SWITCH REGISTER ON FLY	

ADDRESS	HEX	ASCII	COMMENT
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0029	0029		
0030	0030		
0031	0031		
0032	0032		
0033	0033		

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434 PAL10 V142A 15-APR-76 13:24 PAGE 1-8
435 0566 5335 RYRETR, DCA CASETS /CLEAR THE SORT FLAG
436 0507 4765 JMS CAGET /RESTOR REGISTERS
437 0510 5736 JMP I CORETR /EXIT TO ADDRESS SET BY CONTROL S
438
439 /CONTROL R
440 /GO TO THE QUESTION CASWIT
441 0511 3764 CNTRLR, DCA TTYLPT /CLEAR THE TYPE FLAG SET TO TTY
442 0512 3335 DCA CASETS /CLEAR SORT FLAG FOR CNTRL S
443 0513 3772 DCA INMODE /PRINT THE ^ AND CACHAR
444 0514 4765 JMS UPAROW /CLEAR FLAG FOR CNTRL D OR R
445 0515 3762 DCA CASWST
446 0516 6203 CTF CNF R
447 0517 5720 JMP I XROSW /GO TO ADORS OF CASWIT
448 0520 0200 /DOSW IS LABEL FOR CASWIT QUESTION
449
450 /CONTROL S
451 /STOP SENDING CHAR. TO DISPLAY UNTIL A *Q IS RECEIVED
452
453 /CNTRL S, T40
454 0521 1335 S2A CLA /IF1 NO NOT STORF IN CORETR
455 0522 7640 JMP C0807 /DOOT SET UP CORETR
456 0523 5327 JMS TAO /MAKE RETURN CALL PLUS 2
457 0524 7001 JMS TAO /GET RETURN FOR THIS CALL
458 0525 1200 JMS DCA XCACNT /STORE IT HERE FOR USE BE CNTRL D
459 0526 3336 JMS DCA CASETS /SET FLAG TO SAVE CALL
460 0527 2335 JMS T40 XCATYI /LOOK FOR THE INPUT
461 0528 4765 JMS C08ET /GET REGISTERS
462 0531 4765 JMS XCACNT /CHECK FOR THE CONTROL CHAR
463 0532 4200 CLA
464 0533 7200 JMP CNTRL S /IF NOT A CNTRL Q R C REAK
465 0534 5321 CASETS, R
466 0535 0000 CORETR, R
467 0536 0000
468
469 /SWITCH OUTPUT FROM ONE OUTPUT DEVICE TO ANOTHER -THE TWO OUTPUTS ARE THE
470 /CONSOLE AND THE PRINTER WITH DEVICE CODE 66.
471
472 /CNTRL, T40
473 0537 1764 TTYLPT /GET PRESENT CASWIT INDICATOR
474 0540 7000 CMA /COMPLEMENT IT
475 0541 3764 DCA TTYLPT /STOR NEW CASWIT
476 0542 4765 JMS UPAROW /CAPUT ^ AND CHAR ON NEW DEVICE
477 0543 4765 JMS C08ET /RESTORE THE REGISTERS
478 0544 5600 JMP I XCACNT /EXIT
479
480 /CONTROL E
481 /CONTINUE RUNNING FROM A INQUIRE OR ERROR
482
483 /CNTRL, JMS
484 0545 4765 UPAROW /PRINT THE CONTROL CHAR
485 0546 3765 DCA C08WST /CLEAR ENTRY FLAG.
486 0547 4765 JMS C08ET /GET THE REGISTERS
487 0550 5600 JMP I XCACNT /RETURN TO CALL PLUS ONE
488

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489 PAL10 V142A 15-APR-76 13:24 PAGE 1-9
490
491 /CONTROL C
492 /RETURN TO MONITOR CONTROL C
493 0551 3764 CNTRL, DCA TTYLPT /CLEAR THE LPT FLAG TO PRINT ON DISPLAY
494 0552 3765 DCA C08WST /CLEAR ENTRY FLAG.
495 0553 4765 JMS UPAROW /CAPUT ^ AND LETTER IN CHAR
496 0554 6203 CTF CNF /GO TO 0 FLD
497 0555 6007 JMP I (7600 /CLEAR THE WORLD
498 0556 5760 /GO TO DIAGNOSTIC MONITOR
499
500 /*****
501
502 0560 7600
503 0561 0272
504 0562 0745
505 0563 0615
506 0564 1121
507 0565 0624
508 0566 0100
509 0567 1023
510 0570 0277
511 0571 1077
512 0572 1076
513 0573 1075
514 0574 1346
515 0575 1347
516 0576 1200
517 0577 1343
518 0577 1343
519
520 PAGE
521 /CONTROL D
522 /CHANGE THE SWITCH REGISTER ANYTIME CNTRL D AND RETURN TO
523 /THE PROGRAM RUNNING.
524
525 /CNTRL, JMS
526 0600 4215 UPAROW /CHECK IF THE RETURN ADORS IS SAFE
527 0601 1213 CASETO /DO NOT CHANGE THE RETURN ADORS
528 0602 7640 S2A CLA /GET THE RETURN ADORS AND SAVE IT
529 0603 5207 JMP XCACNT /SAVE THE RETURN HERE
530 0604 1777 JMS TAO /INDICATE RETURN SAVED DONOT DESTROY
531 0605 3214 DCA CASETO /GO CHANGE THE SWITCH REGISTER
532 0606 2213 JMS DCA C08WST /CLEAR THE FLAG
533 0607 4256 JMS C08ET /RESTORE THE AC NO LINK ETC
534 0610 3213 JMP I CORETR /RETURN TO THE PROGRAM
535 0611 4224
536 0612 5614
537
538 0613 0000 CASETO, R
539 0614 0000 CORETR, R
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	0652	1124
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PAL10  VI424  15-APR-76  13124  PAGE 1-12

649  0712  2106  15Z  TMRNT  /AUMP COUNT
650  0713  5303  JMD  GETCH1  /JMD BACK+GET NEXT CHAR
651  0714  5342  JMD  ENDT1  /END A CHAR CATYPED IN
652  0715  0000  TSTCHA, 0
653  0716  7041  CTA  /CMPL CHAR IN AC
654  0717  1356  TAD  (215  /TEST IF IT IS A CARRIAGE RETURN
655  0720  7650  JMD  ENDT  /SKIP IN NOT CR
656  0721  5342  JMD  ENDT  /HAS CARRIAGE RETURN
657  0722  1774  TAD  ENDT  /NOT CR, GET CHAR
658  0723  1355  TAD  (250  /CHECK IF IT IS IN RANGE
659  0724  7710  STA  CLA  /IF NOT POSITIVE CARR CHAR SMALLER THEN 260
660  0725  5316  JMD  ENDT  /CARR -CHAR TOO SMALL
661  0727  1774  TAD  ENDT  /GET CHAR
662  0727  1354  TAD  (270  /GET A -270+CHECK IF IT IS LARGER THEN 7
663  0732  7720  SWA  CLA  /SKIP IF LESS THEN 7
664  0733  5316  JMD  ENDT  /CARR ON CHAR NOT IN RANGE
665  0732  1774  TAD  ENDT  /GET CHAR
666  0733  2313  AND  (7  /MASK FOR RIGHT BYTE
667  0734  3774  DCA  CACHAR  /STORE IN CHAR
668  0735  5316  JMD  ENDT  /GET CHAR IN AC
669  0736  1352  JMD  I  /EXIT
670  0737  4774  JMS  XCATYPE  /XCATYPE
671  0740  4774  JMS  XCATYPE  /
672  0741  5286  JMD  ENDT  /EXIT+ASK AGAIN
673  0742  4774  JMS  XCATYPE  /ON A CR LF
674  0743  7305  ENDT,  JMS  XCATYPE  /CLEAR THE PSW ENTRY FLAG
675  0744  7305  ENDT,  JMS  XCATYPE  /EXIT ROUTINE
676  0745  5556  JMD  I  /
677  0746  2000  CASHST, 0
678  0746  2000  CASHST, 0
679  0747  2322  TMRNT, 0
680  0750  7340  MESA,  TEXT  "SR"
681  0751  0000
682  0752  0277  0752  0277  0752  0277
683  0753  0007  0753  0007  0753  0007
684  0754  7512  0754  7512  0754  7512
685  0755  7520  0755  7520  0755  7520
686  0756  0215  0756  0215  0756  0215
687  0757  7774  0757  7774  0757  7774
688  0758  1863  0758  1863  0758  1863
689  0761  1876  0761  1876  0761  1876
690  0762  0040  0762  0040  0762  0040
691  0763  1000  0763  1000  0763  1000
692  0764  0515  0764  0515  0764  0515
693  0765  0072  0765  0072  0765  0072
694  0766  0303  0766  0303  0766  0303
695  0767  1200  0767  1200  0767  1200
696  0770  1345  0770  1345  0770  1345
697  0771  1347  0771  1347  0771  1347
698  0772  1346  0772  1346  0772  1346
699  0773  1023  0773  1023  0773  1023
700  0774  1073  0774  1073  0774  1073

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PAL12  VI424  15-APR-76  13124  PAGE 1-13

702  0775  1077  0775  1077  0775  1077
703  0776  0336  0776  0336  0776  0336
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/PRINT THE ERROR MESSAGE
/PRINT THE PC STATEMENT
/GET PC

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15-APR-76	13:24	PAGE 1-18
971 1244 4770*	JMS	XCRCTA
972 1245 4771*	JMS	XCRPNT
973 1246 1333	MESAC	
974 1247 1345	TAD	ACSAVE
975 1250 4770*	JMS	XCRCTA
976 1251 4771*	JMS	XCRPNT
977 1252 1336	MESMG	
978 1253 1346	TAD	MOSAVE
979 1254 4770*	JMS	XCRCTA
980 1255 4771*	JMS	XCRPNT
981 1256 1341	MESFL	
982 1257 1347	TAD	FLSAVE
983 1260 4770*	JMS	XCRCTA
984 1261 4772*	JMS	XCRPNT
985 1262 4775*	JMS	XCRCTA
986 1263 4774*	JMS	XCRPNT
987 1264 7610	SXP CLA	
988 1265 5302	JMS	CARVP
989 1266 4767*	JMS	CARVP
990 1267 5300	JMS	CARVP
991 1270 4775*	JMS	CARVP
992 1271 4774*	JMS	XCRSW
993		
994 1322 7610	SXP CLA	
995 1323 5607	JMS	XCRPNT
996 1324 1366	TAD	(7402)
997 1325 3744	NCA I	PCSAVE
998 1326 4775*	JMS	PCSAVE
999 1327 5744	JMS	PCSAVE
1000 1330 4775*	JMS	PCSAVE
1001 1331 5607	JMS	XCRPNT
1002		
1003		
1004 1332 7402	ROUTNS, NLT	
1005 1333 7200	NOP	
1006 1334 4317	NCA	WYAC
1007 1335 6201	CONF	
1008 1336 1020	TAD	SXP
1009 1337 3765	NCA I	(SMR)
1010 1338 1776	TAD I	(CLASTX)
1011 1339 3315	NCA	CLTRN
1012 1342 1317	NCA	WYAC
1013 1343 6202	CIF	
1014 1344 5715	JMS I	CLTRN
1015		
1016 1345 0000	CLTRN, N	
1017 1346 0000	REALPC, N	
1018 1347 0000	WYAC, N	
1019		
1020 1322 0410	FRAMES, TEXT	"CHKRAF FAILED"
1021 1321 2213		
1022 1322 0206		
1023 1323 4040		
1024 1324 0601		
1025 1325 1114		

/PUT INSTRUCTION TO EXECUTE HERE.
/SAVE AC
/RETURN TO FIELD 0.

15-APR-76	13:24	PAGE 1-19
1026 0500		
1027 4000	MESPC, TEXT	"PC:"
1028 2203		
1029 7200	MESAC, TEXT	"AC:"
1030 2103		
1031 7200	MESMG, TEXT	"MG:"
1032 1521		
1033 4240	MESFL, TEXT	"FL:"
1034 0614		
1035 7200	PCSAVE, 7777	
1036 7777	ACSAVE, 7777	
1037 7777	MOSAVE, 7777	
1038 7777	FLSAVE, 7777	
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FIELD 2

0000 00000000 00000000 11111111 11111111 11000000 00000000 00000000 00000000
0100 00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 10000001 11111111 11111111
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/ SWITCH REGISTER.
/ CONTROL WORD 1
/ CONTROL WORD 2

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	PAL10	V1424	15-APR-76
1152	0226	5628	K01SKG, 018KG
1153	0027	6041	XPFCHK, MFCCHK
1154	0030	5656	XFIGRE, FIGURE
1155	0031	5047	XFLRUF, FLRUF
1156	0032	5435	XKLRFJ, KLRFJ
1157	0033	5134	XSKLAT, SKLAT
1158	0034	4000	XWT142, WT142
1159	0035	0222	TUSFLD, PRSFLD
1160	0036	6151	XPR147, PR147
1161	0037	6008	XPRR01, PRR01
1162	0047	5208	XPRR01, PRR01
1163	0041	4727	XTONWT, TONWT
1164	0042	4557	XCOMB1, COMB1
1165	0043	3628	XCOMP2, COMP2
1166	0044	5154	XOST, OST
1167	0046	6412	XOST, OST
1168	0046	6412	XOST, OST
1169	0047	1122	XSCAR, SCAR
1170	0047	1122	XSCAR, SCAR
1171	0051	4764	XSCAR, SCAR
1172	0052	4164	XSCAR, SCAR
1173	0053	4571	XSCAR, SCAR
1174	0054	4700	XSCAR, SCAR
1175	0055	4131	XSCAR, SCAR
1176	0056	5028	XSCAR, SCAR
1177	0057	6111	XSCAR, SCAR
1178	0057	6265	XSCAR, SCAR
1179	0061	4236	XSCAR, SCAR
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1206	0061	4236	XSCAR, SCAR

	PAL10	V1424	15-APR-76
1207	0115	5200	K5000, 5000
1208	0116	7771	K7771, 7771
1209	0117	0217	K0017, 0017
1210	0118	0037	K0037, 0037
1211	0121	6291	K0037, 0037
1212	0132	7740	K7740, 7740
1213	0133	7400	K7400, 7400
1214	0134	7600	K7600, 7600
1215	0135	1355	XLOAD, LOADCT
1216			/
1217			DECIMAL
1218			/
1219	0126	7764	M12, -12
1220			/
1221			CTAL
1222	0127	7103	KAFR00, AFRR0
1223	0127	7103	XTRCK, TRCK
1224	0127	7103	REGG, 0
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/
PAL10 15-APR-76 13124 PAGE 1-26
1282 0175 7777 KCNT, =1
1283 0200
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1285 0200
1286 0201 5206 JNP +6
1287 0202 5777 JNP MANUAL
1288 0203 5775 JNP CHANGE
1289 0204 5774 JNP MANRO
1290 0205 5773 JNP AUTO
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1372 0324 0301 TST2 /SCOPE LOOP POINTER
1373 0325 0707 /TEXT POINTER
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1375 /FORCE TIMING ERROR
1376
1377 /VERIFY A "TIMING ERROR" DOES OCCUR IN STATUS REGISTER
1378 /IF A FLAG IS ISSUED WITH THE COMMAND REGISTER IS SET TO
1379 /A FUNCTION OF "7".
1380
1381 0326 1137 TST3, TAN K7000 /SET CURRENT DRIVE
1382 0327 1156 TAN HOMEHA /LOAD COMMAND REGISTER
1383 0328 1172 TAN DRIVEN /SETUP TEXT POINTER
1384 0331 4454 LDCND K8006 /DLAC, LOAD DISK ADDRESS
1385 0333 3557 DCA T3T /WAIT FOR FROM SKIP
1386 0335 4452 JMP L0A0N /ERROR, NO SKIP OCCURRED
1387 0336 4452 JMP T3E /SETUP TEXT POINTER
1388 0337 1164 TAN K5100 /DLAC, LOAD DISK ADDRESS
1389 0340 1172 TAN K5100 /WAIT FOR FROM SKIP
1390 0342 1173 DCA CLC CLR BAR /ERROR, NO SKIP OCCURRED
1391 0343 1143 DCA K0000 /SETUP TEXT POINTER
1392 0344 1143 DCA K0000 /SETUP TEXT POINTER
1393 0345 4444 RNSAT /SETUP EXPECTED STATUS
1394 0346 4444 ACCM01 /READ STATUS REGISTER
1395 0347 7617 JMP CLA T3E /CHECK RESULTS
1396 0348 5355 JMP CLA T3E /STATUS IS O.K.
1397 0349 5355 JMP CLA T3E /FROM STATUS INCORRECT
1398 0350 5355 JMP CLA T3E /CLEAR STATUS
1399 0351 3143 DCA GOREN2 /SETUP EXPECTED STATUS
1400 0352 4444 RNSAT /READ STATUS
1401 0353 4437 ACCM01 /CHECK RESULTS
1402 0354 4437 NERR01 /ALL IS O.K.
1403 0355 4447 NERR01 /ERROR, TIMING SKIP OR STATUS
1404 0356 4447 TST3 /SCOPE LOOP POINTER
1405 0357 4447 JMP J ++1 /TEXT POINTER
1406 0358 4447 TST4
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1426 0400 4425 TST4, RECAL /RECALIBRATE
1427 0401 4406 TAT /TEXT POINTER
1428 0402 5204 JMP T4E /ERROR, SKIP OR STATUS
1429 0403 4437 NERR01 /O.K. TO NEXT TEST
1430 0404 4437 TST4 /ERROR, DISK SKIP OR STATUS
1431 0405 4406 TST4 /SCOPE LOOP POINTER
1432 0406 4406 TST4 /TEXT POINTER
1433 0407 5614 JMP J ++1
1434 0408 4410 TST5
1435 0409 4410 TST5
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1426 0400 4425 TST4, RECAL /RECALIBRATE
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1428 0402 5204 JMP T4E /ERROR, SKIP OR STATUS
1429 0403 4437 NERR01 /O.K. TO NEXT TEST
1430 0404 4437 TST4 /ERROR, DISK SKIP OR STATUS
1431 0405 4406 TST4 /SCOPE LOOP POINTER
1432 0406 4406 TST4 /TEXT POINTER
1433 0407 5614 JMP J ++1
1434 0408 4410 TST5
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1426 0400 4425 TST4, RECAL /RECALIBRATE
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1428 0402 5204 JMP T4E /ERROR, SKIP OR STATUS
1429 0403 4437 NERR01 /O.K. TO NEXT TEST
1430 0404 4437 TST4 /ERROR, DISK SKIP OR STATUS
1431 0405 4406 TST4 /SCOPE LOOP POINTER
1432 0406 4406 TST4 /TEXT POINTER
1433 0407 5614 JMP J ++1
1434 0408 4410 TST5
1435 0409 4410 TST5
1436 0410 4410 TST5
1437 0411 4410 TST5
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1402 1403 0441 7340 /
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1406 1407 0445 0425 /
1407 1408 0446 0493 /
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SEC 0053

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820 0055

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1700 0722 1015 TAD K0200 /ENABLE SET DONE
1701 0723 1104 TAD K3000 /FUNCTION SEEK ONLY
1702 0724 4450 LOCHD /LOAD COMMAND
1703 0725 4452 LOADD /LOAD AND GO
1704 0726 4444 ROSTAT /READ STATUS
1705 0727 4442 ACCM1 /CHECK RESULTS
1706 0730 7610 JMP CLA /O.K.
1707 0731 5353 JMP CLA T13E /ERROR, STATUS
1708 0732 4443 CLRALL /CLEAR STATUS
1709 0733 1157 TAD STCON /EXPECTED STATUS
1710 0734 3153 DCA GOREG2 /SETUP COMPARE
1711 0735 4444 ROSTAT /READ STATUS
1712 0736 4442 ACCM1 /CHECK RESULTS
1713 0737 7610 JMP CLA /O.K.
1714 0740 5353 JMP CLA T13E /ERROR, STATUS
1715 0741 7301 CLRALL /CLEAR CONTROL
1716 0742 3143 OCA GOREG2 /SETUP COMPARE
1717 0743 3143 OCA GOREG2 /READ STATUS
1718 0744 4444 ROSTAT /EXPECTED STATUS
1719 0745 7610 JMP CLA T13E /SETUP COMPARE
1720 0746 5353 JMP T13E /READ STATUS
1721 0747 2154 ISZ TCNTR1 /STATUS SHOULD BE 0000
1722 0748 2154 ISZ TCNTR1 /ERROR, STATUS
1723 0751 5307 JMP T13R /O.K.
1724 0752 4437 NERROR /TRY NEXT DRIVE
1725 0753 4440 T13E /O.K. 4096 LOOPS
1726 0754 4440 ERROR /ERROR, STATUS
1727 0755 0702 T113 /SCOPE LOOP POINTER
1728 0756 5300 5300 /TEXT POINTER
1729 0757 5757 /TO NEXT TEST
1730 0757 1005 TST14P, TST14-3
1731 /
1732 NRES1, TEXT "RKE0F DRIVE CONTROL TEST"

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1733 0760 2213 0761 7005 /
1734 0776 4374 /
1735 0777 6110 PAGE
1736 /
1737 /SURROUTINE TO ISSUE DSKP NISK SKIP TOT
1738 /
1739 1000 0000 DSKP 0
1740 1001 6741 TOT1,
1741 1002 7410 SKP
1742 1003 2200 ISZ DSKP

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/
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1743 1004 5600 JMP I DSKP
1744 /
1745 /SELECT ERROR TEST
1746 /
1747 /VERIFY THAT DISK CAPACITY EXCEEDED DOES OCCUR
1748 /
1749 /
1750 1005 2131 ISZ REGR
1751 1006 7344 DCA N1775
1752 1007 3777 DCA CNT
1753 1010 1066 TST14, TAD TRK12
1754 1011 1012 TAD K0020
1755 1012 3134 DCA TCNTR1
1756 1013 7301 T14R, CLA CLL IAC
1757 1014 4453 CLRALL
1758 1015 7330 CLA CLL CML RAR
1759 1016 1074 TAD K0002
1760 1017 3143 DCA GOREG2
1761 1020 7301 CLA CLL IAC
1762 1021 1104 TAD K1000
1763 1022 1072 TAD DRIVNO
1764 1023 4450 LOCHD
1765 1024 1134 TAD TCNTR1
1766 1025 4452 TAD SKPMAT
1767 1026 4433 JMP T14KE
1768 1027 5267 ROSTAT
1769 1030 4444 ACCM1
1770 1031 4442 ROSTAT
1771 1032 7610 JMP CLA
1772 1033 5263 JMP CLA T14SE
1773 1034 7301 CLA CLL IAC
1774 1035 4453 CLRALL
1775 1036 1150 TAD CMREG
1776 1037 1015 TAD K0200
1777 1040 4450 LOCHD
1778 1041 4433 SKPMAT
1779 1042 5267 JMP T14KE
1780 1043 7330 DCA CML RAR
1781 1044 3143 OCA GOREG2
1782 1045 4444 ROSTAT
1783 1046 4442 ACCM1
1784 1047 7610 JMP CLA
1785 1050 5263 JMP T14SE
1786 1051 1072 TAD DRIVNO
1787 1052 4450 LOCHD
1788 1053 3143 DCA GOREG2
1789 1054 4444 ROSTAT
1790 1055 4442 ACCM1
1791 1056 7610 JMP CLA
1792 1057 5263 JMP T14SE
1793 1060 2134 ISZ TCNTR1
1794 1061 2134 JMP T14R
1795 1062 4437 NERROR
1796 1063 4440 ERROR
1797 1064 1010 TST14

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/RETURN.
/ADDRESS POINTER
/ENABLE CLEAR CONTROL BIT
/CLEAR CONTROL
/EXPECTED STATUS
/SETUP COMPARE REGISTER
/EXTENDED TRACK BIT
/FUNCTION SEEK ONLY
/CURRENT DRIVE
/LOAD COMMAND
/LOAD AND GO
/WAIT FOR SKIP
/ERROR, NO SKIP
/READ STATUS
/CHECK RESULTS
/STATUS O.K.
/ERROR, STATUS
/ENABLE CLEAR CONTROL BIT
/CLEAR CONTROL
/GET LAST COMMAND
/GET ENABLE SEEK DONE BIT
/LOAD COMMAND
/WAIT FOR DISK SKIP
/ERROR, SKIP
/EXPECTED STATUS
/READ STATUS
/CHECK RESULTS
/STATUS O.K.
/ERROR
/LOOP
/O.K. TO NEXT TEST
/ERROR, NISK CAPACITY EXCEEDEN
/SCOPE LOOP POINTER

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1798 1065 5300 5300 /MODIFIED TEXT POINTER
1799 1066 5372 JMP *+ /TO NEXT TEST
1800 1067 4440 T14KE, ERROR /ERROR, DISK SKIP
1801 1070 1010 T5716 /SCOPE LOOP POINTER
1802 1071 0006 /TEXT POINTER

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1803

/STATUS TEST

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1805 /
1806 /VERIFY THAT SKIP AND STATUS DOES OCCUR
1807 /AFTER 256 WRITE ALL AND READ ALL BREAKS.
1808 /THIS SHOULD WRITE ALL ZEROS ON AND
1809 /READ ALL ZEROS OFF THE DISK SECTOR 00000.

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1810 1011 1072 4432 K1BUF /ZERO WRITE BUFFER
1811 1012 1073 1115 T5715, TAD /WRITE ALL FUNCTION
1812 1013 1074 3150 DCA CMREG /SETUP COMMAND
1813 1014 1075 4426 DISKGO /DISK WRITE ALL
1814 1015 1076 5306 T15T /TEXT POINTER
1815 1016 1077 5306 JMP T15E /ERROR, SKIP OR STATUS
1816 1017 1078 1017 TAD K1000 /FUNCTION READ ALL
1817 1018 1079 3150 DCA CMREG /SETUP COMMAND REGISTER
1818 1019 1080 4426 DISKGO /DISK READ ALL
1819 1020 1081 1110 T15T /TEXT POINTER
1820 1021 1082 1110 JMP T15E /ERROR, SKIP OR STATUS
1821 1022 1083 4437 MERROR /O.K. TO NEXT TEST
1822 1023 1084 4440 T5715 /ERROR, WRITE ALL
1823 1024 1085 1073 T15T /SCOPE LOOP POINTER
1824 1025 1086 5300 /MODIFIED TEXT POINTER

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1825

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1826 1027 /STATUS TEST
1827 1028 /
1828 1029 /VERIFY THAT SKIP AND STATUS DOES OCCUR AFTER
1829 1030 /128 WRITE ALL AND READ ALL BREAKS.
1830 1031 /THIS SHOULD WRITE ALL ZEROS ON AND READ ALL
1831 1032 /ALL ZEROS OFF THE DISK SECTOR 00000.

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1832 1033 1111 1115 T5716, TAD /FUNCTION WRITE ALL
1833 1034 1112 1014 TAD K0100 /HALF BIT
1834 1035 1113 3150 DCA CMREG /SETUP COMMAND
1835 1036 1114 4426 DISKGO /DISK WRITE ALL
1836 1037 1115 1130 T16T /TEXT POINTER
1837 1038 1116 5320 JMP T16E /ERROR, DISK SKIP OR STATUS
1838 1039 1117 1017 TAD K1000 /FUNCTION READ ALL
1839 1040 1118 1014 TAD K0100 /HALF BIT
1840 1041 1119 3150 DCA CMREG /SETUP COMMAND
1841 1042 1120 4426 DISKGO /DISK READ ALL
1842 1043 1121 1130 T16T /TEXT POINTER
1843 1044 1122 4426 MERROR /ERROR, SKIP OR STATUS
1844 1045 1123 1130 T16E /O.K. TO NEXT TEST
1845 1046 1124 5326 MERROR /ERROR, WRITE ALL
1846 1047 1125 4437 T5716 /SCOPE LOOP POINTER
1847 1048 1126 4440 T16T, 5300 /MODIFIED TEXT POINTER
1848 1049 1127 1111
1849 1050 1130 5300

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1850 /VERIFY ALL SECTORS CAN BE ACCESSED.
1851 /
1852 /

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1853 /VERIFY A WRITE ALL TO ALL OF CYLINDER 0
1854 /AND USE DATA PATTERN 2525+5252.
1855 /MAKE THE FIRST TWO WORDS IN THE BUFFER
1856 /EQUAL THE DISK ADDRESS, CHECK THE DATA WITH
1857 /READ ALL.

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1858 1058 1131 4525 JMS I /WILL SET UP COUNTERS FOR NEXT TESTS
1859 1059 1132 7771 TAD K7740 /SETUP SECTOR COUNTER
1860 1060 1133 1122 T5717, DCA TCNTR1 /FILL OUTBOUND BUFFER
1861 1061 1134 3134 DCA K2525 /FUNCTION WRITE ALL
1862 1062 1135 1113 T173, TAD K2525 /SETUP COMMAND
1863 1063 1136 4431 TAD K5000 /MASK OFF SECTORS
1864 1064 1137 1115 DCA CMREG /SETUP ADDRESS WORD IN BUFFER
1865 1065 1138 3150 TAD TCNTR1 /GET DRIVE NUMBER
1866 1066 1139 1134 AND K0037 /SETUP ADDRESS WORD IN BUFFER
1867 1067 1140 0120 TAD I /DISK WRITE ALL
1868 1068 1141 3463 DCA I /TEXT POINTER
1869 1069 1142 1072 TAD I /ERROR, SKIP OR STATUS
1870 1070 1143 3463 TAD I /WILL DATA BUFFER
1871 1071 1144 3463 DCA I /FUNCTION READ ALL
1872 1072 1145 3463 TAD I /ERROR, STATUS OR SKIP
1873 1073 1146 1063 T17T /TEXT POINTER
1874 1074 1147 4426 T17T /ERROR, SKIP OR STATUS
1875 1075 1148 1173 JMP T17E /WILL DATA BUFFER
1876 1076 1149 5371 K1BUF /FUNCTION READ ALL
1877 1077 1150 1017 TAD K1000 /SETUP COMMAND
1878 1078 1151 1017 DCA CMREG /MASK OF SECTORS
1879 1079 1152 4432 TAD TCNTR1 /DISK READ ALL
1880 1080 1153 1017 TAD K1000 /TEXT POINTER
1881 1081 1154 3150 DCA CMREG /ERROR, STATUS OR SKIP
1882 1082 1155 1134 AND K0037 /WILL DATA BUFFER
1883 1083 1156 0120 TAD K0037 /FUNCTION READ ALL
1884 1084 1157 4426 DISKGO /MASK OF SECTORS
1885 1085 1158 1173 T17T /DISK READ ALL
1886 1086 1159 5371 JMP T17E /TEXT POINTER
1887 1087 1160 1115 TAD K2525 /ERROR, STATUS OR SKIP
1888 1088 1161 1115 TAD K2525 /TEXT POINTER
1889 1089 1162 1115 TAD K2525 /TEXT POINTER
1890 1090 1163 4430 FIGURE /WORD BY WORD COMPARE OF DATA
1891 1091 1164 7617 JMP KMP CLA /THIS SECTOR O.K.
1892 1092 1165 5371 JMP T17E /ERROR, DATA
1893 1093 1166 2134 ISZ TCNTR1 /UPDATE SECTOR COUNTER
1894 1094 1167 5375 JMP T173 /SKIP NEXT SECTOR
1895 1095 1168 5375 JMP T173 /O.K. TO NEXT TEST
1896 1096 1169 4440 MERROR /ERROR, READ ALL
1897 1097 1170 4440 T5717 /SCOPE LOOP POINTER
1898 1098 1171 5373 T17T, 5373 /TEXT POINTER
1899 1099 1172 1173 /
1900 1099 1173 5373 /
1901 1099 1174 5375 /
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1977 1099 1250 5375 /
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1999 1099 1272 5375 /
2000 1099 1273 5375 /

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1899 /PAGE
1900 /
1901 /VERIFY ALL SECTORS CAN BE ACCESSED.
1902 /
1903 /VERIFY A WRITE DATA TO ALL OF CYLINDER 0
1904 /AND USE DATA PATTERN 5252+5252.
1905 /MAKE THE FIRST TWO WORDS OF THE BUFFER
1906 /EQUAL THE DISK ADDRESS, CHECK THE
1907 /DATA WITH READ DATA.

```

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/VERIFY HALF BLOCK TRANSFERS.
/VERIFY THAT DISK STOPS BREAK AFTER 128
/IF THE HALF BIT IS SET. THE REMAINDER OF THE
/IF THE BUFFER SHOULD BE 0000.
/IF THE FIRST TWO WORDS OF THE BUFFER SHOULD
/EQUAL THE ABSOLUTE DISK ADDRESS.
/IF THE DATA PATTERN USED IS 5252+5253.

1272 1114 T5720, TAD K5252 /FILL BUFFER WITH DATA
1273 4031 FILBUF
1274 1002 TAD DRIVNO /MAKE DISK ADDRESS WORD
1275 3648 DCA I XSTRK /MAKE DISK ADDRESS WORD
1276 3443 DCA I XSTRK /FUNCTION WRITE ALL
1277 1115 TAD K5000 /SETUP COMMAND
1278 3150 DCA CMREG /DISK WRITE ALL
1279 4426 T20T /TEXT POINTER
1301 1301 JUMP T20E /TEXT POINTER
1302 1301 JUMP T20E /TEXT POINTER
1303 5313 JUMP T20E /TEXT POINTER
1304 5437 CLRALL /CLEAR STATUS
1305 4433 KILBUF /CLEAR BUFFER
1306 4432 KILBUF /FUNCTION READ ALL
1307 1014 TAD K1000 /HALF BIT
1308 1217 TAD K0100 /SETUP COMMAND
1310 3150 DCA CMREG /SETUP COMMAND
1311 4426 T20T /TEXT POINTER
1312 1301 JUMP T20E /TEXT POINTER
1313 5313 JUMP T20E /TEXT POINTER
1314 1114 TAD K5252 /WORD BY WORD COMPARE DATA
1315 4427 MAFCHK /O.K. TO NEXT TEST
1316 4437 MERROR /ERROR, DATA BREAK
1317 4440 T20E /ERROR, DATA BREAK
1318 1272 T5720 /SCOPE LOOP POINTER
1319 1272 T20T /TEXT POINTER
1321 5313 T20T, 5313 /TEXT POINTER

/VERIFY HALF BLOCK TRANSFERS.
/VERIFY A WRITE ALL THEN READ ALL 128 WORDS.
/IF THE FIRST TWO WORDS OF THE BUFFER SHOULD
/EQUAL THE ABSOLUTE DISK ADDRESS.
/IF THE DATA PATTERN USED IS 5252+5253.

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9E0 0063

/SETUP SECTOR COUNTRY

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/
PAL10  V1424  15-APR-76  13124  PAGE 1-42

2126  1450  4431  FILBUF  TCNTR1  /FILL BUFFER WITH DATA
2127  1451  1134  TAD  /MASK SECTOR BITS
2128  1453  0120  DCA I  /SETUP ADDRESS WORD IN BUFFER
2129  1453  3463  DCA I  /GET DRIVE NUMBER
2130  1454  1072  TAD  /SETUP ADDRESS WORD IN BUFFER
2131  1455  3464  TAD  /FUNCTION WRITE DATA
2132  1456  1105  TAD  /SETUP COMMAND
2133  1457  3150  DCA I  /SETUP TO LOAD
2134  1460  1463  TAD I  /DISK WRITE ALL
2135  1461  4426  DISKGO  /TEXT POINTER
2136  1462  1510  T23T  /ERROR, STATUS OR SKIP
2137  1463  5306  JMP  /UPDATE SECTOR COUNTER
2138  1464  2134  T23T  /MORE SECTORS TO GO
2139  1465  5251  JMP  /TEXT POINTER
2140
2141  /VERIFY THAT THE DATA WRITTEN ABOVE
2142  /ON CYLINDER N WAS O.K. CHECK WITH READ DATA.
2143
2144  1466  1122  TAD  /COUNTER FOR 37 SECTORS
2145  1467  3134  DCA  /CLEAR DATA BUFFER
2146  1470  4432  T23R2,  /SETUP COMMAND
2147  1471  3150  DCA  /SETUP COMMAND
2148  1472  1134  TAD  /TEXT POINTER
2149  1473  0120  AND  /ERROR, STATUS OR SKIP
2150  1474  4426  DISKGO  /WORD BY WORD COMPARE OF DATA
2151  1475  1510  T23T  /DATA O.K.
2152  1476  5306  JMP  /ERROR, DATA
2153  1477  1114  TAD  /UPDATE SECTOR COUNTER
2154  1500  4430  FIGURE  /MORE SECTORS TO CHECK
2155  1501  7610  JMP CLA  /O.K. TO NEXT TEST
2156  1502  5306  T23E  /ERROR, WRITE ALL
2157  1503  2134  T23R1  /SCOPE LOOP POINTER
2158  1504  5270  JMP  /TEXT POINTER
2159  1505  4437  NERROR  /TEXT POINTER
2160  1506  4437  ERROR  /TEXT POINTER
2161  1507  1445  T23T,  /TEXT POINTER
2162  1508  1445  5373
2163
2164  /VERIFY ALL SECTORS CAN BE ACCESSED
2165
2166  /VERIFY A WRITE ALL TO ALL OF CYLINDER 1450
2167  /AND USE DATA PATTERN 2525+5252.
2168  /THE FIRST TWO WORDS OF THE SECTOR SHOULD
2169  /EQUAL THE DISK ADDRESS. CHECK THE DATA
2170  /WITH READ ALL.
2171
2172  /
2173  1511  1122  T23T,  TAD  /TEXT POINTER
2174  1512  3134  DCA  /SETUP SECTOR COUNTER
2175  1513  1113  TAD  /FILL OUTROUND BUFFER
2176  1514  4431  FILBUF  /TEXT POINTER
2177  1515  7301  CLA CL  /CLEAR DATA BUFFER
2178  1516  1272  TAD  /EXTENDEN BIT
2179  1517  3464  DCA I  /FUNCTION READ ALL
2180  1520  7301  CLA CL  /SETUP COMMAND
2181  1521  1115  TAD  /MASK OFF SECTORS
2182
2183  /DISK READ ALL
2184  /TEXT POINTER
2185  /ERROR, SKIP OR STATUS
2186  /CLEAR DATA BUFFER
2187  /EXTENDEN BIT
2188  /FUNCTION READ ALL
2189  /SETUP COMMAND
2190  /MASK OFF SECTORS
2191  /DISK READ ALL
2192  /TEXT POINTER
2193  1531  1556  T24T  /DISK READ ALL
2194  1532  5354  JMP  /TEXT POINTER
2195  1533  4432  T24E  /ERROR, STATUS OR SKIP
2196  1534  7301  CLA CL  /WORD BY WORD COMPARE OF DATA
2197  1535  1017  TAD  /THIS SECTOR O.K.
2198  1536  3150  DCA  /ERROR, DATA
2199  1537  1134  TAD  /UPDATE SECTOR COUNTER
2200  1540  0120  AND  /TRY NEXT SECTOR
2201  1541  1065  TAD  /O.K. TO NEXT TEST
2202  1542  4426  TAD  /ERROR, READ ALL
2203  1543  1556  DISKGO  /SCOPE LOOP POINTER
2204  1544  5354  T24T  /TEXT POINTER
2205  1545  1113  TAD  /TO NEXT TEST
2206  1546  4430  FIGURE  /TEXT POINTER
2207  1547  7610  JMP CLA  /ERROR, STATUS OR SKIP
2208  1548  4437  T24E  /TEXT POINTER
2209  1550  5373  T24T,  /TEXT POINTER
2210  1551  1511  5373
2211  /
2212  1557  5760  JMP I  /TO NEXT TEST
2213  1558  1600  T24T5
2214
2215  /
2216  1561  0000  /DISK READ ALL
2217  1562  0000  /TEXT POINTER
2218  1563  0000  /ERROR, STATUS OR SKIP
2219  1564  0000  /CLEAR DATA BUFFER
2220  1565  0000  /EXTENDEN BIT
2221  1566  0000  /FUNCTION READ ALL
2222  1567  0000  /SETUP COMMAND
2223  1568  0000  /MASK OFF SECTORS
2224  1569  0000  /DISK READ ALL
2225  1570  0000  /TEXT POINTER
2226  1571  0000  /ERROR, SKIP OR STATUS
2227  1572  0000  /CLEAR DATA BUFFER
2228  1573  0000  /EXTENDEN BIT
2229  1574  0000  /FUNCTION READ ALL
2230  1575  0000  /SETUP COMMAND
2231  1576  0000  /MASK OFF SECTORS
2232  1577  0000  /DISK READ ALL
2233  1578  0000  /TEXT POINTER
2234  1579  0000  /ERROR, STATUS OR SKIP
2235  1580  0000  /UPDATE SECTOR COUNTER
2236  1581  0000  /MORE SECTORS TO GO
2237  1582  0000  JMP  /TEXT POINTER
2238  1583  0000  /TEXT POINTER
2239  1584  0000  /TEXT POINTER
2240  1585  0000  /TEXT POINTER
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2706  2
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2236 1603 4431 FILBUF
2237 1604 7301 CLA CLL IAC
2238 1605 1072 TAD DRIVNO
2239 1606 3464 DCA I XMITRK
2240 1607 7301 CLA CLL IAC
2241 1610 1105 TAD K0000
2242 1611 3150 DCA CMERG
2243 1612 1134 TAD TCNTR1
2244 1613 0127 AND K0037
2245 1614 1065 TAD CYL450
2246 1615 3463 DCA I XLOTRK
2247 1616 1463 TAD I XLOTRK
2248 1617 4426 DISKGD
2249 1622 1644 T2ST
2250 1623 1644 JMP
2251 1624 4242 T2SE
2252 1625 7301 KILBUF
2253 1626 7301 CLA CLL IAC
2254 1627 3150 DCA CMERG
2255 1628 1134 TAD TCNTR1
2256 1629 7122 AND K0037
2257 1630 1065 TAD CYL450
2258 1631 1644 DISKGD
2259 1632 5242 T2ST
2260 1633 1114 JMP
2261 1634 4430 FIGURE
2262 1635 5242 T2SE
2263 1636 7610 SKIP CLA
2264 1637 2144 JMP CLA
2265 1642 5202 T255
2266 1641 4437 JMP
2267 1642 4440 ERROR
2268 1643 1600 T825
2269 1644 5373 T25T,
2270 1645 5373
2271
2272
2273
2274
2275
2276
2277
2278
2279 1645 1122 T5T26, TAD
2280 1646 3134 DCA TCNTR1
2281 1647 1114 TAD K5252
2282 1648 4431 FILBUF
2283 1651 1134 T26R1, TAD TCNTR1
2284 1652 0120 AND K0037
2285 1653 1065 TAD CYL450
2286 1654 3463 DCA I XLOTRK
2287 1655 7301 CLA CLL IAC
2288 1656 1072 TAD DRIVNO
2289 1657 3464 DCA I XMITRK
2290 1662 7301 CLA CLL IAC

```

```

/FILL OUTROUND BUFFER
/GET DRIVE NUMBER
/SETUP ADDRESS WORD IN BUFFER
/EXTENDED BIT
/FUNCTION WRITE DATA
/SETUP COMMAND
/SETUP COUNTER
/MASK OFF SECTOR BITS
/ADD IN CYLINDER
/SETUP ADDRESS WORD IN BUFFER
/DISK WRITE DATA
/TEXT POINTER
/ERROR, SKIP OR STATUS
/CLEAR DATA BUFFER
/EXTENDED BIT
/SETUP COMMAND
/SETUP COUNTER
/MASK OFF SECTORS
/DISK READ DATA
/TEXT POINTER
/ERROR, STATUS OR SKIP
/WORD BY WORD COMPARE OF DATA
/THIS SECTOR O.K.
/ERROR, DATA
/UPDATE SECTOR COUNTER
/TRY NEXT SECTOR
/O.K. TO NEXT TEST
/ERROR, DATA BREAK
/SCORE, LOOP POINTER
/TEXT POINTER

```

```

/VERIFY ALL SECTORS CAN BE ACCESSED INDIVIDUALLY.
/VERIFY A WRITE ALL TO ALL OF CYLINDER 1450
/USE DATA PATTERN 5252+2525
/CHECK FOR NO ERRORS IN STATUS.
/MAKE FIRST TWO WORDS OF EVERY SECTOR
/EQUAL TO ADDRESS OF SECTOR.

```

```

/SETUP SECTOR COUNTER
/FILL BUFFER WITH DATA
/MASK SECTOR BITS
/SETUP ADDRESS WORD IN BUFFER
/GET DRIVE NUMBER
/SETUP ADDRESS WORD IN BUFFER
/EXTENDED BIT

```

```

2291 1661 1115 TAD
2292 1662 3150 DCA K5000
2293 1663 1463 DCA CMERG
2294 1664 4426 TAD I XLOTRK
2295 1665 1716 DISKGD
2296 1666 5314 T26T
2297 1667 2134 JMP
2298 1668 5314 JMP TCNTR1
2299 1670 5251 JMP T6R1
2300
2301
2302
2303 1671 1122 TAD
2304 1672 3134 DCA TCNTR1
2305 1673 4432 T26R2, TAD TCNTR1
2306 1674 7301 KILBUF
2307 1675 1017 CLA CLL IAC
2308 1676 3150 DCA CMERG
2309 1677 1134 TAD TCNTR1
2310 1678 0120 AND K0037
2311 1679 1065 TAD CYL450
2312 1681 1065 DISKGD
2313 1682 4426 T26T
2314 1683 1716 JMP
2315 1684 5314 JMP T6E
2316 1685 1114 TAD K5252
2317 1686 4430 FIGURE
2318 1687 7610 SKIP CLA
2319 1688 5314 JMP CLA
2320 1689 2134 T26E
2321 1690 5273 TCNTR1
2322 1691 4437 JMP T26R2
2323 1692 4437 ERROR
2324 1693 4440 T26E,
2325 1694 4440 ERROR
2326 1695 5373 T26T,
2327 5373
2328
2329
2330
2331
2332
2333
2334 1696 1122 T5T27, TAD
2335 1697 3134 DCA TCNTR1
2336 1698 1113 TAD K5252
2337 1699 4431 FILBUF
2338 1700 1134 T27R1, TAD TCNTR1
2339 1701 1134 AND K0037
2340 1702 0127 TAD CYL450
2341 1703 1065 DCA I XLOTRK
2342 1704 7301 CLA CLL IAC
2343 1705 1072 TAD DRIVNO
2344 1706 3464 DCA I XMITRK
2345 1732 7301 CLA CLL IAC

```

```

/FUNCTION WRITE ALL
/SETUP COMMAND
/GET TRACK AND SECTOR
/DISK WRITE ALL
/TEXT POINTER
/ERROR, STATUS OR SKIP
/UPDATE, STATUS OF SKIP
/SETUP SECTOR COUNTER
/SETUP ADDRESS WORD IN BUFFER
/SETUP ADDRESS WORD IN BUFFER
/EXTENDED BIT
/SETUP COMMAND
/SETUP COUNTER
/MASK SECTOR BITS
/SETUP ADDRESS WORD IN BUFFER
/GET DRIVE NUMBER
/SETUP ADDRESS WORD IN BUFFER
/EXTENDED BIT

```

```

2346 1733 1105 TAD K4000 /FUNCTION WRITE DATA
2347 1734 1150 DCA K4000 /SETUP COMMAND
2348 1735 1463 TAD I KLOTRK /SECTOR TO LOAD
2349 1736 4426 DISKGO /DISK WRITE ALL
2350 1737 1167 T27E /TEXT POINTER
2351 1740 5365 JMP T27E /ERROR, STATUS OR SKIP
2352 1741 2134 ISZ TCNTR1 /UPDATE SECTOR COUNTER
2353 1742 5353 JMP T27E1 /MORE SECTORS TO GO

```

```

2354 1743 1122 TAD K7740 /COUNT FOR 37 SECTORS
2355 1744 3134 DCA TCNTR1 /CLEAR DATA BUFFER
2356 1745 4432 T27R2, KILBUF /FUNCTION READ DATA
2357 1746 7301 CLA CLL IAC /SETUP COMMAND

```

```

2358 1747 3150 DCA CMREG /FUNCTION READ DATA
2359 1748 1134 TAD TCNTR1 /SETUP COMMAND
2360 1751 0120 AND K0037
2361 1752 1065 TAD CYL450

```

```

2362 1753 4426 DISKGO /DISK READ DATA
2363 1754 1747 T27E /TEXT POINTER
2364 1755 5365 JMP T27E /ERROR, STATUS OR SKIP
2365 1756 1113 TAD K525 /ERROR, STATUS OR SKIP
2366 1757 4430 FIGURE /WORD BY WORD COMPARE OF DATA
2367 1758 7610 SKP CLA /DATA O.K.
2368 1759 5365 JMP T27E /ERROR, DATA
2369 1760 3134 TAD TCNTR1 /UPDATE SECTOR COUNTER
2370 1761 2134 ISZ TCNTR1 /MORE SECTORS TO CHECK
2371 1762 5305 JMP T27R2 /O.K. TO NEXT TEST
2372 1763 4437 NERROR /ERROR, WRITE ALL
2373 1764 4437 NERROR /SCOPE LOOP POINTER
2374 1765 1717 T27T, 5373 /TEXT POINTER

```

```

2375 1766 5373 T27T, 5373 /FUNCTION TIMING TEST, VERIFY CONSECUTIVE SECTORS.
2376 1767 1717 T27T, 5373 /VERIFY THAT WRITE AND READ ALL ARE ACTUALLY DOING CONSECUTIVE
2377 1768 1717 T27T, 5373 /SECTORS. WHEN DOING CONSECUTIVE SECTORS IN WRITE OR READ
2378 1769 1717 T27T, 5373 /ALL MODE, SECTOR TRANSFERS SHOULD OCCUR EVERY 2.5 MILLI-
2379 1770 1717 T27T, 5373 /SECONDS. THE PROGRAM WILL REPORT A STATUS ERROR OF
2380 1771 1717 T27T, 5373 /AND DONE FLAG IF THIS DOES NOT OCCUR.

```

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2381 1772 1072 TAD HOME4 /SETUP SECTOR COUNTER
2382 1773 3136 DCA TCNTR3 /FUNCTION WRITE ALL
2383 1774 4525 T3720, JMS I XLOAD /SETUP COMMAND
2384 1775 1120 TAD K7740 /SETUP SECTOR COUNTER
2385 1776 1120 TAD K7740 /FUNCTION WRITE ALL
2386 1777 1120 TAD K7740 /SETUP COMMAND
2387 1778 1120 TAD K7740 /SETUP SECTOR COUNTER
2388 1779 1120 TAD K7740 /FUNCTION WRITE ALL
2389 1780 1120 TAD K7740 /SETUP COMMAND
2390 1781 1120 TAD K7740 /SETUP SECTOR COUNTER
2391 1782 1120 TAD K7740 /FUNCTION WRITE ALL
2392 1783 1120 TAD K7740 /SETUP COMMAND
2393 1784 1120 TAD K7740 /SETUP SECTOR COUNTER
2394 1785 1120 TAD K7740 /FUNCTION WRITE ALL
2395 1786 1120 TAD K7740 /SETUP COMMAND
2396 1787 1120 TAD K7740 /SETUP SECTOR COUNTER
2397 1788 1120 TAD K7740 /FUNCTION WRITE ALL
2398 1789 1120 TAD K7740 /SETUP COMMAND
2399 1790 1120 TAD K7740 /SETUP SECTOR COUNTER
2400 1791 1120 TAD K7740 /FUNCTION WRITE ALL

```

```

2401 2006 1166 TAD K5300 /MODIFY TEXT POINTER
2402 2007 3237 DCA T20T
2403 2008 1134 T20R, TAD TCNTR1
2404 2009 0073 AND K0001

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```

2405 2010 0073 CLL RTR /MAKE READ ALL OR WRITE ALL
2406 2011 1017 TAD K1000 /GET FIELD-DRIVE
2407 2012 1017 TAD TCNTR3 /LOAD COMMAND REGISTER
2408 2013 1136 T2010, DLOC /GET BEGINNING OF BUFFER POINTER
2409 2014 1067 TAD AGNBUF /LOAD CURRENT ADDRESS
2410 2015 6746 T2010R, DCA TCNTR1 /MASK SECTOR BITS
2411 2016 1134 TAD TCNTR1 /LOAD AND GO
2412 2017 0120 AND K0037 /TIME COUNTER
2413 2018 0120 T2010C, DLAG /READ STATUS REGISTER
2414 2019 1174 TAD TCNTR2 /TIME COUNTER
2415 2020 3135 DCA TCNTR2 /READ STATUS REGISTER
2416 2021 6745 T2010D, DRST /YES, GOT TRANSFER DONE
2417 2022 1105 TAD K4000 /UPDATE TIME COUNTER
2418 2023 7450 SNA T200K /WAIT FOR GOOD STATUS
2419 2024 5252 JMP T200K /SUBTRACT, RESET STATUS
2420 2025 2135 ISZ TCNTR2 /SAVE FOR ERROR PRINTER
2421 2026 5225 JMP T2010 /MAKE READ ALL OR WRITE ALL
2422 2027 1105 TAD K4000 /UPDATE TIME COUNTER
2423 2028 1105 TAD K4000 /WAIT FOR GOOD STATUS
2424 2029 1134 DCA STREG /SUBTRACT, RESET STATUS
2425 2030 1134 TAD TCNTR1 /SAVE FOR ERROR PRINTER
2426 2031 0073 AND K0001
2427 2032 1112 CLL RTR
2428 2033 1017 TAD K1000 /MAKE READ ALL OR WRITE ALL
2429 2034 3150 DCA CMREG /SAVE FOR ERROR PRINTER
2430 2035 1067 TAN BGNBUF /GET START OF BUFFER
2431 2036 3152 DCA CAREG /SAVE FOR PRINTER
2432 2037 1134 TAD TCNTR1
2433 2038 0120 AND K0037
2434 2039 1134 DCA DARG
2435 2040 5247 DARG /MAKE SECTOR ADDRESS
2436 2041 5255 JMP T20E /SAVE FOR ERROR PRINTER
2437 2042 2134 T200K, JMP T20R /ERROR, HAVE TO WAIT FOR FLAG
2438 2043 5210 T20R, JMP T20R /HANG IF NO SKIP
2439 2044 4437 T20R, JMP T20R /ERROR, SECTOR RESPONSE NOT FOUND
2440 2045 4437 T20R, JMP T20R /UPDATE SECTOR COUNTER
2441 2046 4437 T20R, JMP T20R /MORE TO NEXT TEST
2442 2047 4437 T20R, JMP T20R /O.K. TO NEXT TEST
2443 2048 4437 T20R, JMP T20R /ERROR, WRITE OR READ ALL
2444 2049 4437 T20R, JMP T20R /SCOPE LOOP POINTER
2445 2050 5300 T20T, 5300 /TEXT POINTER

```

```

2446 2051 4525 JMS I XLOAD /SECTOR TIMING TEST, VERIFY NON-CONSECUTIVE SECTORS.
2447 2052 7775 T775 /VERIFY THAT READ AND WRITE DATA ARE NOT DOING CONSECUTIVE
2448 2053 7775 T775 /SECTORS. WHEN TRYING TO DO CONSECUTIVE SECTORS IN READ DATA
2449 2054 7775 T775 /OR WRITE DATA MODE, SECTOR TRANSFERS SHOULD OCCUR EVERY 2.5 MILLI-
2450 2055 7775 T775 /REVOLUTION, APPROX. EVERY 40 MILLISECONDS. THE PROGRAM WILL
2451 2056 7775 T775 /REPORT AN ERROR OF A DONE FLAG IF THIS DOES NOT OCCUR
2452 2057 7775 T775
2453 2058 7775 JMS I XLOAD
2454 2059 7775 T775
2455 2060 7775 T775
2456 2061 7775 T775
2457 2062 7775 T775
2458 2063 7775 T775
2459 2064 7775 T775
2460 2065 7775 T775
2461 2066 7775 T775
2462 2067 7775 T775
2463 2068 7775 T775
2464 2069 7775 T775
2465 2070 7775 T775
2466 2071 7775 T775
2467 2072 7775 T775
2468 2073 7775 T775
2469 2074 7775 T775
2470 2075 7775 T775
2471 2076 7775 T775
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2479 2084 7775 T775
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2513 2118 7775 T775
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2916 2521 7775 T775
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/SETUP COMMAND
/ISK READ DATA
/TEXT POINTER
/ERRPR, SKIP OR STATUS
/MODIFY TEXT POINTER
/EXPECTED STATUS

/MAKE READ OR WRITE
/GET FIELD+DRIVE
/LOAD COMMAND REGISTER
/LOAD CURRENT ADDRESS
/MASK SECTOR BITS
/LOAD AND GO
/TIME COUNTER
/READ STATUS REGISTER
/STATUS D.K.?
/WAIT FOR CORRECT RESPONSE (00000)
/NO, SAVE STATUS FOR PRINTER

/MAKE READ OR WRITE
/SAVE FOR ERROR POINTER
/GET START OF BUFFER
/SAVE FOR ERROR POINTER

/MAKE SECTOR ADDRESS
/SAVE FOR ERROR POINTER
/ERRPR, SECTOR RESPONSE NOT FOUND
/UPDATE TIME COUNTER
/WAIT FOR GOOD STATUS
/ERRPR, HAVE TO WAIT FOR FLAG
/FLAG IF NO SKIP
/UPDATE SECTOR COUNTER
/MORE TO TEST
/D.K. TO NEXT TEST
/ERRPR, STATUS
/SCOPE LOOP POINTER
/MODIFIED TEXT POINTER

C WORD IN
CRC SHOULD BE

/SETUP SECTOR COUNTER
/CLEAR CONTROL

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/CLEAR BUFFER AREA
/FUNCTION WRITE ALL
/SETUP COMMAND

/MASK SECTOR BITS
/DISK WRITE ALL
/TEXT POINTER
/ERROR, STATUS OR SKIP
/SETUP COMMAND

/MASK SECTOR BITS
/DISK READ ALL
/TEXT POINTER
/ERROR, STATUS OR SKIP

/MODIFY TEXT POINTER
/ENABLE CLEAR CONTROL
/JAN CLEAR PRK ENABLE FLUP
/STORE IN COMPARE REGISTER
/STORE IN COMPARE REGISTER

/READ CRC REGISTER
/CHECK RESULTS
/O.K.
/ERROR, CRC
/JUDGE SECTOR COUNTER
/MORE SECTIONS TO TEST
/O.K. TO NEXT TEST
/ERROR, CRC
/SCOPE LOOP POINTER
/TEXT POINTER

BUFFER,
5252.

/SETUP SECTOR COUNTER
/CLEAR CONTROL

/FILL DATA BUFFER
/FUNCTION WRITE ALL
/SETUP COMMAND

/MASK SECTOR BITS

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2585 2224 1110 TAD K7760 /DISK WRITE ALL
2586 2225 4426 DSKGO /TEXT POINTER
2587 2226 2261 T31T /ERROR, STATUS OR SKIP
2588 2227 5257 T31E /FUNCTION READ ALL
2589 2230 1017 TAD K1000 /SETUP COMMAND
2590 2231 3150 DCA CMREG
2591 2232 1134 TAD TCNTR1
2592 2233 0117 AND K0017 /MASK SECTOR BITS
2593 2234 1110 TAD K7760 /DISK READ ALL
2594 2235 4426 DSKGO /TEXT POINTER
2595 2236 2261 T31T /ERROR, STATUS OR SKIP
2596 2237 5257 T31E /FUNCTION CLEAR CONTROL AND
2597 2240 1167 TAD K6354 /CLEAR BRK ENABLE FLDP.
2598 2241 7261 DCA CLL IAC /GET GOOD CRC
2599 2242 7301 CLA ALL /STORE IN COMPARE REGISTER
2600 2243 4453 TAD CMR01 /GET GOOD CRC
2601 2244 1160 TAD CMR01 /STORE IN COMPARE REGISTER
2602 2245 3142 TAD CMR02 /STORE IN COMPARE REGISTER
2603 2246 1161 TAD CMR02 /READ CRC REGISTER
2604 2247 1143 DCA CMREG /CHECK RESULTS
2605 2250 4453 ACC02 /O.K.
2606 2251 4443 SKP CLA /ERROR, CRC
2607 2252 7610 SKP CLA /UPDATE SECTOR COUNTER
2608 2253 5257 T31E /MORE SECTORS TO TEST
2609 2254 2134 T31T /O.K. TO NEXT TEST
2610 2255 5214 JMP T31P /ERROR, CRC
2611 2256 4437 NERROR /SCORE LONG POINTER
2612 2257 4440 T31E /TEXT POINTER
2613 2258 2261 T31T
2614 2259 4374 T31Y, 4374

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/VERIFY HEAD MOTION AND CAPABILITY
/OF SELECTING TWO TRACKS INDIVIDUALLY.

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2594 2594 /VERIFY A WRITE ALL TO ALL OF CYLINDER 1450
2595 2599 /AND THEN CYLINDER 0. USE DATA PATTERN 5252+2525 ON
2600 2607 /CYLINDER 1450 AND 2525+5252 ON CYLINDER 0.
2601 2607 /CHECK FOR NO ERRORS IN STATUS.
2602 2607 /MAKE FIRST TWO WORDS OF EVERY SECTOR
2603 2604 /EQUAL TO ADDRESS OF SECTOR.
2604 2605 /FIRST WRITE CYLINDER 1450
2605 2606 /FIRST WRITE CYLINDER 1450
2606 2607 /FIRST WRITE CYLINDER 1450
2607 2608 /FIRST WRITE CYLINDER 1450
2608 2609 /FIRST WRITE CYLINDER 1450
2609 2610 /FIRST WRITE CYLINDER 1450
2610 2611 /FIRST WRITE CYLINDER 1450
2611 2612 /FIRST WRITE CYLINDER 1450
2612 2613 /FIRST WRITE CYLINDER 1450
2613 2614 /FIRST WRITE CYLINDER 1450
2614 2615 /FIRST WRITE CYLINDER 1450
2615 2616 /FIRST WRITE CYLINDER 1450
2616 2617 /FIRST WRITE CYLINDER 1450
2617 2618 /FIRST WRITE CYLINDER 1450
2618 2619 /FIRST WRITE CYLINDER 1450
2619 2620 /FIRST WRITE CYLINDER 1450

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2602 2262 1122 T31Z2, TAD K7760 /SETUP SECTOR COUNTER
2603 2263 3134 DCA TCNTR1 /SETUP SECTOR COUNTER
2604 2264 1114 TAD K5252 /FILL BUFFER WITH DATA
2605 2265 4431 FILBUF /FILL BUFFER WITH DATA
2606 2266 7391 TAD K0037 /GET DRIVE NUMBER
2607 2267 1272 TAD K0037 /SETUP ADDRESS WORD IN BUFFER
2608 2268 1134 T3201, TAD TCNTR1 /MASK SECTOR BITS
2609 2269 1134 TAD K0037 /LOWER CYLINDER
2610 2270 1134 TAD K0037 /SETUP WORD IN BUFFER
2611 2271 1134 TAD K0037 /SETUP WORD IN BUFFER
2612 2272 1134 TAD K0037 /SETUP WORD IN BUFFER
2613 2273 1134 TAD K0037 /SETUP WORD IN BUFFER
2614 2274 1134 TAD K0037 /SETUP WORD IN BUFFER
2615 2275 1134 TAD K0037 /SETUP WORD IN BUFFER
2616 2276 1134 TAD K0037 /SETUP WORD IN BUFFER
2617 2277 1134 TAD K0037 /SETUP WORD IN BUFFER
2618 2278 1134 TAD K0037 /SETUP WORD IN BUFFER
2619 2279 1134 TAD K0037 /SETUP WORD IN BUFFER
2620 2280 1134 TAD K0037 /SETUP WORD IN BUFFER

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2620 2276 1115 TAD K5000 /FUNCTION WRITE ALL
2621 2277 3150 DCA CMREG /SETUP COMMAND
2622 2278 1463 TAD I XLOTRK /SETUP TO GO
2623 2279 4426 DSKGO /DISK WRITE ALL
2624 2280 2374 T32T /TEXT POINTER
2625 2281 5372 JMP T32E /ERROR, STATUS OR SKIP
2626 2282 2374 TAD TCNTR1 /UPDATE SECTOR COUNTER
2627 2283 2305 JMP T3201 /MORE SECTORS TO GO
2628 2284 5271 JMP T3201 /MORE SECTORS TO GO
2629 2285 5271 JMP T3201 /MORE SECTORS TO GO
2630 2286 5271 JMP T3201 /MORE SECTORS TO GO
2631 2287 5271 JMP T3201 /MORE SECTORS TO GO
2632 2288 5271 JMP T3201 /MORE SECTORS TO GO
2633 2289 5271 JMP T3201 /MORE SECTORS TO GO
2634 2290 5271 JMP T3201 /MORE SECTORS TO GO
2635 2291 5271 JMP T3201 /MORE SECTORS TO GO
2636 2292 5271 JMP T3201 /MORE SECTORS TO GO
2637 2293 5271 JMP T3201 /MORE SECTORS TO GO
2638 2294 5271 JMP T3201 /MORE SECTORS TO GO
2639 2295 5271 JMP T3201 /MORE SECTORS TO GO
2640 2296 5271 JMP T3201 /MORE SECTORS TO GO
2641 2297 5271 JMP T3201 /MORE SECTORS TO GO
2642 2298 5271 JMP T3201 /MORE SECTORS TO GO
2643 2299 5271 JMP T3201 /MORE SECTORS TO GO
2644 2300 5271 JMP T3201 /MORE SECTORS TO GO
2645 2301 5271 JMP T3201 /MORE SECTORS TO GO
2646 2302 5271 JMP T3201 /MORE SECTORS TO GO
2647 2303 5271 JMP T3201 /MORE SECTORS TO GO
2648 2304 5271 JMP T3201 /MORE SECTORS TO GO
2649 2305 5271 JMP T3201 /MORE SECTORS TO GO
2650 2306 5271 JMP T3201 /MORE SECTORS TO GO
2651 2307 5271 JMP T3201 /MORE SECTORS TO GO
2652 2308 5271 JMP T3201 /MORE SECTORS TO GO
2653 2309 5271 JMP T3201 /MORE SECTORS TO GO
2654 2310 5271 JMP T3201 /MORE SECTORS TO GO
2655 2311 5271 JMP T3201 /MORE SECTORS TO GO
2656 2312 5271 JMP T3201 /MORE SECTORS TO GO
2657 2313 5271 JMP T3201 /MORE SECTORS TO GO
2658 2314 5271 JMP T3201 /MORE SECTORS TO GO
2659 2315 5271 JMP T3201 /MORE SECTORS TO GO
2660 2316 5271 JMP T3201 /MORE SECTORS TO GO
2661 2317 5271 JMP T3201 /MORE SECTORS TO GO
2662 2318 5271 JMP T3201 /MORE SECTORS TO GO
2663 2319 5271 JMP T3201 /MORE SECTORS TO GO
2664 2320 5271 JMP T3201 /MORE SECTORS TO GO
2665 2321 5271 JMP T3201 /MORE SECTORS TO GO
2666 2322 5271 JMP T3201 /MORE SECTORS TO GO
2667 2323 5271 JMP T3201 /MORE SECTORS TO GO
2668 2324 5271 JMP T3201 /MORE SECTORS TO GO
2669 2325 5271 JMP T3201 /MORE SECTORS TO GO
2670 2326 5271 JMP T3201 /MORE SECTORS TO GO
2671 2327 5271 JMP T3201 /MORE SECTORS TO GO
2672 2328 5271 JMP T3201 /MORE SECTORS TO GO
2673 2329 5271 JMP T3201 /MORE SECTORS TO GO
2674 2330 5271 JMP T3201 /MORE SECTORS TO GO

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2675 2352 3134 15-APR-76 13124 PAGE 1-52
2676 2353 4032 733R4, OCA TCNTR1 /COUNTER FOR 37 SECTORS
2677 2354 1017 733R4, KILBUF /CLEAR DATA BUFFER
2678 2355 3150 DCA K1000 /READ ALL FUNCTION
2679 2356 1134 DCA TCNTR1 /SETUP COMMAND
2680 2357 0120 AND K0037 /DISK READ ALL
2681 2360 4026 DISKGO T32T /TEXT POINTER
2682 2361 2374 T32T /ERROR, STATUS OR SKIP
2683 2362 5372 JMP T32E K2525 /WORD BY WORD COMPARE OF DATA
2684 2363 1113 TAO K2525 /DATA O.K.
2685 2364 4430 FIGURE JMP T32E TCNTR1 /ERROR, DATA
2686 2365 7610 SKP CLA 152 TCNTR1 /UPDATE SECTOR COUNTER
2687 2366 5372 152 TCNTR1 /MORE SECTORS TO CHECK
2688 2367 2134 JMP T32R4 /O.K. TO NEXT TEST
2689 2370 5353 ERROR T32R4 /ERROR, WRITE ALL
2690 2371 4437 T32E, ERROR T3132 /SCOPE LOOP POINTER
2691 2372 4400 T32T, 5373 /TEXT POINTER
2692 2373 2262 JMP I ++1 /TO NEXT TEST
2693 2374 5373 T3133
2694 2375 5776
2695 2376 2400
2696 2377 2142
2697 2378 2400
2698 2379 2400
2699 2400 1122
2700 2401 3134
2701 2402 1114
2702 2403 4031
2703 2404 7500 733R1, CLA CLL
2704 2405 1134 TAO TCNTR1
2705 2406 0120 K0037
2706 2407 3463 DCA I X10TRK
2707 2408 1072 TAO X10TRK
2708 2409 1072 DCA I X10TRK
2709 2410 1105 TAO K0008
2710 2411 3464 DCA K0008
2711 2412 1105 DCA TCNTR1
2712 2413 1463 TAO I X10TRK
2713 2414 4026 DISKGO
2714 2415 2511 T33T
2715 2416 5307 JMP T33E
2716 2417 2134 TCNTR1
2717 2418 13124
2718 2419 13124
2719 2420 13124
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2723 2424 13124
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2729 2421 5504 JMP T33R1 /WRITE DATA TO ALL OF CYLINDER 145R
2730 2422 1132 TAO K7740 /SETUP SECTOR COUNTER
2731 2423 3134 DCA TCNTR1 /FILL BUFFER WITH DATA
2732 2424 1113 TAO K2525 /GET DRIVE NUMBER
2733 2425 4031 FIGURE TAO K2525 /SETUP ADDRESS WORD IN BUFFER
2734 2426 7501 CLA CLL IAC /MASK OFF SECTOR BITS
2735 2427 1072 DCA I X10TRK /SETUP ADDRESS WORD IN BUFFER
2736 2428 3464 DCA I X10TRK /SETUP ADDRESS WORD IN BUFFER
2737 2429 1134 TAO K0037 /FUNCTION WRITE DATA
2738 2430 0120 TAO CYL50 /SETUP COMMAND
2739 2431 1065 DCA I X10TRK /SECTOR TO LOAD
2740 2432 1065 DCA I X10TRK /DISK WRITE DATA
2741 2433 1065 TAO I X10TRK /TEXT POINTER
2742 2434 1065 DCA K0008 /ERROR, STATUS OR SKIP
2743 2435 1065 DCA K0008 /UPDATE SECTOR COUNTER
2744 2436 1065 DCA K0008 /MORE SECTORS TO GO
2745 2437 1105 TAO K0008 /SETUP COMMAND
2746 2438 1105 DCA K0008 /DISK READ DATA
2747 2439 1105 TAO K0008 /TEXT POINTER
2748 2440 1463 DCA K0008 /ERROR, STATUS OR SKIP
2749 2441 4426 DISKGO T33T /SCOPE LOOP POINTER
2750 2442 2511 JMP T33E /WORD BY WORD COMPARE OF DATA
2751 2443 5307 JMP T33E /DATA O.K.
2752 2444 2134 TCNTR1 /ERROR, DATA
2753 2445 5231 JMP T33R2 /UPDATE SECTOR COUNTER
2754 2446 5231 TAO T33R2 /MORE SECTORS TO CHECK
2755 2447 1122 TAO K7740 /COUNTER FOR 37 SECTORS
2756 2448 3134 DCA TCNTR1 /CLEAR DATA BUFFER
2757 2449 4032 733R3, KILBUF /SETUP COMMAND
2758 2450 1134 DCA K1000 /DISK READ ALL
2759 2451 3150 DCA TCNTR1 /TEXT POINTER
2760 2452 1134 TAO K0037 /ERROR, STATUS OR SKIP
2761 2453 0120 AND K0037 /WORD BY WORD COMPARE OF DATA
2762 2454 4026 DISKGO T32T /TEXT POINTER
2763 2455 2511 TAO K2525 /ERROR, DATA
2764 2456 5307 JMP T32E TCNTR1 /UPDATE SECTOR COUNTER
2765 2457 1114 SKP CLA 152 TCNTR1 /O.K. TO NEXT TEST
2766 2458 4430 FIGURE JMP T32R4 /ERROR, WRITE ALL
2767 2459 7610 SKP CLA 152 TCNTR1 /SCOPE LOOP POINTER
2768 2460 5307 T32T, 5373 /TEXT POINTER
2769 2461 2374 T32T, 5373 /TO NEXT TEST
2770 2462 2134
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2772 2464 5250
2773 2465 1122
2774 2466 3134
2775 2467 4032 733R4, OCA TCNTR1 /COUNTER FOR 37 SECTORS
2776 2468 1017 733R4, KILBUF /CLEAR DATA BUFFER
2777 2469 3150 DCA K1000 /READ ALL FUNCTION
2778 2470 1134 DCA TCNTR1 /SETUP COMMAND
2779 2471 0120 AND K0037 /DISK READ ALL
2780 2472 4026 DISKGO T32T /TEXT POINTER
2781 2473 2374 T32T /ERROR, STATUS OR SKIP
2782 2474 5372 JMP T32E K2525 /WORD BY WORD COMPARE OF DATA
2783 2475 7610 SKP CLA 152 TCNTR1 /DATA O.K.
2784 2476 4430 FIGURE JMP T32E TCNTR1 /ERROR, DATA
2785 2477 5372 152 TCNTR1 /UPDATE SECTOR COUNTER
2786 2478 2134 JMP T32R4 /MORE SECTORS TO CHECK
2787 2479 5353 ERROR T32R4 /O.K. TO NEXT TEST
2788 2480 4437 T32E, ERROR T3132 /ERROR, WRITE ALL
2789 2481 4400 T32T, 5373 /SCOPE LOOP POINTER
2790 2482 2262 JMP I ++1 /TEXT POINTER
2791 2483 5776 /TO NEXT TEST
2792 2484 2400
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2803 2495 3463 DCA I X10TRK
2804 2496 1072 TAO X10TRK
2805 2497 1072 DCA I X10TRK
2806 2498 1105 TAO K0008
2807 2499 3464 DCA K0008
2808 2500 1105 DCA TCNTR1
2809 2501 1463 TAO I X10TRK
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2784	2475	4426	DISKGO		/DISK READ DATA
2785	2476	5311	TST3		/TEXT POINTER
2786	2477	5307	JMP		/ERROR, STATUS OR SKIP
2787	2500	1113	TAD	K2525	
2788	2501	4439	FIGURE		/WORD BY WORD COMPARE OF DATA
2789	2502	7610	SKP CLA		/DATA O.K.
2790	2503	5307	JMP	T33E	/ERROR, DATA
2791	2504	2134	182	TCTNTR1	/UPDATE SECTOR COUNTER
2792	2505	5267	JMP	T3384	/MORE SECTORS TO CHECK
2793	2506	4427	NEPROR		/O.K. TO NEXT TEST
2794	2507	4440	ERROR		/ERROR, WRITE DATA
2795	2517	2400	TST33		/SCOPE LOOP POINTER
2796	2511	5373	TST37,	5373	/TEXT POINTER
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2805	2512	4525	JMS I	XLOAD	
2806	2513	0001	1		
2807	2514	7301	TST34,	CLA CLL IAC	/CLEAR CONTROL
2808	2515	4453	SEK	CLEARLL	/SEEK ONLY TRACK N
2809	2516	4424	TAD		/TEXT POINTER
2810	2517	2574	TAD		/ERROR, SKIP OR STATUS
2811	2520	5346	JMP	T34E	
2812	2521	7301	CLA CLL IAC		
2813	2522	1156	TAD	HOMEWA	/ERROR, DATA
2814	2523	1072	TAD	DRIVND	/UPDATE SECTOR COUNTER
2815	2524	1125	TAN	K4200	/MORE SECTORS TO CHECK
2816	2525	4450	LDCCMD		/O.K. TO NEXT TEST
2817	2526	7301	CLA CLL IAC		/ERROR, WRITE DATA
2818	2527	1105	TAD	K4000	/SCOPE LOOP POINTER
2819	2537	1143	DCA	GDREG2	
2820	2531	1066	TAD	TRK212	
2821	2532	4452	LDADN		
2822	2533	7330	CLA CLL CML RAR		/TOTAL COMMAND WRITE DATA.
2823	2534	4455	LDMAN		/LOAD COMMAND REGISTER
2824	2535	7010	RAR		
2825	2536	4455	LDMAN		/EXPECTED STATUS
2826	2537	7010	RAR		/LOAD AND GO READ
2827	2540	4455	LDMAN		/ENTER MAINTENANCE
2828	2541	4447	DSKSKP		/SET DR4 FOR ENABLE SHIFT
2829	2542	5341	JMP		/SHIFT CRC
2830	2543	4444	RDSTAT	--1	/WAIT FOR FLAG
2831	2544	4442	ACCPM1		/READ STATUS REGISTER
2832	2545	4437	NEPROR		/CHECK RESULTS
2833	2546	4400	ERROR		/O.K. TO NEXT TEST
2834	2547	2514	TST34		/ERROR, CYLINDER ADDRESS
2835	2550	5300	T34T,	5300	/SCOPE LOOP POINTER
2836					/TEXT POINTER
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2897 2630 3444 DCA I XMITRK
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2902 2635 1767 TAD RGNBUF
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2904 2637 6530 DCA CMREG
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2910 2645 6746 TAD CMREG
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2918 2655 5273 JMP T36E
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2920 2657 4262 JMP *-3
2921 2660 2150 CMREG
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2925 2664 7620 SNL CLA
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2927 2666 1463 TAD I XLOTRK
2928 2667 1172 TAD ENDRK
2929 2670 7440 SZA CLA
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2938 2701 4447 MERROR
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PAL10	VI02A	15-APR-76	13124	PAGE 1-58	
3004	2772	2775	MPR1		/TEXT PRINTER
3005	2773	0000	TMPTOT, 0000		
3006	2774	4005	CLASSIC		
3007	2775	4456	COERR		
3008	2776	7402	MPHLT2, WLT		
3009					/SUCCESSFUL WRITE PROTECT
3010					/TO REPEAT TEST: IF ON
3011					/CLASSIC CONSOLE PACKAGE
3012					/BIT CONTROL: IF NOT THEN
3013					/PRESS KEY CONTINUE.
3014					
3015	2777	5306			
3016	3000		PAGE	JMP	MANPRO
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3030	3001	7770	JMS I	YLOAD	
3031	3002	3134	TS37,	DCA	TCNTR1
3032	3003	1017	TAD	K1000	
3033	3004	1156	TAD	H0004	
3034	3005	1072	TAD	DRIVN	
3035	3006	3150	DCA	CMREG	
3036	3007	1211	TAD	**2	
3037	3008	7410	SKP		
3038	3009	3102	TS37		
3039	3010	3172	DCA	SAVAPT	
3040	3012	1067	TAD	AGNBUF	
3041	3014	3152	DCA	CAREG	
3042	3015	4530	TICK		
3043	3016	7340	TS3R,	TICK	
3044	3017	3171	CLA	CLL	CHA
3045	3020	4432	DCA	SOERR	
3046	3021	1134	K1LUF		
3047	3022	3151	TAD	TCNTR1	
3048	3023	1067	DCA	DAREG	
3049	3024	6144	TAD	BNBUF	
3050	3025	1150	CMREG		
3051	3026	6746	TS37A,	DCA	CLL
3052	3027	1134	TAD	TCNTR1	
3053	3030	6743	IO73A2,	DLAG	
3054	3031	6741	IO71A2,	DSKP	
3055	3032	5231	JMP		
3056	3033	6745	IO75A2,	DRST	
3057	3034	3146	DCA		
3058	3035	1146	TAD	STREG	

/TEXT PRINTER

/SUCCESSFUL WRITE PROTECT

/TO REPEAT TEST: IF ON

/CLASSIC CONSOLE PACKAGE

/BIT CONTROL: IF NOT THEN

/PRESS KEY CONTINUE.

/VERIFY THAT THE DATA ON DISK IS CORRECT

/CHECK THE COMPLETE SURFACE

/THE DATA ON THE COMPLETE DISK SHOULD BE 2525+5252.

/HOWEVER, THE TWO FIRST WORDS OF EVERY SECTOR

/SHOULD EQUAL THE ABSOLUTE DISK ADDRESS.

JMS I YLOAD

TS37,

DCA TCNTR1

TAD K1000

TAD H0004

TAD DRIVN

DCA CMREG

TAD **2

SKP

TS37

DCA SAVAPT

TAD AGNBUF

DCA CAREG

TICK

CLA CLL CHA

DCA SOERR

K1LUF

TAD TCNTR1

DCA DAREG

TAD BNBUF

CMREG

TS37A,

DCA CLL

TAD TCNTR1

IO73A2,

DLAG

IO71A2,

DSKP

JMP

IO75A2,

DRST

DCA

STREG

/FUNCTION READ ALL

/CURRENT FIELD

/CURRENT DRIVE

/SETUP COMMAND

/GET TEXT POINTER

/TEXT POINTER

/SAVE FOR CRC ERROR

/GET START OF BUFFER

/SAVE FOR ERROR PRINTER

/ABT TIMING

/SETUP CRC ERROR POINTER

/CLEAR DATA BUFFER

/LOWER DISK ADDRESS

/SAVE FOR PRINTER

/GET START OF BUFFER

/LOAD CURRENT ADDRESS

/GET COMMAND

/LOAD COMMAND REGISTER

/GET DISK ADDRESS

/LOAD DISK ADDRESS AND GO

/DISK SKIP TO

/WAIT FOR DISK SKIP

/READ STATUS

/SAVE FOR ERROR PRINTER

PAL10	VI02A	15-APR-76	13124	PAGE 1-59	
3059	3036	1105	TAD	K4000	/ADD IN FUDGE FACTOR
3060	3037	7450	JMP	SNA	CLA
3061	3040	5854	JMP	TS37A	
3062	3041	7330	CLA	CLL	CHL
3063	3042	3143	DCA	GOREG2	
3064	3043	1146	TAD	STREG	
3065	3044	0011	AND	K0010	
3066	3045	7640	SZA	CLA	
3067	3046	5252	JMP	**4	
3068	3047	1146	TAD	K5300	
3069	3048	3302	DCA	TS37	
3070	3051	5300	JMP	TS37	
3071	3052	3171	DCA	TS37	
3072	3053	5256	JMP	TS37	
3073	3054	7301	DCA	SOERR	
3074	3055	6742	JMP	**3	
3075	3056	1165	IO72A2,	DCA	CLL
3076	3057	3302	TAD	K5373	
3077	3060	1113	DCA	TS37	
3078	3061	4430	TAD	K2525	
3079	3062	7610	SKP	CLL	
3080	3063	5300	JMP	TS37	
3081	3064	2134	TS2	TCNTR1	
3082	3065	7610	SKP	CLL	
3083	3066	2150	IS2	CMREG	
3084	3067	1156	TAD	CMREG	
3085	3070	0073	AND	K0001	
3086	3071	7650	SNA	CLA	
3087	3072	5215	JMP	TS37	
3088	3073	1134	TAD	TCNTR1	
3089	3074	1170	TAD	ENTR1	
3090	3075	7640	SZA	CLA	
3091	3076	5215	JMP	TS37	
3092	3077	4437	NONROR		
3093	3078	4440	ERROR		
3094	3100	3002	TS37		
3095	3102	5300	TS37,	5300	
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3110	3103	4525	JMS I	XLOAD	
3111	3104	7770	TS37,	DCA	CLL
3112	3105	7340	DCA	CLL	CHA
3113	3106	3171	DCA	SOERR	

/BIG ADDRESSING CHECK!

/IF A DATA ERROR SHOULD HAPPEN TO OCCUR

/WITH THE FIRST TWO WORDS OF THE BUFFER, YOU

/SHOULD REALIZE THAT THE PROBLEM COULD BE

/ADDRESSING.

/READ ALL SECTORS ON THE DISK AND CHECK

/THE STATUS. IF STATUS ERROR OCCURS THEN CHECK THE DATA.

/THE DATA ON THE COMPLETE DISK SHOULD BE 2525+5252.

/HOWEVER, THE TWO FIRST WORDS OF EVERY SECTOR

/SHOULD EQUAL THE ABSOLUTE DISK ADDRESS.

JMS I XLOAD

TS37,

DCA CLL CHA

DCA SOERR

/SETUP CRC ERROR POINTER

THE DATA ON THE COMPLETE DISK SHOULD BE 2525+5252.
HOWEVER, THE TWO FIRST WORDS OF EVERY SECTOR

75740, TAD K7000

3E0 0005

	PAL10	V102A	15-APR-76	13124	PAGE 1-63
3279	3353	1137	TAD	TCNTRA	/EXTENDED BIT
3280	3354	1155	TAD	K4800	/FUNCTION WRITE DATA
3281	3355	3150	DCA	CMRG	/SETUP COMMAND
3282	3356	1136	TAD	TCNTR3	/DISK ADDRESS
3283	3357	4426	DISKGO		/DISK WRITE DATA
3284	3358	3361	TAD		/TEXT POINTER
3285	3359	5357	JMP	TAIE	/ERROR SKIP OR STATUS
3286	3360	1155	TAD	TCNTR2	/GET EXTENDED BIT
3287	3361	3150	DCA	CMRG	/SETUP COMMAND REGISTER
3288	3362	1134	TAD	TCNTR1	/GET DISK ADDRESS
3289	3363	4424	SECK		/GO SEEK ONLY
3290	3364	3361	TAD		/TEXT POINTER
3291	3365	5357	JMP	TAIE	/ERROR, SECK SKIP OR STATUS
3292	3366	1137	TAD	TCNTRA	/GET EXTENDED BIT
3293	3367	3150	DCA	CMRG	/SETUP READ DATA COMMAND
3294	3368	1136	TAD	TCNTR3	/DISK ADDRESS
3295	3369	4426	DISKGO		/DISK READ DATA
3296	3370	5357	TAD		/TEXT POINTER
3297	3371	3361	JMP	TAIE	/ERROR, SKIP OR STATUS
3298	3372	5357	JMP	K2555	
3299	3373	1113	TAD		/WORD BY WORD COMPARE OF DATA
3300	3374	4430	FIGURE		/DATA O.K.
3301	3375	7610	SKC CLA		/DATA ERROR
3302	3376	5357	JMP	TAIE	/COUNT TO SAME TRACKS
3303	3377	2141	IS2	TCNTR6	/REPEAT
3304	3378	5306	IS3	TCNTR7	/PASS COUNTER
3305	3379	5272	JMP	TCNTR5	/LOOP
3306	3380	4437	NEROR		/O.K. TO NEXT TEST
3307	3381	4440	ERROR		/ERROR
3308	3382	3276	TAD	TA11	/SCOPE LOOP POINTER
3309	3383	5373	TAD	5373	/TEXT POINTER
3310	3384	5763	JMP	1	
3311	3385	3420	TSY42		
3312			/		
3313			/VERIFY A RECALIBRATE THEN A RANDOM WRITE DATA.		
3314			/THEN A RECALIBRATE THEN RANDOM READ DATA.		
3315			/THE DATA PATTERN WRITTEN IS 2525+5252 AND		
3316			/THE FIRST TWO WORDS OF EVERY SECTOR		
3317			/		
3318			/SHOULD EQUAL THE ABSOLUTE DISK ADDRESS.		
3319			/		
3320			/		
3321	3377	3173	PAGE		
3322	3400	3400			
3323			/		
3324	3400	1111	TSY42, TAD	K7700	/PASS COUNTER
3325	3401	5425	DCA	TCNTR5	/RANDOM DISK ADDRESS
3326	3402	4425	TAD, RAMADD		/SAVE
3327	3403	3134	DCA	TCNTR1	/LINK IS EXTENDED BIT
3328	3404	7004	RAL		/SAVE
3329	3405	3135	DCA	TCNTR2	
3330	3406	1113	TAD	K2525	
3331	3407	4431	FILBUF		/FILL BUFFER
3332	3408	1135	TAD	TCNTR2	/GET EXTENDED BIT
3333	3410	1135	TAD		/GET DRIVE NUMBER
3334	3411	1072	TAD	DRIVNO	

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/
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3333 3412 1464 DCA I XMITR /DISK ADDRESS WORD IN BUFFER
3334 3413 1134 TAD TCNTR1 /LOWER DISK ADDRESS
3335 3414 1463 NCA I XLOTRK /DISK ADDRESS WORD IN BUFFER
3336 3415 4425 RECAL /RESTORE DRIVE
3337 3416 1451 T42T /TEXT POINTER
3338 3417 5247 JMP T42T /ERROR SKIP OR STATUS
3339 3420 1135 TAD TCNTR2 /EXTENDED BIT
3340 3421 1125 TAD K4000 /FUNCTION WRITE DATA
3341 3422 1157 DCA TCNTR1 /SETUP COMMAND
3342 3423 1134 TAD TCNTR1 /DISK ADDRESS
3343 3424 4426 T42T /DISK WRITE DATA
3344 3425 5247 JMP T42T /TEXT POINTER
3345 3426 5247 JMP T42T /ERROR SKIP OR STATUS
3346 3427 4425 RECAL /RESTORE DRIVE
3347 3428 1451 T42T /TEXT POINTER
3348 3429 5247 JMP T42T /ERROR, SKIP OR STATUS
3349 3430 1135 TAD TCNTR2 /GET EXTENDED BIT
3350 3431 1135 TAD TCNTR1 /SETUP READ DATA COMMAND
3351 3432 1157 DCA TCNTR1 /DISK ADDRESS
3352 3433 4426 T42T /DISK READ DATA
3353 3434 5247 JMP T42T /TEXT POINTER
3354 3435 5247 JMP T42T /ERROR, SKIP OR STATUS
3355 3436 1133 TAD K2525 /WORD BY WORD COMPARE OF DATA
3356 3437 4426 T42T /DATA O.K.
3357 3438 7617 JMP CL4 /DATA ERROR
3358 3439 5247 JMP T42T /PASS COUNTER
3359 3440 2140 T42T /PASS COUNTER
3360 3441 5247 JMP T42T /LOOP
3361 3442 4427 T42T /O.K. TO NEXT TEST
3362 3443 4427 T42T /ERROR
3363 3444 4427 T42T /SCOPE LUMP POINTER
3364 3445 5373 T42T, 5373 /TEXT POINTER
3365 3446 5373 /
3366 3447 5373 /SINGLE DRIVE VIBRATION TEST
3367 3448 5373 /
3368 3449 5373 /TRY TO CAUSE CYLINDER ADDRESS ERRORS BY
3369 3450 5373 /DOING A FEW RANDOM SEKS THEN A READ DATA.
3370 3451 5373 /
3371 3452 1341 T43T3, TAD TIMSTP /SETUP PASS COUNTER
3372 3453 3140 DCA TCNTR5 /CLEAR BUFFER
3373 3454 4425 T43R1, KILRUP /GET RANDOM NUMBER
3374 3455 4423 RANDO /
3375 3456 0120 AND K0037 /SETUP COUNTER FOR SEKS
3376 3457 1122 TAD K7740 /GET RANDOM SEEK ADDRESS
3377 3458 3137 DCA TCNTR4 /SAVE IT
3378 3459 4423 RANDO /LINK IS EXTENDED BIT
3379 3460 3136 DCA TCNTR3 /SAVE IT
3380 3461 3136 RAL /SETUP COMMAND
3381 3462 7004 T43E, TAD TCNTR2 /SEEK ONLY A RANDOM TRACK
3382 3463 3135 TAD TCNTR2 /TEXT POINTER
3383 3464 1135 DCA TCNTR3 /ERROR, SKIP OR STATUS
3384 3465 1136 T43T /
3385 3466 4424 T43E /
3386 3467 3514 JMP T43E /
3387 3468 5312 JMP T43E /

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3388 3471 2137 1S2 TCNTR4 /COUNT NUMBER TO DO
3389 3472 5261 JMP T43R2 /
3390 3473 1135 TAD TCNTR2 /SETUP FOR READ DATA
3391 3474 3136 DCA CMREG /
3392 3475 1136 TAD TCNTR3 /LOAD AND GO READ DATA
3393 3476 4426 T43T /TEXT POINTER
3394 3477 1136 DSKGO /ERROR SKIP OR STATUS
3395 3478 5312 JMP T43E /CHECK DATA READ
3396 3479 5312 TAD K2525 /ALL O.K.
3397 3480 4430 T43E, TAD TCNTR5 /ERROR, DATA
3398 3481 7610 JMP CL4 /MORE TO NEXT
3399 3482 5312 T43E, TAD TCNTR5 /P.K. TO NEXT TEST
3400 3483 2140 T43E, TAD TCNTR5 /ERROR, SKIP, STATUS, OR DATA
3401 3484 5254 T43E, TAD TCNTR5 /SCOPE LOOP POINTER
3402 3485 2140 T43T, 0000 /TEXT POINTER
3403 3486 4426 T43E, TAD TCNTR5 /
3404 3487 5312 T43T, 0000 /CHECK DISK HEADER WORDS WITH READ DATA
3405 3488 5312 /IF STATUS ERROR OCCURS THEN CHECK DATA.
3406 3489 5312 /THE DATA ON THE COMPLETE DISK SHOULD BE 2525+5252.
3407 3490 5312 /HOWEVER, THE TWO FIRST WORDS OF EVERY SECTOR
3408 3491 5312 /SHOULD EQUAL THE ABSOLUTE DISK ADDRESS.
3409 3492 5312 /
3410 3493 5312 JMS I XLOAD
3411 3494 5312 T43T, 7775
3412 3495 5312 T43T, 7775
3413 3496 5312 T43T, 7775
3414 3497 5312 T43T, 7775
3415 3498 5312 T43T, 7775
3416 3499 5312 T43T, 7775
3417 3500 5312 T43T, 7775
3418 3501 5312 T43T, 7775
3419 3502 5312 T43T, 7775
3420 3503 5312 T43T, 7775
3421 3504 5312 T43T, 7775
3422 3505 5312 T43T, 7775
3423 3506 5312 T43T, 7775
3424 3507 5312 T43T, 7775
3425 3508 5312 T43T, 7775
3426 3509 5312 T43T, 7775
3427 3510 5312 T43T, 7775
3428 3511 5312 T43T, 7775
3429 3512 5312 T43T, 7775
3430 3513 5312 T43T, 7775
3431 3514 5312 T43T, 7775
3432 3515 5312 T43T, 7775
3433 3516 5312 T43T, 7775
3434 3517 5312 T43T, 7775
3435 3518 5312 T43T, 7775
3436 3519 5312 T43T, 7775
3437 3520 5312 T43T, 7775
3438 3521 5312 T43T, 7775
3439 3522 5312 T43T, 7775
3440 3523 5312 T43T, 7775
3441 3524 5312 T43T, 7775
3442 3525 5312 T43T, 7775

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/SETUP CRC ERROR POINTER
/SETUP LOWER ADDRESS
/SETUP EXTENDED
/CURRENT DRIVE
/CURRENT FIELD
/SETUP COMMAND
/START OF BUFFER
/LOAD CURRENT ADDRESS
/LOAD COMMAND ISSUED
/LOAD COMMAND
/LOWER ADDRESS
/LOAD AND GO
/DISK SKIP TOT
/HANG IF NO SKIP
/READ STATUS
/SHOULD ONLY BE DONE
/JUST DONE FLAG ?
/STATUS ERROR
/UPDATE ADDRESS
/DON'T SET EXTENDED TRACK
/YES, SET IT
/IS EXTENDED SET
/NO, CONTINUE

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/
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3443  3533  1170  TAD  ENDRK  /ADD IN FUDGE FACTOR
3444  3534  7440  STA CLA  /DONE WITH DISK
3445  3535  5386  JMP  TADR  /NO, MORE TO GO
3446  3536  5387  JMP  TADR  /ALL O.K.
3447  3537  1113  TAD  K2825
3448  3538  4430  FIGURE
3449  3539  5340  JMP  *3  /WORD BY WORD COMPARE OF DATA
3450  3540  1155  TAD  K5373  /ERROR, JUST STATUS
3451  3541  7410  SLP  K5308  /TEXT POINTER
3452  3542  1166  TAD  K5308  /ERROR
3453  3543  3372  DCA  TADR  /STATUS ERROR POINTER
3454  3544  7610  SKP  CLA  /SETUP
3455  3545  3507  4437  TADR,  /O.K. TO NEXT TEST
3456  3546  3571  4440  ERROR  /ERROR, READ DATA
3457  3547  3517  3517  TADR  /SCOPE LOOP POINTER
3458  3548  5382  5380  TADR,  /TEXT POINTER
3459  3549  5374  5374  JMP I  *+1
3460  3550  3616  3616  PAGE  /NEXT TEST
3461  3551  3600
3462  3552  3600
3463  3553  3600
3464  3554  3600
3465  3555  3600
3466  3556  3600
3467  3557  3600
3468  3558  3600
3469  3559  3600
3470  3560  3600
3471  3561  3600
3472  3562  3600
3473  3563  3600
3474  3564  3600
3475  3565  3600
3476  3566  3600
3477  3567  3600
3478  3568  3600
3479  3569  3600
3480  3570  3600
3481  3571  3600
3482  3572  3600
3483  3573  3600
3484  3574  3600
3485  3575  3600
3486  3576  3600
3487  3577  3600
3488  3578  3600
3489  3579  3600
3490  3580  3600
3491  3581  3600
3492  3582  3600
3493  3583  3600
3494  3584  3600
3495  3585  3600
3496  3586  3600
3497  3587  3600

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/VERIFY THAT WRITING ON A TRACK DOES NOT AFFECT
/AN ADJACENT TRACK, THE TEST SEQUENCE IS AS FOLLOWS:
/WRITE TRACKS 00000-00100-00000 THEN READ AND CHECK
/TRACKS 00000-00000-00100, WRITE TRACKS 00020-00120-00000
/THEN READ AND CHECK TRACKS 00000-00020-00120, ETC.
/THE CENTER TRACK IS SET TO A DATA PATTERN OF
/255+255. THE LOWER AND UPPER TRACKS ARE
/SET TO A DATA PATTERN OF 450+255. THE FIRST TWO
/WORDS OF EVERY SECTOR ARE SET TO THE ABSOLUTE
/DISK ADDRESS.

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/ESTABLISH PROPER COUNT
CCL CLA CMA RTL
DCA KCNT
JMS I XLOAD
7750

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/
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3498  3602  1012  TSTG5,  TAD  K0020  /GET STARTING POINTER
3499  3603  3134  DCA  TCNTR1  /SAVE IT
3500  3604  1172  TAD  K7156  /COUNTER FOR TRACKS TO GO
3501  3605  3100  DCA  TCNTR5  /COUNTER FOR TRACKS TO GO
3502  3606  7346  T453C,  CLA CCL CMA RTL  /THREE TRACK COUNTER POINTER
3503  3607  3137  DCA  TCNTR4  /WRITE CENTER TRACK FIRST
3504  3608  1134  TAD  TCNTR1  /DATA PATTERN FOR CENTER TRACK
3505  3609  3136  DCA  TCNTR3  /GO WRITE CENTER TRACK
3506  3610  1113  TAD  K2525  /GET POINTER
3507  3611  5244  JMP  T45A1  /WRITE UPPER OR LOWER???
3508  3612  1137  TAD  TCNTR4  /NO LOWER
3509  3613  7110  CCL PAR  /REDUCE OR UPDATE
3510  3614  7630  STL CLA  /SAVE TRACK TO DO
3511  3615  1162  TAD  K0020  /FILL BUFFER WITH DATA
3512  3616  1012  TAD  TCNTR1  /GET SECTOR COUNTER POINTER
3513  3617  1134  TAD  TCNTR3  /SETUP COUNTER
3514  3618  3136  DCA  K5552  /GET SECTOR POINTER
3515  3619  1114  TAD  K5552  /MASK SECTORS
3516  3620  4431  FILBUF  /SETUP ADDRESS WORD IN BUFFER
3517  3621  1110  TAD  K7760  /GET DISK ADDRESS
3518  3622  3135  DCA  TCNTR2  /PUT EXTENDED BIT IN LINK
3519  3623  3141  TAD  TCNTR6  /ADD IN SECTORS
3520  3624  1141  AND  T45R2,  /SETUP ADDRESS WORD IN BUFFER
3521  3625  0117  DCA  TCNTR3  /SET EXTENDED BIT???
3522  3626  3443  TAD  K7760  /YES!!
3523  3627  1136  TAD  K7760  /ADD IN CURRENT DRIVE
3524  3628  7104  CCL RAL  /SETUP ADDRESS WORD IN BUFFER
3525  3629  0110  AND  K7760  /GET EXTENDED BIT
3526  3630  1403  TAD I  XLOTRK  /FUNCTION WRITE DATA
3527  3631  3443  DCA I  XLOTRK  /SETUP COMMAND REGISTER POINTER
3528  3632  7630  DCA I  XLOTRK  /GET CYL., SURFACE, AND SECTOR
3529  3633  7001  TAD  T45R2  /WRITE ALL
3530  3634  1072  TAD  T45R2  /TEXT POINTER
3531  3635  3443  DCA I  XLOTRK  /ERROR, WRITE SKIP OR STATUS
3532  3636  1404  TAD I  XLOTRK  /UPDATE SECTOR POINTER
3533  3637  1105  DCA  K0000  /UPDATE SECTOR COUNTER
3534  3638  3150  DCA  K0000  /DO REST OF TRACK
3535  3639  1403  DCA  K0000  /UPDATE TRACK COUNTER
3536  3640  4426  TAD I  XLOTRK  /DO OTHERS
3537  3641  3767  TAD  T45R1  /SETUP FIRST TIME POINTER
3538  3642  5365  JMP  T45R1  /TRACK COUNTER POINTER
3539  3643  1141  TAD  T45R1  /SETUP FOR READ CENTER FIRST
3540  3644  1075  DCA  K0000  /SETUP FOR READ CENTER FIRST
3541  3645  3141  DCA  K0000  /SETUP FOR READ CENTER FIRST
3542  3646  2133  DCA  K0000  /SETUP FOR READ CENTER FIRST
3543  3647  5250  DCA  K0000  /SETUP FOR READ CENTER FIRST
3544  3648  2137  DCA  K0000  /SETUP FOR READ CENTER FIRST
3545  3649  3701  DCA  K0000  /SETUP FOR READ CENTER FIRST
3546  3650  5234  DCA  K0000  /SETUP FOR READ CENTER FIRST
3547  3651  7100  DCA  K0000  /SETUP FOR READ CENTER FIRST
3548  3652  3144  DCA  K0000  /SETUP FOR READ CENTER FIRST
3549  3653  7104  DCA  K0000  /SETUP FOR READ CENTER FIRST
3550  3654  3137  DCA  K0000  /SETUP FOR READ CENTER FIRST
3551  3655  1134  DCA  K0000  /SETUP FOR READ CENTER FIRST
3552  3656  3136  DCA  K0000  /SETUP FOR READ CENTER FIRST

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3553	3710	5320
3554	3711	1137
3555	3712	7119
3556	3713	7610
3557	3714	1122
3558	3715	1012
3559	3716	1134
3560	3717	5136
3561	3720	1119
3562	3721	3135
3563	3722	3141
3564	3723	1136
3565	3724	7194
3566	3725	5105
3567	3726	3105
3568	3727	7650
3569	3728	7691
3570	3729	5159
3571	3731	1141
3572	3733	0117
3573	3734	1145
3574	3735	4426
3575	3736	3767
3576	3737	5164
3577	3740	1144
3578	3741	7650
3579	3742	1113
3580	3743	1113
3581	3744	4430
3582	3745	7610
3583	3746	5165
3584	3747	1141
3585	3750	1277
3586	3751	3141
3587	3752	2135
3588	3753	5135
3589	3754	5144
3590	3755	2137
3591	3756	5311
3592	3757	1134
3593	3758	1111
3594	3761	3134
3595	3762	5226
3596	3763	5226
3597	3764	4437
3598	3765	4440
3599	3766	7632
3600	3767	0202
3601	3770	5771
3602	3771	4062
3603	3772	7156
3604		
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3604	4000	0000
3605	4001	7320
3606	4002	1122
3607	4003	3344
3608	4004	3331
3609	4005	2331
3610	4006	5205
3611	4007	2140
3612	4010	5205
3613	4011	5600
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3663 4081 0110 AND K7760 /MASK OFF CYLINDER AND SURFACE
3664 4082 4492 LOAD0 /LOAD AND GO SEEK
3665 4083 4447 DAKSKIP /WAIT FOR DONE
3666 4084 5253 JMP *-1 /
3667 4085 4493 CLRALL /CLEAR STATUS
3668 4086 4444 RDSTAT /READ STATUS
3669 4087 7600 82A CLA /DRIVE DONE?
3670 4088 5255 JMP *-3 /NO, WAIT
3671 4081 5225 JMP REDEK+3 /CHECK FOR NEW ADDRESS
3672 3673 /
3674 3675 /IF ALL DRIVES HAVE BEEN TESTED INDIVIDUALLY
3676 3677 /THEN RUN OVERLAP SEEKS AND OVERLAP SEEKS, WRITES,
3678 3679 /AND READS ON ALL DRIVES SELECTED. ALSO CHECK FOR HALT AT PASS
3680 3681 /COMPLETION. AFTER OVERLAP TESTS START AT FIRST
3682 3683 /DISK DRIVE ON SYSTEM.
3684 4086 3071 DCA DRYCNV /SETUP NO. OF DRIVES COUNTER.
3685 4087 4763 JMS I XOVERD /PERFORM OVERLAP SEEKS
3686 4071 3776 DCA DENT2 /OVERLAP SEEKS+WRITES+READS
3687 4072 4777 JMS I (GETDRV /START OVER AT 0
3688 4073 4405 SAMDSK, CLASSIC /SELECT FIRST DRIVE
3689 4074 4424 CAPAS3 /CHECK FOR CLASSIC ACTIVE
3690 4075 7610 JMP CLP CLA /PASS COMPLETE
3691 4076 5302 JMP *-4 /
3692 4077 1025 TAD 22 /SFF IF ON APT
3693 4100 0105 AND K000R /APRT?
3694 4101 7650 SNA CLA /NO
3695 4102 5307 JMP *-5 /CLFAR APT TIMING COUNTER
3696 4103 3775 DCA I (CLKCNT /
3697 4104 7340 CLL CLA CMA /
3698 4105 3175 DCA KCNT /
3699 4106 5323 JMP NEXDSK /LOOP PROGRAM
3700 4107 4462 CRLF /PRINT PASS COMPLETE
3701 4114 4457 PRINTER /
3702 4111 0760 NEXSI /
3703 4112 4457 PRINTER /
3704 4113 7715 TEND /
3705 4114 4404 LAS /
3706 4115 0076 AND K0004 /
3707 4116 7650 JMP *-4 /SWITCH 9 SFT?
3708 4120 4405 /
3709 4121 4437 CLASIC /YES, STOP PROGRAM
3710 4121 4437 CAINQU /DCLR
3711 4122 7402 ENDLT, MLT /
3712 4123 7301 NEXDSK, CLA CLL IAC /
3713 4124 4453 /
3714 4125 3131 DCA REG0 /DCLR
3715 4126 3132 DCA REG1 /
3716 4127 5730 JMP I *-1 /LOOP ON PROGRAM
3717 4138 0240 TST0 /

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/
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3718 4131 0000 /SUBROUTINE TO ISSUE "DMAN" MAINTENANCE IOT
3719 3720 /
3721 4132 0000 LDNM, 0 /
3722 4132 6747 IOT7, DMAN /
3723 4133 5731 JMP I LDNM /"DMAN" MAINTENANCE IOT
3724 4134 4405 CLASSIC /EXIT
3725 4135 4436 COERR /
3726 4136 7402 FRMLT7, MLT /SKIP TRAP ERROR
3727 4137 5334 JMP *-3 /
3728 /
3729 /SUBROUTINE TO SHIFT, THEN READ DISK ADDRESS
3730 /INTO DATA BUFFER. 12 SHIFTS
3731 /
3732 4140 0000 ROAD, 0 /
3733 4141 7300 TAD M12 /
3734 4142 1126 TAD DCA SRCNT1 /SET MAIN(1) ENABLE BIT
3735 4143 3133 DCA SRCNT1 /LOAD MAINTENANCE
3736 4144 7330 CLA CLL CML PAR /LOAD MAINTENANCE
3737 4145 4455 LDNM /LOAD MAINTENANCE
3738 4146 7010 PAR /
3739 4147 4455 LDNM /LOAD MAINTENANCE
3740 4150 7300 CLA CLL /
3741 4151 1015 TAD K0200 /SHIFT TRACK ADDRESS BIT
3742 4152 4455 LDNM /LOAD MAINTENANCE IOT
3743 4153 2133 IS2 SRCNT1 /SHIFT 12 BITS
3744 4154 5352 JMP *-2 /
3745 4155 7300 TAD CLA CLL /
3746 4156 1012 TAD K0220 /
3747 4157 4455 LDNM /READ DATA BUFFER
3748 4160 3151 DCA DAREG /SAVE RESULTS
3749 4161 1151 TAD DAREG /
3750 4162 5740 JMP I ROAD /EXIT
3751 /
3752 4163 4200 XLAP, OVERLAP /
3753 4164 4400 XOVERD, OVERD /
3754 /
3755 4165 0411 NMESS3, TEXT "DISK" /
3756 4166 2313 /
3757 4167 0000 /
3758 /
3759 4175 7162 /PAGE
3760 4177 4345 /
3761 4200 /
3762 /ROUTINE TO DO OVERLAP SEEKS ON EXISTING DRIVES
3763 /AFTER ALL HAVE RUN THE COMPLETE DIAGNOSTIC
3764 /
3765 4200 0000 OVERLAP, 0 /
3766 4201 1105 TAD K0004 /PASS COUNTER
3767 4202 3140 DCA DRYCNV /SET COUNTER FOR NO. OF DRIVES.
3768 4203 1070 TAD DRYCNV /
3769 4204 3137 DCA TCNTR4 /

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377C 4205 3371 DCA DCNT2 /START WITH DRIVE 0
3771 4206 4777 NVPR2, JMS I /GET NEXT DRIVE.
3772 4207 1972 TAD /GET DRIVE NO.
3774 4210 7110 CLR RAR /SELECT A RANDOM ADDRESS
3775 4211 4423 RANAND /SEND DISK OUT
3776 4212 4406 NSKOUT /CLEAR STATUS
3777 4213 4453 CLRALL /UPDATE DISK COUNTER
3778 4214 2137 TSZ /NO ALL EXISTING DISKS
3779 4215 5206 JMP NVPR2 /CLEAR FOR 0
3780 4216 3371 DCA DCNT2 /NO. OF DRIVES.
3781 4217 1972 TAD /SETUP COUNTER
3782 4220 3137 DCA /GET NEXT DRIVES.
3783 4221 4777 NVPR3, JMS I /GET SELECTED DRIVE.
3784 4222 1972 TAD /NO. OF DRIVES.
3785 4223 7110 CLR RAR /CHECK FOR DRIVE DONE
3786 4224 4407 NSKIN /DRIVE NOT DONE
3787 4225 5232 JMP NVPR2 /DRIVE DONE AND NO ERRORS
3788 4226 5233 JMP NVPR2 /DRIVE ERRORS
3789 4227 5254 JMP NVPR2 /UPDATE NO. OF DRIVE COUNTER.
3790 4231 5221 TSZ /NO. NO TEST
3791 4232 5217 JMP NVPR2 /YES, RESET
3792 4233 7340 NVPR4, JMS I /UPDATE PASS COUNTER, DONE ?
3793 4234 1137 DCA CLC CMA /NO, SEND OUT
3794 4235 2147 TSZ /SET FOR 0
3795 4236 4227 JMP NVPR2+1 /NO. OF DRIVES ON SYSTEM.
3796 4237 3371 DCA DCNT2 /GET NEXT DRIVE.
3797 4240 1972 TAD /CHECK FOR DRIVE DONE
3798 4241 3137 DCA /WAIT FOR THIS DRIVE
3799 4242 4777 ALLBAK, JMS I /WAIT FOR NEXT
3800 4243 1072 TAD /LAST DRIVE HOME YET
3801 4244 7110 CLR RAR /WAIT FOR ALL
3802 4245 4407 DSKIN /O.K. TO NEXT
3803 4246 5242 JMS ALLBAK /ERROR, OVERLAP SEKS
3804 4247 7616 SKP CLA /SCOPE LOOP POINTER
3805 4250 5254 JMP NVPR2 /TEXT POINTER
3806 4251 2137 TSZ /TO NEXT TEST
3807 4252 5242 JMP NVPR2
3808 4253 4437 NVPRR, ERROR
3809 4254 4448 NVPRR, ERROR
3810 4255 4201 NVPR2+1
3811 4256 5300 JMP I
3812 4257 5600 JMP I
3813
3814 /ROUTINE TO GET DRIVES FROM OPERATOR.
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3825 4271 1776* TAD M0 /NO. OF POSSIBLE DRIVES.
3826 4272 3371 DCA DCNT2
3827 4273 4462 CRLF /NO. OF POSSIBLE DRIVES.
3828 4274 4457 NVPR2, JMS I /GET NEXT DRIVE.
3829 4275 4165 NVPR3, JMS I /SELECT A RANDOM ADDRESS
3830 4276 1370 TAD /SEND DISK OUT
3831 4277 1374 TAD /CLEAR STATUS
3832 4278 3372 DCA DCNT2 /UPDATE DISK COUNTER
3833 4279 1370 TAD /NO ALL EXISTING DISKS
3834 4281 1364 TAD /CLEAR FOR 0
3835 4282 4436 TYPE /NO. OF DRIVES.
3836 4283 1366 TAD /SETUP COUNTER
3837 4284 1366 TAD /GET NEXT DRIVES.
3838 4285 4436 TYPE /GET SELECTED DRIVE.
3839 4286 6031 KSF /NO. OF DRIVES.
3840 4287 5306 JMP NVPR2+1 /CHECK FOR DRIVE DONE
3841 4288 6036 KRA /DRIVE NOT DONE
3842 4289 0367 AND /DRIVE DONE AND NO ERRORS
3843 4291 1215 TAD /UPDATE NO. OF DRIVE COUNTER.
3844 4293 3373 DCA DCNT2 /NO. NO TEST
3845 4294 1373 TAD /YES, RESET
3846 4295 4436 TYPE /SELECT A RANDOM ADDRESS
3847 4296 1373 TAD /SEND DISK OUT
3848 4297 7041 CTA /CLEAR STATUS
3849 4298 1365 TAD /UPDATE DISK COUNTER
3850 4299 7100 CLR /NO ALL EXISTING DISKS
3851 4301 7650 SNA CLA /CLEAR FOR 0
3852 4302 3772 DCA CLC CMA /NO. OF DRIVES.
3853 4303 7630 DCA T /SETUP COUNTER
3854 4304 7630 S2L CLA /GET NEXT DRIVES.
3855 4305 2070 NVPR4, JMS I /GET SELECTED DRIVE.
3856 4306 1775* TYPE /NO. OF DRIVES.
3857 4307 4436 TYPE /CHECK FOR DRIVE DONE
3858 4308 2370 TSZ /DRIVE NOT DONE
3859 4309 2371 ISZ /DRIVE DONE AND NO ERRORS
3860 4310 5274 JMP NVPR2 /UPDATE NO. OF DRIVE COUNTER.
3861 4311 1070 TAD /NO. NO TEST
3862 4312 5261 SNA CLA /YES, RESET
3863 4313 7650 JMP NVPR2+1
3864 4314 7041 TAD
3865 4315 3070 DCA /CHECK FOR DRIVE DONE
3866 4316 3371 DCA /DRIVE NOT DONE
3867 4317 4405 DCA DCNT2 /DRIVE DONE AND NO ERRORS
3868 4318 4405 JMS /UPDATE NO. OF DRIVE COUNTER.
3869 4319 5600 JMP I /NO. NO TEST
3870
3871 /ROUTINE TO SELECT DRIVES ON SYSTEM.
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3880 4355 3972 DCA DRIVE NO. /SETUP DRIVE NO.
3881 4356 3971 /UPDATE TO NEXT DRIVE.
3882 4357 7000 NOP
3883 4360 1700 YAD I DCNT1 /GET BUFFER FLAG.
3884 4361 7600 52A CLA /DISK ON SYSTEM?
3885 4362 5745 JMP I GETDRV /YES, USE DRIVE NO.
3886 4363 5346 JMP I GETDRV*1 /SELECT NEXT.
3887
3888 4364 0260 /K260, 0260
3889 4365 0331 K0331, 0331
3890 4366 0277 K0277, 0277
3891 4367 0177 K0177, 0177
3892 4370 0000 DCNT1, 0
3893 4371 0000 DCNT2, 0
3894 4372 0000 DCNT3, 0
3895 4373 0000 DCNT4, 0
3896 4374 1561 DSKON, DSKON
3897
3898 4375 6064 /
3899 4376 6110 /
3900 4377 4345 /
3901
3902 PAGE
3903 /ROUTINE TO PERFORM RANDOM OVERLAP SEKS, WRITES AND,
3904 /REANS ON ALL EXISTING DRIVES AFTER THEY HAVE RUN THE
3905 /COMPLETE DIAGNOSTIC.
3906
3907 OVERED, 0
3908 CLA CLL CML RAR
3909 DCA TCMTR5
3910 DCA DRYHAY
3911 DCA TCMTR4
3912 DCA DCNT2
3913 JMS I GETDRV
3914 TAD DRIVNO
3915 CLL RAR
3916 RANDO
3917 DSKOUT
3918 CLRALL
3919 TCMTR4
3920 JMP OVERD2
3921 DCA DCNT2
3922 TAD DRYHAY
3923 TAD TCMTR4
3924 JMS I GETDRV
3925 CLL RAR
3926 DSKIN
3927 DSKOUT
3928 JMP OVERDOK
3929 TAD K530R
3930 DCA TOVRNT
3931 JMP OVERD
3932 TCMTR4
3933 OVERD3
3934
3935 /YES, RESET
3936
3937 /COMPUTER WAY TO BUFFER.
3938 /GET DISK ADDRESS.
3939 /SAVE IT
3940 /GET POINTER
3941 /ADD IN FUDGE FACTOR
3942 /MAKE ADDRESS
3943 /GET DATA PATTERN TO USE
3944 /FILL DATA BUFFER
3945 /GET EXTENDED BIT
3946 /ADD IN DRIVE NUMBER
3947 /SETUP ADDRESS WORD IN BUFFER
3948 /GET CYL., SURFACE, AND SECTOR
3949 /SETUP ADDRESS WORD IN BUFFER
3950 /GET EXTENDED BIT
3951 /ADD IN WRITE FUNCTION
3952 /SETUP COMMAND POINTER
3953 /GET ADDRESS
3954 /DISK WRITE DATA
3955 /TEXT POINTER
3956 /ERROR, WRITE SKIP OR STATUS
3957 /CLEAR DATA BUFFER
3958 /GET EXTENDED BIT
3959 /SETUP COMMAND REGISTER
3960 /GET DISK ADDRESS
3961 /GO, READ DATA
3962 /TEXT POINTER
3963 /ERROR
3964
3965 /WORD BY WORD COMPARE DATA
3966 /DATA O.K., CONTINUE
3967 /DATA ERROR
3968 /GET DRIVE NO., SELECTED
3969
3970 /GENERATE RANDOM ADDRESS
3971 /SEND DRIVE BACK OUT
3972 /UPDATE PASS COUNTER, DONE ?
3973 /CHECK FOR NEXT DRIVE
3974 /SET FOR 0 OF DRIVES ON SYSTEM.
3975 /SELECT DRIVE
3976 /GET DRIVE SELECTED.
3977
3978 /CHECK THIS DRIVE
3979 /WAIT FOR DRIVE
3980 /CHECK FOR NEXT
3981 /ERROR
3982 /LAST DRIVE HOME YET
3983 /WAIT FOR ALL
3984 /O.K. TO NEXT
3985 /OVERLAP SEKS+READ DATA
3986
3987 OVERD,
3988 ERROR

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3934 4434 5217 /YES, RESET
3935 4435 1072
3936 4436 7110
3937 4437 1377
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3942 4444 1076
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3962 4470 4486
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3989 4530 0000 /ROUTINE TO CHECK DRIVE IN AC
3990 4531 7104 RTN,
3991 4532 0450 CLL RAL
3992 4533 1150 LDCMO
3993 4534 1015 TAD CHRG
3994 4535 0450 TAD K0200
3995 4536 7332 LDCMO
3996 4537 3143 CLA CLL CML PTR
3997 4540 0444 DCA GOREG2
3998 4541 0447 DOSTAT
3999 4542 5353 DSKAMP
4000 4543 7357 JMP NOIN
4001 4544 3143 CLA CLL CML RAR
4002 4545 0444 DCA GOREG2
4003 4546 1105 TAN K0000
4004 4547 7640 SZA CLA
4005 4550 2330 ISZ DIN
4006 4551 2330 ISZ DIN
4007 4552 6730 JMP I DIN
4008 4553 1106 JMP I K0000
4009 4554 7640 SZA CLA
4010 4555 5500 JMP I DIN
4011 4556 5730 JMP I DIN
4012 4557 0000 /ROUTINE TO COMPARE AC TO GOREG2
4013 4558 0000 COMP1,
4014 4559 3155 DCA ACREG
4015 4560 1155 TAD ACREG
4016 4561 7041 CIA
4017 4562 1143 TAD GOREG2
4018 4563 7640 SZA CLA
4019 4564 2357 ISZ COMP1
4020 4565 5757 JMP I COMP1
4021 4566 5757 /
4022 4567 0000 /
4023 4568 0000 /MANUAL FUNCTION TEST
4024 4569 3155 DCA ACREG
4025 4570 1155 TAD ACREG
4026 4571 7041 CIA
4027 4572 1143 TAD GOREG2
4028 4573 7640 SZA CLA
4029 4574 2357 ISZ COMP1
4030 4575 5757 JMP I COMP1
4031 4576 0000 /
4032 4577 0000 /MANUAL FUNCTION TEST
4033 4578 0000 /LOAD ADDRESS 0201 OR "MANUAL".
4034 4579 0000 /SET SWITCHES TO FUNCTION
4035 4580 0000 /PRESS START
4036 4581 0000 /MACHINE SHOULD HALT
4037 4582 0000 /SET SWITCHES TO 0000
4038 4583 0000 /PRESS START
4039 4584 0000 /INCREASE OF FAILURES USE NORMAL SCORE SWITCHES
4040 4585 0000 /IF LOOP IS DESTROYED USE NORMAL SCORE SWITCHES
4041 4586 0000 /
4042 4587 0000 /MANUAL, CLASSIC
4043 4588 0000 /COSWIT
4044 4589 0000 /NOP
4045 4590 0000 /LAS
4046 4591 0000 /AND
4047 4592 0000 /K7707
4048 4593 0000 /DCA TCNTR1
4049 4594 0000 /CLA CLL CMA
4050 4595 0000 /REGD
4051 4596 0000 /RTF
4052 4597 0000 /TAD TCNTR1
4053 4598 0000 /DCA TCNTR1
4054 4599 0000 /TAD TCNTR1
4055 4600 0000 /AND
4056 4601 0000 /K0006
4057 4602 0000 /DCA DRIVND
4058 4603 0000 /CLASSIC
4059 4604 0000 /CEPRR
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4740 528
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/SHARROUTINF TO WAIT FOR INTERRUPTS

AUTPRO, CLASSIC /CHECK FOR CLASSIC.

	PAL10	V1424	15-APR-76	13:24	PAGE 1-80
4200	5201	4431	CASHIT		/ROUTINE TO EXECUTE.
4209	5202	7700	NOP		/GET THE SWITCHES
4210	5203	4424	LAS		/MASK DRIVE NUMBER
4211	5204	7124	CLL RAL		/SAVE DRIVE NUMBER
4212	5205	0100	AND		/SETUP REPEAT POINTER
4213	5206	1972	NCA		/DATA PATTERN TO WRITE
4214	5207	7344	CLA CLL		/FILL OUTROUND BUFFER
4215	5208	3132	NCA		
4216	5209	3131	NCA		
4217	5210	1113	REG		
4218	5211	4431	REG		
4219	5212	1272	K2525		
4220	5213	3464	FILE		
4221	5214	1463	FILE		
4222	5215	1115	FILE		
4223	5216	1115	FILE		
4224	5217	1115	FILE		
4225	5218	1115	FILE		
4226	5219	1115	FILE		
4227	5220	1115	FILE		
4228	5221	1115	FILE		
4229	5222	1115	FILE		
4230	5223	1115	FILE		
4231	5224	1115	FILE		
4232	5225	1115	FILE		
4233	5226	1115	FILE		
4234	5227	1115	FILE		
4235	5228	1115	FILE		
4236	5229	1115	FILE		
4237	5230	1115	FILE		
4238	5231	1115	FILE		
4239	5232	1115	FILE		
4240	5233	1115	FILE		
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4256	5249	1115	FILE		
4257	5250	1115	FILE		
4258	5251	1115	FILE		
4259	5252	1115	FILE		
4260	5253	1115	FILE		
4261	5254	1115	FILE		
4262	5255	1115	FILE		

	PAL10	V1424	15-APR-76	13:24	PAGE 1-81
4263	5256	2200	TAPROT, 0200		/TEXT POINTER
4264	5257	4425	CLASSIC		/SUCCESSFUL WRITE PROTECT. TO
4265	5258	4436	CLASSIC		/REPEAT: IF ON CLASSIC CONSOLE
4266	5259	4436	CLASSIC		/PACKAGE HIT CONTROL F. IF NOT
4267	5260	4436	CLASSIC		/PRESS KEY CONTINUE.
4268	5261	4436	CLASSIC		
4269	5262	4436	CLASSIC		
4270	5263	4436	CLASSIC		
4271	5264	4436	CLASSIC		
4272	5265	4436	CLASSIC		
4273	5266	4436	CLASSIC		
4274	5267	4436	CLASSIC		
4275	5268	4436	CLASSIC		
4276	5269	4436	CLASSIC		
4277	5270	4436	CLASSIC		
4278	5271	4436	CLASSIC		
4279	5272	4436	CLASSIC		
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4286	5279	4436	CLASSIC		
4287	5280	4436	CLASSIC		
4288	5281	4436	CLASSIC		
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4307	5300	4436	CLASSIC		
4308	5301	4436	CLASSIC		
4309	5302	4436	CLASSIC		
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4312	5305	4436	CLASSIC		
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4316	5309	4436	CLASSIC		
4317	5310	4436	CLASSIC		

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PAL10 VI42A 15-APR-76 13124 PAGE 1-62

4310 5140 3275 DCA MYLAS
4310 5141 3302 DCA CLASSIK
4320 5142 4447 DSKKMP
4321 5143 7610 JMP CLA
4322 5144 5352 JMP **6
4323 5145 2302 132 CLASSIK
4324 5146 5342 JMP **4
4325 5147 2275 132 MYLAS
4326 5148 5342 JMP **6
4327 5151 7610 SKP CLA
4328 5152 2310 132 SKMAT
4329 5153 5734 JMP I SKMAT
/EXIT

/NO SKIP OCCURRED

/ROUTINE TO READ STATUS REGISTER
/
4330 5154 0000 ROST, 0
4331 5155 6745 ROST, ROST
4332 5156 5363 JMP **5
4333 5157 4405 CLASSIC
4334 5158 7402 CAREP
4335 5162 5357 FAULTS, MVT
4336 5163 5357 JMP **3
4337 5164 1146 DCA STREG
4338 5165 5744 JMP I STREG
4339 5173 1302 JMP I ROST
4340 5174 0022
4341 5175 0021
4342 5176 0020
4343 5177 0000
4344 5177 0000
4345 5177 0000
4346 5177 0000
4347 5177 0000
4348 5200 0000
4349 5201 4527
4350 5202 1600
4351 5203 3173
4352 5204 4404
4353 5205 7700
4354 5206 5217
4355 5207 4404
4356 5210 7806
4357 5211 7710
4358 5212 5215
4359 5213 1356
4360 5214 4436
4361 5215 1600
4362 5216 5757
4363 5217 1600
4364 5220 3369
4365 5221 2204
4366 5222 7391
4367 5223 1200
4368 5223 1200
4369 5223 1200
4370 5223 1200
4371 5223 1200

PAGE
/ROUTINE FOR "ERRORS," SCOPE LOOPS, AND
/ERROR TYPEOUTS.
/
FERR, 0
JMS I KERR0
TAD I ERRO
DCA RSTRT
LAS
SKP CLA
JMP ERR01
LAS
RPL
SPA CLA
JMP **3
JMP K0207
TYPE
TAD I ERRO
JMP I ESCAPE
ERR01, DCA RETRN2
132 ERRO
CLA CLL IAC
TAD ERRO
/NEXT TEST POINTER

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/
PAL10 VI42A 15-APR-76 13124 PAGE 1-63

4372 5224 3361 DCA
4373 5225 4462 CRLF INHINT
4374 5226 4462 CRLF
4375 5227 1600 TAD I ERRO
4376 5231 0101 AND K0007
4377 5231 1367 TAD MEDIAN
4378 5233 3233 DCA **1
4379 5233 7402 MVT **2
4380 5236 1236 DCA
4381 5237 4457 PRINTER
4382 5237 7402 MVT
4383 5237 4462 CRLF
4384 5240 4457 PRINTER
4385 5241 5750 TXPC
4386 5242 7346 CLA CLL CMA
4387 5243 1200 TAD ERRO
4388 5244 4460 OCTEL
4389 5245 1600 TAD I ERRO
4390 5246 7104 CLL BAL
4391 5247 7430 SWL
4392 5250 5264 JMP NTCN
4393 5250 5264
4394 5251 3200 DCA ERRO
4395 5252 4457 PRINTER
4396 5253 5752 TXGN
4397 5254 1200 TAD ERRO
4398 5255 7700 SKA CLA
4399 5256 5261 JMP **1
4400 5257 1142 TAD GREG1
4401 5260 4461 TAD TWOCT
4402 5261 1103 TAD GOREG2
4403 5262 4460 OCTEL
4404 5263 7610 SKP CLA
4405 5264 3200 DCA ERRO
4406 5265 1200 TAD BAL
4407 5267 7402 SWL
4408 5267 7402 NTCRC
4409 5270 5391 JMP DCA ERRO
4410 5271 1200 DCA
4411 5272 4457 PRINTER
4412 5273 5754 TXGN
4413 5274 1144 TAD CAREG1
4414 5275 4461 TAD TWOCT
4415 5276 1145 TAD CAREG2
4416 5277 4460 OCTEL
4417 5278 7610 SKP CLA
4418 5279 3200 DCA ERRO
4419 5280 1365 TXEXT
4420 5281 1366 PCNTR2
4421 5282 1364 TAN XREG
4422 5283 1010 DCA AUTOLO
4423 5284 1116 TAD K7771
4424 5285 3365 DCA PCNTR1
4425 5286 1200 STRAUT, TAD
4426 5286 1200

/STORE FOR SPECIAL RETURN
/GET TEXT POINTER
/MAKE ERROR HEADER TAD
/MODIFIED HEADER TAD
/MODIFIED HEADER POINTER
/PRINT PC:
/GET PC POINTER
/PRINT PC STORED
/GET TEXT POINTER
/NOT GO: REGISTER
/PRINT GN:
/MAS IT A & BIT OCTAL BYTE
/GET DATA
/PRINT TWO OCTAL
/PRINT FOUR OCTAL
/GET TEXT POINTER
/PRINT CR:
/PRINT
/PRINT FOUR OCTAL
/COUNTER FOR # OF HEADS
/GET TEXT POINTER

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PAGE 15-APR-76 13124 PAGE 1-64

4427 5311 7500 SNA
4428 5312 5350 JMP NUTEX
4429 5313 7104 CLL RAL
4430 5314 3200 DCA ERRO
4431 5315 1366 TAN PCNTR2
4432 5316 2366 TSZ PCNTR2
4433 5317 2366 ISZ PCNTR2
4434 5320 1322 DCA PCNTR2
4435 5321 4457 NCA PRNTR
4436 5322 7402 MLY TAN I
4437 5323 1410 OCTFL PCNTR1
4438 5324 2366 ISZ STRAUT
4439 5325 5312 AGAIN,
4440 5326 4460 LMS
4441 5327 4454 BUT
4442 5331 7224 AND K0000
4443 5332 7652 SNA CLA
4444 5333 5340 JMP CMRFR
4445 5334 7337 JNP
4446 5335 5340 JNP CMRFR
4447 5336 5340 JNP CMRFR
4448 5337 1361 TAN INHIBIT
4449 5338 5757 JNP I ESCAPE
4450 5339 1360 TAN ESCAPE
4451 5341 5757 JNP I ESCAPE
4452 5342 4404 CMRFR, CLASIC
4453 5343 4436 CMRFR
4454 5344 7422 F0MLT9, MLY
4455 5345 4762 JMS I XOTREG
4456 5346 5760 JNP I RETRN2
4457 5347 5264 JNP I NYCN
4458 5350 7104 NOTFX, CLL RAL
4459 5351 3200 DCA ERRO
4460 5352 2366 ISZ PCNTR2
4461 5353 2366 ISZ PCNTR2
4462 5354 2010 ISZ AUTO10
4463 5355 5325 JNP AGAIN
4464
4465 5356 0207 K0207, 0207
4466 5357 5470 ESCAPE, SCOPE
4467 5358 0000 RETRN2, 0
4468 5361 0000 INHIBIT, 0
4469 5362 5527 XOTREG, GTRG
4470 5363 5756 TEXT, TEXT
4471 5364 0145 X0FG, CMRFR2
4472 5365 0000 PCNTR1, 0
4473 5366 0000 PCNTR2, 0
4474 5367 1370 HENTAD, TAN
4475 5370 6671 HENTAD, FRTX1
4476 5371 6704 FRTX1
4477 5372 6720 FRTX2
4478 5373 6736 FRTX3
4479 5374 6746 FRTX4
4480 5375 6760 FRTX5
4481 5376 6772 FRTX6
4482 5377 6772 FRTX7

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PAGE 15-APR-76 13124 PAGE 1-65

4482 5377 7002 PAL10 V1424
4483 5378 7002 FRTX8
4484
4485 5400 PAGE
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4489 5400 0200 R0AF, 0
4490 5401 7330 CLL CIL CIL RAR
4491 5402 4455 L0MAN
4492 5403 1012 TAN K0020
4493 5404 4455 L0MAN
4494 5405 3147 NCA D0REG
4495 5406 1147 TAN D0REG
4496 5407 3154 DCA D0REG
4497 5408 1154 TAN D0REG
4498 5411 5600 JNP I R0AF
4499
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4504 5412 0200 R0CM, 0
4505 5413 7330 CLL CIL
4506 5414 1126 TAN M12
4507 5415 3133 NCA SRCNT1
4508 5416 7330 CLL CIL CIL RAR
4509 5417 4455 L0MAN
4510 5421 7017 RAR
4511 5422 4455 L0MAN
4512 5423 7330 CLL CIL
4513 5424 1216 TAN K0020
4514 5425 2133 L0MAN
4515 5426 5224 ISZ SRCNT1
4516 5427 7330 JNP --2
4517 5428 1012 CLL CIL
4518 5431 4455 TAN K0020
4519 5432 3154 L0MAN
4520 5433 1150 DCA CMRFR
4521 5434 1150 TAN CMRFR
4522 5435 5612 JNP I R0CM
4523
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4526 5435 0200 ROUTINE TO ZERO WORK BUFFER
4527 5436 7340 KLRUF, 0
4528 5437 1067 TAN
4529 5440 3010 R0N0UF
4530 5441 1123 DCA AUTO10
4531 5442 3162 TAN K7400
4532 5443 3410 DCA DATCNT
4533 5444 2162 DCA I AUTO10
4534 5445 5203 JNP --2
4535 5446 5635 JNP I KLRUF
4536

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PAL10 15-APR-76 13:24 PAGE 1-06

/THE COMPLEMENT DATA THATS IN THE AC.

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4537 5447 0000 15-APR-76 13:24 PAGE 1-06
4538 5450 3163
4539 5451 7340
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4541 5453 3212
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/ROUTINE TO CHECK FOR WAIT AND RECALIBRATE

```

SCOPE, DCA TOTST
LAS
AND K0020
SZA CLA
WAIT LOOP?
YES
GET SWITCH 6
/IS IT CLEAR DISK
/NO, DOWN?
/ENABLE CLEAR CONTROL
/GET LAST COMMAND
/ASK OUT SFT DONE
/LOAD COMMAND
/ENABLE RECALIBRATE
/WAIT FOR FIRST DONE
/LAST COMMAND
/LOAD COMMAND
/WAIT FOR SECOND DONE
/MASK SET DONE
/ENABLE CLEAR CONTROL
/CLEAR CONTROL
/GO TO TEST

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SEE 0111

4702	5656	0000	FIGURE,	15-APR-76	13124	PAGE 1-60
4703	5657	3143	OCA			/SAVE FOR ERROR PRINTER
4704	5660	1067	TAD			/GET START OF BUFFER
4705	5661	3153	NCA			/SAVE FOR ERROR PRINTER
4706	5662	1150	ADREG			/GET DISK NO. AND EXT. BIT
4707	5663	0101	AND			/MASK THEM
4708	5664	7041	CIA			
4709	5665	1553	TAD I			/GET FIRST TRACK WORD
4710	5666	7650	SNV CLA			/WAS IT O.K.?
4711	5667	5273	JMP			/YES, CHECK NEXT TRACK WORD
4712	5670	1150	TAD			/GET DISK NO. AND EXT. BIT
4713	5671	0101	AND			/MASK THEM
4714	5672	5333	JMP			/DATA ERROR
4715	5673	2153	ADREG			/UPDATE ADDRESS
4716	5674	1553	TAD I			/GET SECOND WORD
4717	5675	7041	CIA			
4718	5676	1151	TAD			/COMPARE TO ADDRESS
4719	5677	7650	SNV CLA			/WAS SECOND TRACK WORD O.K.
4720	5700	5303	JMP			/YES, NOW CHECK DATA
4721	5701	1151	TAD			/GET GOOD INFO
4722	5702	5343	JMP			/DATA ERROR
4723	5703	7326	CLA			
4724	5704	1123	CLA CLL			/SETUP COUNTER
4725	5705	3162	NCA			/UPDATE COUNTER
4726	5706	2153	ADREG			/MORE TO CHECK
4727	5707	1553	TAD I			/UPDATE ADDRESS
4728	5710	7041	CIA			
4729	5711	1143	TAD			/COMPARE TO GOOD ONE
4730	5712	7640	STA CLA			/WAS WORD O.K.?
4731	5713	5340	JMP			/NO, DATA ERROR
4732	5714	1143	TAD			/GET GOOD DATA
4733	5715	7040	CHA			
4734	5716	3143	NCA			/IT IS A COMPLEMENT DATA PATTERN
4735	5717	2162	ADREG			/UPDATE BUFFER COUNTER
4736	5720	5306	JMP			/MORE TO CHECK
4737	5721	2153	ADREG			/UPDATE ADDRESS
4738	5722	1102	TAD			
4739	5723	7041	CIA			
4740	5724	1553	TAD I			/GET WORD IN BUFFER+1
4741	5725	7650	SNV CLA			/WAS IT O.K.?
4742	5726	5331	JMP			/YES ALL DATA O.K.
4743	5727	1102	TAD			
4744	5730	5343	JMP			/WORD LOST IN BUFFER+1
4745	5731	7330	NCA			/EXPECTED STATUS
4746	5732	3143	ADREG			/SETUP COMPARE REGISTER
4747	5733	1171	TAD			/GET CRC ERROR FLAG
4748	5734	7640	CHA			/WAS IT SET
4749	5735	5656	JMP I			/NO THE BUFFER IS O.K.
4750	5736	7340	CLA CLL			/SETUP CRC FLAG
4751	5737	3171	NCA			/RESET FLAG
4752	5740	1166	TAD			/TEXT MESS
4753	5741	3572	NCA I			/SETUP TEXT POINTER
4754	5742	7330	CLA CLL			/EXPECTED STATUS
4755	5743	1143	ADREG			/SETUP COMPARE
4756	5744	1553	TAD I			/GET BAD WORD

SEE 0113

ADDRESS	HEX	ASSEMBLY	DISASSEMBLY	COMMENT
4002	6770	7320	CLA CL	
4003	6031	1212	TAD	K0020
4004	6032	4055	LOMAN	
4005	6033	0117	AND	K0017
4006	6034	3140	DCA	CRPF061
4007	6035	5600	JMP I	ROCR
4008				
4009				
4010				
4011				
4012	6036	0000	TOCT,	0
4013	6037	3133	DCA	SR0CT1
4014	6040	1133	TAD	SR0CT1
4015	6041	7010	RAR	
4016	6042	7012	RTR	
4017	6743	0101	AND	K0007
4018	6044	1777	TAD	K0260
4019	6045	4436	TYPE	
4020	6746	1133	TAD	SR0CT1
4021	6047	0101	AND	K0007
4022	6050	1777	TAD	K0260
4023	6051	4436	TYPE	
4024	6052	5636	JMP I	TOCT
4025				
4026				
4027				
4028				
4029				
4030	6053	0000	UPONE,	0
4031	6054	7300	CLA CL	
4032	6055	1262	TAD	K0215
4033	6056	4436	TYPE	
4034	6057	1263	TAD	K0212
4035	6058	4436	TYPE	
4036	6261	5633	JMP I	UPONE
4037				
4038	6062	0215	K0215,	0215
4039	6063	0212	K0212,	0212
4040	6064	0240	K0240,	0240
4041				
4042				
4043				
4044	6065	0000	ROUTINE TO PRINT FOUR OCTAL	
4045	6066	7006	ROUT,	0
4046	6067	7006	RTL	
4047	6070	3253	DCA	UPONE
4048	6071	1310	TAD	M4
4049	6072	3236	TOCT	
4050	6073	1253	TAD	UPONE
4051	6074	0101	AND	K0007
4052	6075	1777	TAD	K0260
4053	6076	4436	TYPE	
4054	6077	1253	TAD	UPONE
4055	6100	7006	RTL	
4056	6101	7004	RTL	

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4837 6102 3253 OCA UPONE
4838 6103 2236 I82 TOCT
4839 6104 5273 JMP --11
4840 6105 1264 TAD K0240
4841 6106 4436 TYPE
4842 6107 5665 JMP I PROCT
4843 6110 7774 M4, 7774
4844
4845 /SUBROUTINE TO PRINT TEXT
4846 PRN,
4847 CLA CLL
4848 TAD I PRN
4849
4850 /GET POINTER
4851 6111 0000
4852 6112 7300
4853 6113 1711
4854
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4857
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4872 6114 2311
4873 6115 3265
4874 6116 1665
4875 6117 0111
4876 6120 7490 SNA
4877 6121 5305 JMP EXIT
4878 6122 7500 SNA
4879 6123 7020 CML
4880 6124 7001 IAC
4881 6125 7012 RTR
4882 6126 7012 RTR
4883 6127 7012 RTR
4884 6130 4036 TYPE
4885 6131 1665 TAD I
4886 6132 0112 AND K0877
4887 6133 7490 SNA
4888 6134 5345 JMP EXIT
4889 6135 1350 TAD K3740
4890 6136 7500 SNA
4891 6137 1347 TAD K4100
4892 6140 1264 TAD K0240
4893 6141 4036 TYPE
4894 6142 2265 ISZ PROCT
4895 6143 7300 CLA CLL
4896 6144 5316 JMP PRN+5
4897 6145 7300 EXIT, CLA CLL
4898 6146 7300 JMP I PRN
4899
4900 /
4901 K4100, 4100
4902 K3740, 3740
4903
4904 /ROUTINE TO TYPE
4905
4906 PRINT,
4907 CLASIC
4908 CBTYP
4909 SKP I
4910 JMP I PRINT
4911 TLS

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4912 6157 6041 TSF
4913 6160 5357 JMP --1
4914 6161 6042 TCF
4915 6162 7200 CLA
4916 6163 5741 JMP I PRINT
4917
4918 /SUBROUTINE TO LOAD TRACK ADDRESS REGISTER
4919 LOAD,
4920 RCA
4921 6165 3151 DAREG
4922 6166 1151 TAD DAREG
4923 6167 4743 TOT3, DLAG
4924 6170 5764 JMP I LOAD
4925 6171 4005 CLASIC
4926 6172 4436 COERR
4927 6173 7002 ERHLT3, WIT
4928 6174 5371 JMP --3
4929
4930 /
4931 PAGE
4932 /ROUTINE TO RECALIBRATE SELECTED DRIVE OR
4933 /SEEK ONLY POSITION IN AC ON SELECTED DRIVE.
4934
4935 RESTOR, R
4936 CLA CLL
4937 TAD I RESTOR
4938 DCA SAVPC
4939 6200 1600 DCA
4940 6201 3316 ISZ RESTOR
4941 6202 2200 ISZ RESTOR
4942 6205 1200 DCA ONLY
4943 6207 1272 TAD DRIVNO
4944 6210 1156 TAD HOMEWA
4945 6211 4450 LOCMD
4946 6212 7336 CLA CLL CML RTL
4947 6213 4453 CLALL
4948 6214 5232 JMP CHECK
4949
4950 /ONLY,
4951 R
4952 DCA SAVTO
4953 TAD I ONLY
4954 DCA SAVPC
4955 6220 3316 DCA
4956 6221 2215 ISZ ONLY
4957 6222 1150 TAD CREG
4958 6223 0073 AND K0001
4959 6224 1156 TAD HOMEWA
4960 6225 1072 TAD DRIVNO
4961 6226 1104 TAD K3000
4962 6227 1104 LOCMD
4963 6230 1317 TAD
4964 6231 4452 LOAD
4965 6232 4452 SKPMAT
4966 6233 5314 JMP SEER1
4967 6234 7330 CML PAR
4968 6235 3145 DCA CLL GDREG2

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4969 6235 3145 DCA CLL GDREG2
4970
4971 /SAVE OUTBOUND DATA
4972 /LOAD DISK ADDRESS REGISTER
4973 /EXIT
4974 /SKIP TRAP ERROR.
4975
4976 /GET TEXT POINTER
4977 /SAVE FOR ERROR
4978 /UPDATE PC
4979 /GET PC
4980 /SAVE FOR END OF SEEK ROUTINE
4981 /CURRENT DRIVE
4982 /CURRENT FIELD
4983 /LOAD COMMAND
4984 /ENABLE RECALIBRATE BIT
4985 /"RECALIBRATE"
4986 /CHECK FOR ERRORS
4987
4988 /GET LOWER TRACK BITS
4989 /GET TEXT POINTER
4990 /SAVE FOR ERROR
4991
4992 /GET COMMAND
4993 /MASK OFF EXTENDED BIT
4994 /CURRENT FIELD
4995 /CURRENT DRIVE
4996 /SEEK ONLY FUNCTION
4997 /LOAD COMMAND
4998 /GET POSITION
4999 /LOAD AND GO
5000 /WAIT FOR FIRST DONE FLAG
5001 /ERROR, NO SKIP
5002 /EXPECTED STATUS
5003 /SETUP COMPARE REGISTER

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	PAL10	VI42A	15-APR-76	13:24	PAGE 1-98
4986	6236	1122	TAD	K7700	/SETUP SKIP TIMER
4987	6237	3321	DCA	RNAD	/READ STATUS
4988	6240	4844	RNSTAT		
4989	6241	1125	TAD	K8000	/HAS DRIVE DONE?
4970	6242	7650	SNA CLA		/YES DRIVE MUST BE BUSY!
4971	6243	5252	JMP	*7	/NO, DRIVE MUST BE BUSY!
4972	6244	1106	TAD	K8000	/EXPECTED STATUS
4973	6245	3143	DCA	GREGE2	/GET STATUS READ
4974	6246	3143	TAD	K8000	/SET STATUS READ
4975	6247	1103	TAD	K2000	/ADD IN FUDGE FACTOR
4976	6248	7640	SZA CLA		/HAS DRIVE BUSY
4977	6251	5311	JMP	SEKER2	/NO, ERROR
4978	6252	1015	TAD	K0200	/ENABLE SET SECOND DONE FLAG
4979	6253	1150	TAD	CREG	/ORIGINAL COMMAND
4980	6254	4850	LNCMD		/LOAD COMMAND
4981	6255	7332	CLA CLA	CML ATR	/EXPECTED STATUS
4982	6256	3143	DCA	GOREGE2	/APT TITING
4983	6257	4530	TICK		/READ STATUS
4984	6260	4844	RNSTAT		/FLAG SET?
4985	6261	4847	SKP	DSKSKP	/YES GOT IT!
4986	6262	7100	SKP		/NO
4987	6263	5274	JMP	GOTSKP	
4988	6264	1106	TAD	K6000	/DRIVE BUSY?
4989	6265	7640	SZA CLA		/NO, ERROR
4990	6266	5311	JMP	SEKER2	
4991	6267	2365	ISZ	RNWRD4	
4992	6270	5257	JMP	CHKSKP	
4993	6271	2321	ISZ	RNAD	
4994	6272	5257	JMP	CHKSKP	
4995	6273	5314	JMP	SEKER1	
4996	6274	7330	CLA CLA	CML BAR	/ERROR, NO SKIP!
4997	6275	3143	DCA	GOREGE2	/SETUP EXPECTED STATUS
4998	6276	4844	RNSTAT		/READ STATUS
4999	6277	1105	TAD	K4000	/HAS IT ONLY DONE FLAG
5000	6300	7640	SZA CLA		/NO, ERROR STATUS
5001	6301	5311	JMP	SEKER2	/GET LAST COMMAND
5002	6302	1150	TAD	CREG	/MASK OUT
5003	6303	0320	AND	A7577	/CLEAR STATUS
5004	6304	4850	LNCMD	GOREGE2	/SETUP COMPARE REGISTER
5005	6305	3143	DCA	POSTAT	/READ STATUS
5006	6306	4844	RNSTAT		/HAS STATUS 0000?
5007	6307	7650	SNA CLA	ONLY	/UPDATE PC
5008	6310	2215	ISZ		
5009	6311	1166	SEKER2	TAD	/SETUP TEXT POINTER
5010	6312	3716	GORAK	DCA I	/BACK TO TEST
5011	6313	5615	ONLY	JMP I	/SKIP TEXT POINTER
5012	6314	1164	SEKER1	TAD	/EXIT
5013	6315	5312	JMP	GORAK	
5014					
5015	6316	0000	SAVPC		
5016	6317	0000	SAVTD		
5017	6320	7577	A7577	7577	
5018					
5019					
5020					

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5021	6321	0000	RNAD		/SAVE DISK NO. POINTER
5022	6322	3361	DCA	SAVPOT	
5023	6323	7101	CLA IAC		
5024	6324	1363	TAD	RNWRD1	
5025	6325	1364	TAD	RNWRD2	
5026	6326	7106	CLA RTL		
5027	6327	3363	DCA	RNWRD1	
5028	6328	1364	TAD	RNWRD2	
5029	6331	7012	TR		
5030	6332	1363	TAD	RNWRD1	
5031	6333	3364	DCA	RNWRD2	
5032	6334	1364	TAD	RNWRD2	
5033	6335	7420	SNL		
5034	6336	5342	JMP	GOTAD	
5035	6337	1170	TAD	ENDTRK	
5036	6341	1364	CLA		
5037	6342	3365	DCA	RNWRD2	/GET SAME
5038	6343	1362	TAD	DSKSAV	/SAVE WORD
5039	6344	1361	TAD	SAVPOT	/GET POINTER
5040	6345	3361	DCA	SAVPOT	/ADD IN DRIVE NUMBER
5041	6346	1365	TAD	RNWRD4	/MAKE ADDRESS
5042	6347	3761	DCA I	SAVPOT	/GET WORD
5043	6350	1361	TAD	SAVPOT	/STORE IT
5044	6351	1076	TAD	K0004	
5045	6352	3361	DCA	SAVPOT	/ADD IN FUDGE FACTOR
5046	6353	7004	RAL		/MAKE ADDRESS
5047	6354	3761	DCA I	SAVPOT	/GET THE LINK
5048	6355	1761	TAD I	SAVPOT	/SAVE EXTENDED BIT
5049	6356	7110	CLA RAR		/SHIFT
5050	6357	1365	TAD	RNWRD4	/GET WORD
5051	6360	5721	JMP I	RNAD	/EXIT
5052					
5053					
5054	6361	0000	SAVPOT		
5055	6362	6366	DSKSAV	DSKSA	
5056	6363	1234	RNWRD1	1234	
5057	6364	2345	RNWRD2	2345	
5058	6365	0000	RNWRD4		
5059	6366	0000	DSKRAV		
5060	6367	0000	DSK1A		
5061	6370	0000	DSK2A		
5062	6371	0000	DSK3A		
5063	6372	0000	DSK0R		
5064	6373	0000	DSK1B		
5065	6374	0000	DSK2B		
5066	6375	0000	DSK3B		
5067					
5068					
5069	6400		PAGE		
5070					
5071					
5072	6400	0000	/SUBROUTINE FOR "NO ERRORS" AND SCOPE		
5073	6401	2200	/LOOPS. UPDATE UP COUNTER "REGI" ON EVERY ENTRY.		
5074					
5075	6402	7300	NRROD		
			ISZ	NRRO	
			CLA CLA		

```

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5076 6403 4530 TICK
5077 6400 1690 TAD I NERR0
5078 6405 3173 DCA RSTRT
5079 6406 4405 CLASIC
5080 6407 4402 CCRKA
5081 6410 7330 NOP
5082 6411 4404 LAS
5083 6412 0215 AND K0200
5084 6413 7650 SNA CLA
5085 6414 5223 JMP
5086 6415 4405 CLASIC
5087 6416 4437 CTRINH
5088 6417 7442 STOPLT, HIT
5089 6420 4424 LAS
5090 6421 7324 GLE
5091 6422 7720 SNA CLA
5092 6423 5226 JMP
5093 6424 1632 TAD I NERR0
5094 6425 5642 JUP I NSCOPE
5095 6426 1131 TAD RFG0
5096 6427 7642 SZA CLA
5097 6430 5213 JMP NEXTST
5098 6431 2132 ISZ REG1
5099 6432 5513 JMP I RSTRT
5100 6433 7301 JUP I CLRALL
5101 6434 4453 ISZ
5102 6435 2200 NERR0
5103 6436 2200 NERR0
5104 6437 5600 JUP I NERR0
5105 6440 5470 NSCOPE, SCOPE
5106 6441 7330 /
5107 6442 1647 /ROUTINE TO DO HALF BLOCK DATA CHECKS
5108 6443 3153 /
5109 6444 1150 DCA
5110 6445 1150 DCA
5111 6446 0101 AND K0007
5112 6447 7041 CTA
5113 6448 1553 TAD I ADRG
5114 6449 7650 SNA CLA
5115 6452 5256 JMP
5116 6453 1150 TAD
5117 6454 8101 AND K0007
5118 6455 5137 JMP NERR0
5119 6456 2153 ISZ ADRG
5120 6457 1553 TAD I ADRG
5121 6458 7041 CTA
5122 6459 1151 TAD NAREG
5123 6461 1151 SNA CLA
5124 6462 7650 JMP
5125 6463 5246 DAREG
5126 6464 1151 JMP
5127 6465 5337 NERR0

```

```

/GET RSTRT ADDRESS
/STORE

```

```

/GET SWITCH 4
/MASK
/PROGRAM HALT

```

```

/STOP HALT FROM SWRST1
/GET SWITCH 1

```

```

/IS IT SCOPE LOOP
/NO
/GET RETURN POINTER
/CHECK FOR WAIT AND RETURN.

```

```

/1 OR 4096 PASSES
/1 PASS PER TEST
/UPDATE UP-COUNTER
/BACK TO SAME TEST
/ENABLE CLEAR CONTROL
/CLEAR CONTROL
/UPDATE PC STORE
/UPDATE PC STORE
/TO NEXT SEQUENTIAL TEST

```

```

/SETUP FOR ERROR PRINTER
/GET START OF BUFFER
/FOR ERROR PRINTER

```

```

/COMPARE TO BUFFER WORD
/SAME ?
/VFS

```

```

/NO
/UPDATE ADDRESS
/COMPARE TO DISK ADDRESS
/NAME??7?
/VFS
/NO

```

```

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5131 6466 2153 ISZ ADRG
5132 6467 7326 CLA CLL CML RTL
5133 6471 1124 TAD K7600
5134 6472 1162 DCA DATCNT
5135 6473 1553 NERR1, TAD I ADRG
5136 6474 7041 CTA
5137 6475 1143 TAD ADRG2
5138 6476 7640 SZA CLA
5139 6477 5340 JMP NERR+1
5140 6478 2153 ISZ ADRG
5141 6479 1143 TAD
5142 6480 7040 CMA
5143 6482 3143 DCA GDRG2
5144 6483 2142 ISZ DATCNT
5145 6484 5272 JMP NERR1
5146 6485 1124 TAD K7600
5147 6486 3143 DCA DATCNT
5148 6487 3143 DCA GDRG2
5149 6488 1553 NERR2, TAD I ADRG
5150 6489 7640 SZA CLA
5151 6491 2153 JMP NERR
5152 6492 2153 ISZ ADRG
5153 6493 2162 ISZ DATCNT
5154 6494 5310 JMP NERR2
5155 6496 1553 TAD I ADRG
5156 6497 7041 CTA
5157 6498 1102 TAD K1234
5158 6499 7650 SNA CLA
5159 6501 5325 JMP
5160 6502 1172 NERR+3
5161 6503 5337 TAD K1234
5162 6504 5337 JUP NERR
5163 6505 3143 DCA CLL CML RAR
5164 6506 1171 TAD SOFER0
5165 6507 7640 SZA CLA
5166 6511 5641 JMP I MFCMK
5167 6512 7340 CLA CLL CMA
5168 6513 3171 DCA SOFER0
5169 6514 1166 TAD K5300
5170 6515 3572 DCA I SAVDCT
5171 6516 7330 NCA I
5172 6517 3143 CLA CLL CML RAR
5173 6518 1553 NERR0, NCA
5174 6519 3143 DCA GDRG2
5175 6520 1154 TAD I ADRG
5176 6521 2241 DCA
5177 6522 5641 ISZ MFCMK
5178 6523 5641 JMP I MFCMK
5179 6524 0200 /
5180 6525 3150 /SUPEROUTINE TO LOAD COMMAND REGISTER
5181 6526 4405 LCM, 0
5182 6527 4405 DCA CMREG
5183 6528 4405 CLASIC
5184 6529 7020 CCRKA
5185 6531 1150 NOP
5186 6532 1150 TAN CMREG

```

```

/UPDATE ADDRESS

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```

/SETUP COUNTER FOR FIRST HALF

```

```

/COMPARE TO GOOD VALUE
/WERE THEY THE SAME
/UPDATE ADDRESS POINTER

```

```

/NEXT WORD IS COMPLEMENT
/MORE TO TEST IN FIRST HALF
/SETUP COUNTER
/REST OF BUFFER SHOULD BE 0000
/VAS IT 0
/ERROR

```

```

/MORE TO CHECK
/GET WORD IN BUFFER+1
/VFS
/MAS IT 0.K.?

```

```

/ERROR, BUFFER+1
/EXPECT STATUS
/SETUP COMPARE REGISTER
/GET CRC ERROR FLAG
/MAS IT SET
/NO ERRORS
/RESET CRC ERROR FLAG
/TEXT
/SET UP POINTER
/EXPECT STATUS
/SETUP COMPARE
/GET RAR WORD
/SAVE FOR PRINTER

```

```

/SAVE OUTROUND DATA

```



```

6721 2313
6722 4001
6723 0404
6724 2205
6725 2323
6726 4022
6727 0507
6730 1123
6731 2405
6732 2240
6733 3522
6734 2217
6735 2204
6736 2411
6737 2313
6738 4024
6741 0124
6742 0140
6743 0522
6744 2217
6745 2204
6746 0322
6747 0340
6750 2205
6751 0711
6752 2324
6753 0522
6754 4005
6755 2222
6756 1722
6757 0000
6758 0401
6761 2401
6762 4022
6763 0507
6764 1123
6765 2405
6766 2240
6767 0522
6774 2217
6771 2200
6772 0411
6773 2313
6774 4023
6775 1311
6776 2004
6777 0522
7224 2317
7225 2204
7226 0411
7227 2313
7228 4011
7229 1624
7230 0522
7237 2225

5267
FRTX0, TEXT "DATA REGISTER ERROR"
6761 2401
6762 4022
6763 0507
6764 1123
6765 2405
6766 2240
6767 0522
6774 2217
6771 2200
6772 0411
6773 2313
6774 4023
6775 1311
6776 2004
6777 0522
7224 2317
7225 2204
7226 0411
7227 2313
7228 4011
7229 1624
7230 0522
7237 2225

5268
FRTY7, TEXT "DISK SKIP ERROR"
6771 2200
6772 0411
6773 2313
6774 4023
6775 1311
6776 2004
6777 0522
7224 2317
7225 2204
7226 0411
7227 2313
7228 4011
7229 1624
7230 0522
7237 2225

5269
FRTX0, TEXT "DISK INTERRUPT ERROR"
7202 0411
7203 2313
7204 4011
7205 1624
7206 0522
7207 2225

5270
FRTX0, TEXT "PASS COMPLETE"
7215 4020
7216 0123
7217 2340
7220 0317
7221 1520
7222 1405
7223 2405
7224 0004

5271
FRTX0, TEXT "PASS COMPLETE"
7215 4020
7216 0123
7217 2340
7220 0317
7221 1520
7222 1405
7223 2405
7224 0004

5272
/
5273
/
5274
/
5275
/THIS ROUTINE WILL TEST FOR THE AVAILABILITY OF THE
5276 /APT AA TEST SYSTEM AND NOP ANY CONSOLE PACKAGE WHICH
5277 /MIGHT HAVE BEEN SET UP.
5278
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5313 7065 1677 TAN I XNSKON
5314 7066 3357 DCA PCSAV
5315 7067 7240 CLA CMA
5316 7077 1727 NCA I PCSAV
5317 7071 2362 182 CLKCNT
5318 7072 2337 182 KTKCK
5319 7073 5251 JND APTAR
5320 7074 1071 TAN DRVCNT
5321 7075 3076 NCA DRVCNT
5322 7076 5762 JND I TSTOP
5323
5324 7077 4374 XNSKON, DSKON
5325 7100 7377 K7377, 7377
5326 7101 5102 WYLAS, WYLA8+3
5328 7102 0233 TSTOP, TST0-5
5329
5330 /THIS ROUTINE WILL REPORT ERRORS TO THE APT SYSTEM IF REQUIRED.
5331 /IT FIRST TEST FOR APT THEN FIGURES THE ERROR CODING.
5332
5333 AERRO, 0
5334 7103 0000
5335 7104 7200 TAN 22
5336 7105 1022 AND K4000
5337 7106 0105 SWA CLA
5338 7107 7650 JND I AERRO
5339 7110 5703 JND I AERRO
5340 7111 7340 CLC CLA CMA
5341 7112 1725 TAN I PERRO
5342 7113 1327 NCA PCSAV
5343 7114 4002 TFC
5344 7115 4224 WIF
5345 7116 1121 TAN KCOF
5346 7117 3321 OCA +2
5347 7120 1327 TAN PCSAV
5348 7121 7482 HLT
5349 7122 6272 CIF 70
5350 7123 5726 JND I K5520
5351 7124 5703 JND I AERRO
5352 7125 5200 PERRO, ERRO
5353 7126 4520 K5520, 4520
5354 7127 0000 PCSAV, 0
5355
5356 /THIS ROUTINE IS A NOP IF NOT BEING USED ON THE APT LINE.
5357 /IF APT IS ENABLED A TIMING PULSE IS GENERATED AT
5358 /APPROXIMATELY 1.5 SECOND INTERVALS
5359
5360 7130 0000 KTKCK, 0
5361 7131 1022 TAN 22
5362 7132 0105 AND K4000
5363 7133 7650 SWA CLA
5364 7134 4730 JND I KTKCK
5365 7135 2382 182 CLKCNT
5366 7136 5760 JND I KTKCK
5367 7137 1361 TAN COUNT

/SET HARDWARE CONFIGURATION
/TEST FOR APT FACH TIME
/NO TIMING GENERATED
/SEE IF TIMING NEEDS TO BE DONE
/NO. RETURN TO MAIN FLOW
/INIT FIRST CLOCK

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5368 7100 3362 NCA CLKCNT
5369 7101 2175 182 KCNT
5370 7102 5730 JND I KTKCK
5371 7103 6224 BIF
5372 7104 1121 TAN KCOF
5373 7105 3307 NCA +2
5374 7106 4002 TFC
5375 7107 7002 WIF
5376 7108 6272 CIF 70
5377 7109 4763 JND I K4500
5378 7152 7300 CLC CLA
5379 7153 1361 TAN COUNT
5380 7154 3362 NCA CLKCNT
5381 7155 1360 TAN CNT
5382 7156 3175 NCA KCNT
5383 7157 4730 JND I KTKCK
5384
5385 7149 7777 CNT, -1
5386 7161 7777 COUNT, 7777
5387 7162 0000 CLKCNT, 0
5388 7163 4500 K4500, 4500
5389
5390 7177 *7177
5391 7177 WKAUF=.
5392 7177
5393 7177 HITBK=.
5394 7200 1070K*+1
5395 7200
5396 7576 FUNDUF*+377
5397 7576
5398 7577 STPCMK*+400
5399 7577
5400
5401 $$$

/MAKE SURE AC IS CLEAR
/GET CONFIGURATION
/ISOLATE APT BIT
/ON APT
/SET UP FOR GETTING ERROR PC
/GET ERROR PC
/STORE FOR FUTURE USE
/INITIALF INTERRUPT SYSTEM
/SET UP FOR DATA FIELD IN ERROR
/ESTABLISHF DATA FIELD
/GET ERROR ADDRESS
/REPLACF WITH ERROR DATA FFLD
/FIELD OF UVPR0M
/REPORT ERROR
/RETURN TO THE NORMAL REPORTING
/POINTER TO PC IN ERROR
/POINTER TO UV PROM ADDRESS
/PLACE WHERE ERROR PC IS STORED

```

```

/FOR TESTS REQUIRING LONGER TIME OUT ON APT
/RETURN, NOT READY TO NOTIFY APT
/STAY SETUP FOR UV PROM
/WILL ESTABLISH CURRENT DATA FIELD
/CHANGED TO CURRENT DATA FIELD
/LOCATION OF UVPR0M
/LET APT KNOW YOU ARE RUNNING
/MAKE SURE AC AND LINK ARE CLEAR
/INITIALIZE CLOCK COUNTER
/POINTS TO UV PROM

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SEP 0129

6321

PAL10	V1424	15-APR-76	13124	PAGE 1-108			
RNRD1	6363	T149E	1063	T2010A	2100	T40R	3242
RNRD2	6364	T15E	1106	T2010B	2102	T40S	3246
RNRD4	6365	T15T	1118	T2010C	2105	T40T	3265
ROUTN	1302	T16E	1126	T2010D	2110	T41E	3357
ROUTP	5133	T16T	1130	T200K	2132	T41R	3372
RSTR	6630	T17E	1171	T20R	2074	T41S	3366
SANDSK	4073	T17S	1135	T20T	2137	T41T	3361
SAVAC	5132	T17T	1173	T20W	2126	T42E	3447
SAVDAT	0163	T18E	1237	T2E	0325	T42E	3402
SAVPC	6316	T18S	1204	T30D	2200	T42S	3406
SAVPT	0172	T18T	1241	T30E	2207	T42T	3451
SAVOT	6361	T19E	1267	T30R	2142	T43E	3512
SAVTR	5654	T19K	1266	T30T	2211	T43R1	3454
SAVTRK	0133	T19T	1271	T31E	2257	T43R2	3461
SCBNT1	5478	T1E	0275	T31R	2214	T43T	3514
SCOPE	1000	T200E	1317	T31T	2261	T43T	3517
SDKP	4424	T20K	1316	T32E	2372	T440K	3567
SEK	6314	T20T	1321	T32R1	2271	T44R	3526
SEKER1	6311	T21E	1350	T32R2	2312	T44T	3572
SEKER2	4260	T21K	1347	T32R3	2331	T45A1	3604
SETUP1	1233	T21T	1452	T32R4	2353	T45A2	3720
SETUP2	0225	T22E	1442	T33E	2507	T45E	3765
SKPER1	5634	T22R1	1423	T33R	2404	T45R1	3634
SKPER2	4433	T22T	1404	T33R2	2431	T45R2	3650
SKPMAT	5134	T23E	1506	T33R3	2450	T45R3	3711
SOFER1	0171	T23R1	1451	T33R4	2467	T45R4	3723
STAEPR	5636	T23R2	1470	T33T	2511	T45R5C	3620
STCON	0157	T23T	1510	T34E	2546	T45T	3787
STCKK	7577	T24E	1534	T34T	2550	T46E	0484
STPLT	6417	T24S	1513	T35E	2615	T47	0486
STRAUT	5310	T24T	1556	T36E	2673	T5E	0420
STREG	0146	T25E	1642	T36N	2700	T5T	0422
SMR	0020	T25S	1602	T36R	2637	T6E	0435
SMSEK	4012	T26T	1644	T36T	2703	T7E	0437
TPE	0256	T26E	1714	T37A	3054	T7T	0451
T10E	0571	T26R1	1651	T37E	3100	T8E	0477
T10R	0542	T26R2	1673	T37R	3015	T8T	0477
T10T	0573	T26T	1674	T37T	3102	T8T	0454
T11E	0617	T27E	1745	T38DE	3157	T9E	0501
T11R1	0602	T27R1	1723	T38E	3146	T9E	0531
T11R2	0612	T27R2	1745	T39K	3156	T9R	0507
T11R3	0616	T27T	1767	T39T	3115	T9T	0534
T11T	0641	T28E	2055	T39T	3161	TABLA	0461
T12A	0673	T2810A	2015	T39DE	3235	TABLB	0471
T12E	0677	T2810R	2017	T39E	3224	TABROT	5070
T12R	0654	T2810C	2022	T39K	3234	TCUTR1	0134
T13A	0747	T2810D	2025	T39R	3173	TCUTR2	0135
T13E	0753	T280K	2052	T39T	3237	TCUTR3	0136
T13R	0737	T28R	2010	T3E	0355	TCUTR4	0137
T13T	0737	T28T	2057	T3T	0357	TCUTR5	0140
T14E	1013	T29E	2135	T40E	3263	TCUTR6	0141

PAL10	V1424	15-APR-76	13124	PAGE 1-109	
TEXAD	5770	T8T32	2262	XDOLPT	1112
TEXCA	5766	T8T33	2408	XDO3W	0520
TEXCM	5762	T8T34	2514	XDOUT	0006
TEXCR	5754	T8T35	2551	XOSKON	7077
TEXDA	5764	T8T36	2622	XOERR0	0040
TEXD8	5760	T8T37	3002	XFCURE	0030
TEXDNT	5772	T8T38	3105	XFLUR	0031
TEXEND	7045	T8T39	3164	XPROCT	0060
TEXGD	5735	T8T40	0408	XGREG	5362
TEAPC	5750	T8T41	3248	XHCHK	0027
TEB3T	5756	T8T42	3270	XHTRK	0064
TEHPLD	0035	T8T43	3408	XIONW	0041
TICK	4530	T8T44	3452	XKLBUP	0032
TIKSTP	3541	T8T45	3517	XLAS	4163
THANE	4722	T8T46	3622	XLDAN	0052
THANOK	4721	T8T47	0423	XLOCA	0051
THANS	4643	T8T48	0430	XLOCM	0050
THANT	4720	T8T49	0434	XLOHN	0055
THPCNT	0746	T8T50	0436	XLOTR	0063
THPRT	2773	T8T51	0502	XLOTR	0063
TICT	6036	T8T52	0715	XMYLAS	7101
TIO3T	5326	T8T53E	0067	XNERR0	0037
TOURPT	4524	TYLPT	1121	XONLY	0024
TRK212	0066	TWOCT	4461	XOVRD	4164
T310	0240	TYPE	4436	XPRINT	0036
T3TOP	7102	UPAROW	0615	XRN	0057
T3T1	0265	UPONE	6053	XRNAD	0057
T3T10	0540	MAT152	4432	XRNBF	0056
T3T11	0600	MATRES	0631	XRNCH	0045
T3T12	0645	MATRUP	7177	XRNCR	0054
T3T13	0702	MAT32	4000	XRNST	0044
T3T14	1010	XCRCKP	1041	XREG	5164
T3T15	0757	XCRCL	0400	XRESTR	0025
T3T16	1111	XCRCL	1023	XRNAD	0023
T3T17	1133	XCRCL	1063	XSDP	0027
T3T18	1282	XCRCL	1207	XSMAT	0033
T3T19	1242	XCRCL	1008	XSMAT	0033
T3T20	0301	XCRCL	0200	XSMAT	0033
T3T21	1272	XCRCL	0337	XSMAT	0033
T3T22	1352	XCRCL	0335	XSMAT	0033
T3T23	1405	XCRCL	0656	XSMAT	0033
T3T24	1445	XCRCL	0656	XSMAT	0033
T3T25	1511	XCRCL	0656	XSMAT	0033
T3T26	1600	XCRCL	0656	XSMAT	0033
T3T27	1645	XCRCL	0656	XSMAT	0033
T3T28	1717	XCRCL	0656	XSMAT	0033
T3T29	1773	XCRCL	0656	XSMAT	0033
T3T30	2062	XCRCL	0656	XSMAT	0033
T3T31	2140	XCRCL	0656	XSMAT	0033
T3T32	2212	XCRCL	0656	XSMAT	0033

CORPNT	72#	673	539#	538#	461	467#	4070	4085	4208	5200	4067	4073
COROPS	688#	536	468#	538#	461	467#	4070	4085	4208	5200	4067	4073
CORPTD	531	468#	538#	461	467#	4070	4085	4208	5200	4067	4073	4073
CORTRD	436	468#	538#	461	467#	4070	4085	4208	5200	4067	4073	4073
CORSTC	527	538#	461	467#	4070	4085	4208	5200	4067	4073	4073	4073
CORSTS	429	430	442	455	461	467#	4070	4085	4208	5200	4067	4073
CORSTL	704	1291	2049	3650	3641	4054	427	4085	4208	5200	4067	4073
CORSTT	445	486	494	624	641	675	677#	4085	4208	5200	4067	4073
CORSTP	722	725	729	732	734#	675	677#	4085	4208	5200	4067	4073
COTYPE	68#	4084	2485	2903	3041	4192	924	924	926	950	4053	4067
CAF	82#	4097	2485	2903	3041	4192	924	924	926	950	4053	4067
CAREG	1293#	2430	2485	2903	3041	4192	924	924	926	950	4053	4067
CCNTR1	5295	5233#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHANG	1348	5190#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHANGR	5209#	5217	2485	2903	3041	4192	924	924	926	950	4053	4067
CHECK	4997	4962#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHECLA	134	4957	2485	2903	3041	4192	924	924	926	950	4053	4067
CHKERA	445	4455#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHKEX	3927	3932#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHKSKP	4983#	4992	2485	2903	3041	4192	924	924	926	950	4053	4067
CHMLT	1058	5220#	2485	2903	3041	4192	924	924	926	950	4053	4067
CHNPOT	5227	5230#	2485	2903	3041	4192	924	924	926	950	4053	4067
CKCOUT	1152	159#	171	171	171	171	171	171	171	171	171	171
CLASIC	1115#	1299	2049	3650	3641	4054	427	4085	4208	5200	4067	4073
CLASIK	5966	5182	5184	5189	5218	4207	4204	4275	4336	4452	4448	4073
CLD	931	1010	1127	1127	1127	1127	1127	1127	1127	1127	1127	1127
CLDNT	1173	4637#	4639	5312	5317	5345	5364	5380	5387#	1714	1757	1996
CLCNT	2936	3694	5691	5312	5317	5345	5364	5380	5387#	1714	1757	1996
CLRALL	1166#	1316	1334	1330	1304	1451	1471	1485	1708	1757	1774	1996
CLRTN	2029	2510	2529	2548	2548	2604	2604	2663	2692	2993	3654	3776
CMREG	3917	4252	4569	4574	4586	4610	4946	5101	5101	5101	5101	5101
CNCRN	1011	1014	1014	1014	1014	1014	1014	1014	1014	1014	1014	1014
CNCRG	1291#	1446	1463	1482	1584	1536	1570	1622	1628	1775	1813	1842
CNCRD	1866	1878	1915	1924	1944	1944	1944	1944	1944	1944	1944	1944
CNCRLE	2181	2193	2242	2253	2292	2384	2387	2362	2395	2433	2483	2520
CNCRLL	2582	2570	2621	2621	2657	2674	2733	2746	2760	2760	2801	2849
CNCRLL	2939	2936	3035	3030	3043	3043	3123	3135	3135	3179	3183	3195
CNCRLL	3261	3247	3343	3341	3358	3343	3341	3341	3341	3341	3341	3341
CNCRLL	4110	4120	4133	4223	4230	4230	4230	4230	4230	4230	4230	4230
CNCRLL	4658	4646	4706	4712	4854	4879	5082	5114	5120	5181	5185	5185
CNCRLL	5209	5209	5231#	5381	5385#	5385#	5385#	5385#	5385#	5385#	5385#	5385#
CNCRLL	1332	1566	1752	5381	5385#	5385#	5385#	5385#	5385#	5385#	5385#	5385#
CNCRLL	415	493#	493#	493#	493#	493#	493#	493#	493#	493#	493#	493#
CNCRLL	421	528#	485#	485#	485#	485#	485#	485#	485#	485#	485#	485#
CNCRLL	420	485#	485#	485#	485#	485#	485#	485#	485#	485#	485#	485#
CNCRLL	416	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#
CNCRLL	417	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#
CNCRLL	418	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#
CNCRLL	419	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#	474#
CNCRLL	163	175#	466	466	466	466	466	466	466	466	466	466
CNCRLL	164	4020#	4030	4031	4031	4031	4031	4031	4031	4031	4031	4031

COMP2	1165	1467#	3479	3480	3480	3480	3480	3480	3480	3480	3480	3480
CONSL	41#	59	2055	2060#	2060#	2060#	2060#	2060#	2060#	2060#	2060#	2060#
CONST1	2052	2053	5367	5379	5386#	5386#	5386#	5386#	5386#	5386#	5386#	5386#
COUNT	1339	2054	5367	5379	5386#	5386#	5386#	5386#	5386#	5386#	5386#	5386#
CREPR	3474	3479#	3479#	3479#	3479#	3479#	3479#	3479#	3479#	3479#	3479#	3479#
CRLE	1114#	1246	3700	3817	3822	3822	3822	3822	3822	3822	3822	3822
CRREG1	1237#	3472	3548	3577	3589	4010	4887	4887	4887	4887	4887	4887
CRREG2	1238#	3475	3567	3573	4816	4871	4792	4792	4792	4792	4792	4792
CRREG3	1239#	2541	3567	3573	4816	4871	4792	4792	4792	4792	4792	4792
CRREG4	1240#	2543	3567	3573	4816	4871	4792	4792	4792	4792	4792	4792
CRREG5	1241#	2184	2196	2245	2254	2254	2254	2254	2254	2254	2254	2254
CRREG6	1242#	2433	2488	2935	2947	2947	2947	2947	2947	2947	2947	2947
CRREG7	1243#	4530	4832	4832	4832	4832	4832	4832	4832	4832	4832	4832
CRREG8	1244#	4094	4495	4545	4551	4725	4735	5134	5144	5144	5144	5144
DCLE	1074#	3074	4636	4636	4636	4636	4636	4636	4636	4636	4636	4636
DCNT1	3823	3830	3833	3857	3876	3883	3892#	3892#	3892#	3892#	3892#	3892#
DCNT2	3868	3770	3779	3796	3826	3858	3866	3873	3877	3881	3892#	3911
DCNT3	3882	3842	3846	3894#	3894#	3894#	3894#	3894#	3894#	3894#	3894#	3894#
DCNT4	3893	3844	3846	3894#	3894#	3894#	3894#	3894#	3894#	3894#	3894#	3894#
DIN	1129	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2214#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2215#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2216#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2217#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2218#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2219#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2220#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2221#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2222#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2223#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2224#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2225#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2226#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2227#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2228#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2229#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2230#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2231#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2232#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2233#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2234#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2235#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2236#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2237#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2238#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2239#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2240#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2241#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2242#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2243#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2244#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2245#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2246#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2247#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2248#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2249#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2250#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2251#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2252#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2253#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2254#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2255#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2256#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2257#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2258#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2259#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2260#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2261#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2262#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2263#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2264#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2265#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2266#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2267#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2268#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2269#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2270#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2271#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2272#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2273#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2274#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2275#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2276#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2277#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2278#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2279#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2280#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2281#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2282#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2283#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2284#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2285#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK2	2286#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK3	2287#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK4	2288#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK5	2289#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK6	2290#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK7	2291#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK8	2292#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK9	2293#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK0	2294#	3894#	4014	4015	4016	4020	4020	4020	4020	4020	4020	4020
D1SK1	2295#	3894#	4014									

TCNTR5	TCNTR4	TCNTR3	TCNTR2	TCNTR1	TCNTR0	TCNTR6	TCNTR7	TCNTR8	TCNTR9	TCNTR10	TCNTR11	TCNTR12	TCNTR13	TCNTR14	TCNTR15	TCNTR16	TCNTR17	TCNTR18	TCNTR19	TCNTR20	TCNTR21	TCNTR22	TCNTR23	TCNTR24	TCNTR25	TCNTR26	TCNTR27	TCNTR28	TCNTR29	TCNTR30	TCNTR31	TCNTR32	TCNTR33	TCNTR34	TCNTR35	TCNTR36	TCNTR37	TCNTR38	TCNTR39	TCNTR40	TCNTR41	TCNTR42	TCNTR43	TCNTR44	TCNTR45	TCNTR46	TCNTR47	TCNTR48	TCNTR49	TCNTR50	TCNTR51	TCNTR52	TCNTR53	TCNTR54	TCNTR55	TCNTR56	TCNTR57	TCNTR58	TCNTR59	TCNTR60	TCNTR61	TCNTR62	TCNTR63	TCNTR64	TCNTR65	TCNTR66	TCNTR67	TCNTR68	TCNTR69	TCNTR70	TCNTR71	TCNTR72	TCNTR73	TCNTR74	TCNTR75	TCNTR76	TCNTR77	TCNTR78	TCNTR79	TCNTR80	TCNTR81	TCNTR82	TCNTR83	TCNTR84	TCNTR85	TCNTR86	TCNTR87	TCNTR88	TCNTR89	TCNTR90	TCNTR91	TCNTR92	TCNTR93	TCNTR94	TCNTR95	TCNTR96	TCNTR97	TCNTR98	TCNTR99	TCNTR100	TCNTR101	TCNTR102	TCNTR103	TCNTR104	TCNTR105	TCNTR106	TCNTR107	TCNTR108	TCNTR109	TCNTR110	TCNTR111	TCNTR112	TCNTR113	TCNTR114	TCNTR115	TCNTR116	TCNTR117	TCNTR118	TCNTR119	TCNTR120	TCNTR121	TCNTR122	TCNTR123	TCNTR124	TCNTR125	TCNTR126	TCNTR127	TCNTR128	TCNTR129	TCNTR130	TCNTR131	TCNTR132	TCNTR133	TCNTR134	TCNTR135	TCNTR136	TCNTR137	TCNTR138	TCNTR139	TCNTR140	TCNTR141	TCNTR142	TCNTR143	TCNTR144	TCNTR145	TCNTR146	TCNTR147	TCNTR148	TCNTR149	TCNTR150	TCNTR151	TCNTR152	TCNTR153	TCNTR154	TCNTR155	TCNTR156	TCNTR157	TCNTR158	TCNTR159	TCNTR160	TCNTR161	TCNTR162	TCNTR163	TCNTR164	TCNTR165	TCNTR166	TCNTR167	TCNTR168	TCNTR169	TCNTR170	TCNTR171	TCNTR172	TCNTR173	TCNTR174	TCNTR175	TCNTR176	TCNTR177	TCNTR178	TCNTR179	TCNTR180	TCNTR181	TCNTR182	TCNTR183	TCNTR184	TCNTR185	TCNTR186	TCNTR187	TCNTR188	TCNTR189	TCNTR190	TCNTR191	TCNTR192	TCNTR193	TCNTR194	TCNTR195	TCNTR196	TCNTR197	TCNTR198	TCNTR199	TCNTR200	TCNTR201	TCNTR202	TCNTR203	TCNTR204	TCNTR205	TCNTR206	TCNTR207	TCNTR208	TCNTR209	TCNTR210	TCNTR211	TCNTR212	TCNTR213	TCNTR214	TCNTR215	TCNTR216	TCNTR217	TCNTR218	TCNTR219	TCNTR220	TCNTR221	TCNTR222	TCNTR223	TCNTR224	TCNTR225	TCNTR226	TCNTR227	TCNTR228	TCNTR229	TCNTR230	TCNTR231	TCNTR232	TCNTR233	TCNTR234	TCNTR235	TCNTR236	TCNTR237	TCNTR238	TCNTR239	TCNTR240	TCNTR241	TCNTR242	TCNTR243	TCNTR244	TCNTR245	TCNTR246	TCNTR247	TCNTR248	TCNTR249	TCNTR250	TCNTR251	TCNTR252	TCNTR253	TCNTR254	TCNTR255	TCNTR256	TCNTR257	TCNTR258	TCNTR259	TCNTR260	TCNTR261	TCNTR262	TCNTR263	TCNTR264	TCNTR265	TCNTR266	TCNTR267	TCNTR268	TCNTR269	TCNTR270	TCNTR271	TCNTR272	TCNTR273	TCNTR274	TCNTR275	TCNTR276	TCNTR277	TCNTR278	TCNTR279	TCNTR280	TCNTR281	TCNTR282	TCNTR283	TCNTR284	TCNTR285	TCNTR286	TCNTR287	TCNTR288	TCNTR289	TCNTR290	TCNTR291	TCNTR292	TCNTR293	TCNTR294	TCNTR295	TCNTR296	TCNTR297	TCNTR298	TCNTR299	TCNTR300	TCNTR301	TCNTR302	TCNTR303	TCNTR304	TCNTR305	TCNTR306	TCNTR307	TCNTR308	TCNTR309	TCNTR310	TCNTR311	TCNTR312	TCNTR313	TCNTR314	TCNTR315	TCNTR316	TCNTR317	TCNTR318	TCNTR319	TCNTR320	TCNTR321	TCNTR322	TCNTR323	TCNTR324	TCNTR325	TCNTR326	TCNTR327	TCNTR328	TCNTR329	TCNTR330	TCNTR331	TCNTR332	TCNTR333	TCNTR334	TCNTR335	TCNTR336	TCNTR337	TCNTR338	TCNTR339	TCNTR340	TCNTR341	TCNTR342	TCNTR343	TCNTR344	TCNTR345	TCNTR346	TCNTR347	TCNTR348	TCNTR349	TCNTR350	TCNTR351	TCNTR352	TCNTR353	TCNTR354	TCNTR355	TCNTR356	TCNTR357	TCNTR358	TCNTR359	TCNTR360	TCNTR361	TCNTR362	TCNTR363	TCNTR364	TCNTR365	TCNTR366	TCNTR367	TCNTR368	TCNTR369	TCNTR370	TCNTR371	TCNTR372	TCNTR373	TCNTR374	TCNTR375	TCNTR376	TCNTR377	TCNTR378	TCNTR379	TCNTR380	TCNTR381	TCNTR382	TCNTR383	TCNTR384	TCNTR385	TCNTR386	TCNTR387	TCNTR388	TCNTR389	TCNTR390	TCNTR391	TCNTR392	TCNTR393	TCNTR394	TCNTR395	TCNTR396	TCNTR397	TCNTR398	TCNTR399	TCNTR400	TCNTR401	TCNTR402	TCNTR403	TCNTR404	TCNTR405	TCNTR406	TCNTR407	TCNTR408	TCNTR409	TCNTR410	TCNTR411	TCNTR412	TCNTR413	TCNTR414	TCNTR415	TCNTR416	TCNTR417	TCNTR418	TCNTR419	TCNTR420	TCNTR421	TCNTR422	TCNTR423	TCNTR424	TCNTR425	TCNTR426	TCNTR427	TCNTR428	TCNTR429	TCNTR430	TCNTR431	TCNTR432	TCNTR433	TCNTR434	TCNTR435	TCNTR436	TCNTR437	TCNTR438	TCNTR439	TCNTR440	TCNTR441	TCNTR442	TCNTR443	TCNTR444	TCNTR445	TCNTR446	TCNTR447	TCNTR448	TCNTR449	TCNTR450	TCNTR451	TCNTR452	TCNTR453	TCNTR454	TCNTR455	TCNTR456	TCNTR457	TCNTR458	TCNTR459	TCNTR460	TCNTR461	TCNTR462	TCNTR463	TCNTR464	TCNTR465	TCNTR466	TCNTR467	TCNTR468	TCNTR469	TCNTR470	TCNTR471	TCNTR472	TCNTR473	TCNTR474	TCNTR475	TCNTR476	TCNTR477	TCNTR478	TCNTR479	TCNTR480	TCNTR481	TCNTR482	TCNTR483	TCNTR484	TCNTR485	TCNTR486	TCNTR487	TCNTR488	TCNTR489	TCNTR490	TCNTR491	TCNTR492	TCNTR493	TCNTR494	TCNTR495	TCNTR496	TCNTR497	TCNTR498	TCNTR499	TCNTR500	TCNTR501	TCNTR502	TCNTR503	TCNTR504	TCNTR505	TCNTR506	TCNTR507	TCNTR508	TCNTR509	TCNTR510	TCNTR511	TCNTR512	TCNTR513	TCNTR514	TCNTR515	TCNTR516	TCNTR517	TCNTR518	TCNTR519	TCNTR520	TCNTR521	TCNTR522	TCNTR523	TCNTR524	TCNTR525	TCNTR526	TCNTR527	TCNTR528	TCNTR529	TCNTR530	TCNTR531	TCNTR532	TCNTR533	TCNTR534	TCNTR535	TCNTR536	TCNTR537	TCNTR538	TCNTR539	TCNTR540	TCNTR541	TCNTR542	TCNTR543	TCNTR544	TCNTR545	TCNTR546	TCNTR547	TCNTR548	TCNTR549	TCNTR550	TCNTR551	TCNTR552	TCNTR553	TCNTR554	TCNTR555	TCNTR556	TCNTR557	TCNTR558	TCNTR559	TCNTR560	TCNTR561	TCNTR562	TCNTR563	TCNTR564	TCNTR565	TCNTR566	TCNTR567	TCNTR568	TCNTR569	TCNTR570	TCNTR571	TCNTR572	TCNTR573	TCNTR574	TCNTR575	TCNTR576	TCNTR577	TCNTR578	TCNTR579	TCNTR580	TCNTR581	TCNTR582	TCNTR583	TCNTR584	TCNTR585	TCNTR586	TCNTR587	TCNTR588	TCNTR589	TCNTR590	TCNTR591	TCNTR592	TCNTR593	TCNTR594	TCNTR595	TCNTR596	TCNTR597	TCNTR598	TCNTR599	TCNTR600	TCNTR601	TCNTR602	TCNTR603	TCNTR604	TCNTR605	TCNTR606	TCNTR607	TCNTR608	TCNTR609	TCNTR610	TCNTR611	TCNTR612	TCNTR613	TCNTR614	TCNTR615	TCNTR616	TCNTR617	TCNTR618	TCNTR619	TCNTR620	TCNTR621	TCNTR622	TCNTR623	TCNTR624	TCNTR625	TCNTR626	TCNTR627	TCNTR628	TCNTR629	TCNTR630	TCNTR631	TCNTR632	TCNTR633	TCNTR634	TCNTR635	TCNTR636	TCNTR637	TCNTR638	TCNTR639	TCNTR640	TCNTR641	TCNTR642	TCNTR643	TCNTR644	TCNTR645	TCNTR646	TCNTR647	TCNTR648	TCNTR649	TCNTR650	TCNTR651	TCNTR652	TCNTR653	TCNTR654	TCNTR655	TCNTR656	TCNTR657	TCNTR658	TCNTR659	TCNTR660	TCNTR661	TCNTR662	TCNTR663	TCNTR664	TCNTR665	TCNTR666	TCNTR667	TCNTR668	TCNTR669	TCNTR670	TCNTR671	TCNTR672	TCNTR673	TCNTR674	TCNTR675	TCNTR676	TCNTR677	TCNTR678	TCNTR679	TCNTR680	TCNTR681	TCNTR682	TCNTR683	TCNTR684	TCNTR685	TCNTR686	TCNTR687	TCNTR688	TCNTR689	TCNTR690	TCNTR691	TCNTR692	TCNTR693	TCNTR694	TCNTR695	TCNTR696	TCNTR697	TCNTR698	TCNTR699	TCNTR700	TCNTR701	TCNTR702	TCNTR703	TCNTR704	TCNTR705	TCNTR706	TCNTR707	TCNTR708	TCNTR709	TCNTR710	TCNTR711	TCNTR712	TCNTR713	TCNTR714	TCNTR715	TCNTR716	TCNTR717	TCNTR718	TCNTR719	TCNTR720	TCNTR721	TCNTR722	TCNTR723	TCNTR724	TCNTR725	TCNTR726	TCNTR727	TCNTR728	TCNTR729	TCNTR730	TCNTR731	TCNTR732	TCNTR733	TCNTR734	TCNTR735	TCNTR736	TCNTR737	TCNTR738	TCNTR739	TCNTR740	TCNTR741	TCNTR742	TCNTR743	TCNTR744	TCNTR745	TCNTR746	TCNTR747	TCNTR748	TCNTR749	TCNTR750	TCNTR751	TCNTR752	TCNTR753	TCNTR754	TCNTR755	TCNTR756	TCNTR757	TCNTR758	TCNTR759	TCNTR760	TCNTR761	TCNTR762	TCNTR763	TCNTR764	TCNTR765	TCNTR766	TCNTR767	TCNTR768	TCNTR769	TCNTR770	TCNTR771	TCNTR772	TCNTR773	TCNTR774	TCNTR775	TCNTR776	TCNTR777	TCNTR778	TCNTR779	TCNTR780	TCNTR781	TCNTR782	TCNTR783	TCNTR784	TCNTR785	TCNTR786	TCNTR787	TCNTR788	TCNTR789	TCNTR790	TCNTR791	TCNTR792	TCNTR793	TCNTR794	TCNTR795	TCNTR796	TCNTR797	TCNTR798	TCNTR799	TCNTR800	TCNTR801	TCNTR802	TCNTR803	TCNTR804	TCNTR805	TCNTR806	TCNTR807	TCNTR808	TCNTR809	TCNTR810	TCNTR811	TCNTR812	TCNTR813	TCNTR814	TCNTR815	TCNTR816	TCNTR817	TCNTR818	TCNTR819	TCNTR820	TCNTR821	TCNTR822	TCNTR823	TCNTR824	TCNTR825	TCNTR826	TCNTR827	TCNTR828	TCNTR829	TCNTR830	TCNTR831	TCNTR832	TCNTR833	TCNTR834	TCNTR835	TCNTR836	TCNTR837	TCNTR838	TCNTR839	TCNTR840	TCNTR841	TCNTR842	TCNTR843	TCNTR844	TCNTR845	TCNTR846	TCNTR847	TCNTR848	TCNTR849	TCNTR850	TCNTR851	TCNTR852	TCNTR853	TCNTR854	TCNTR855	TCNTR856	TCNTR857	TCNTR858	TCNTR859	TCNTR860	TCNTR861	TCNTR862	TCNTR863	TCNTR864	TCNTR865	TCNTR866	TCNTR867	TCNTR868	TCNTR869	TCNTR870	TCNTR871	TCNTR872	TCNTR873	TCNTR874	TCNTR875	TCNTR876	TCNTR877	TCNTR878	TCNTR879	TCNTR880	TCNTR881	TCNTR882	TCNTR883	TCNTR884	TCNTR885	TCNTR886	TCNTR887	TCNTR888	TCNTR889	TCNTR890	TCNTR891	TCNTR892	TCNTR893	TCNTR894	TCNTR895	TCNTR896	TCNTR897	TCNTR898	TCNTR899	TCNTR900	TCNTR901	TCNTR902	TCNTR903	TCNTR904	TCNTR905	TCNTR906	TCNTR907	TCNTR908	TCNTR909	TCNTR910	TCNTR911	TCNTR912	TCNTR913	TCNTR914	TCNTR915	TCNTR916	TCNTR917	TCNTR918	TCNTR919	TCNTR920	TCNTR921	TCNTR922	TCNTR923	TCNTR924	TCNTR925	TCNTR926	TCNTR927	TCNTR928	TCNTR929	TCNTR930	TCNTR931	TCNTR932	TCNTR933	TCNTR934	TCNTR935	TCNTR936	TCNTR937	TCNTR938	TCNTR939	TCNTR940	TCNTR941	TCNTR942	TCNTR943	TCNTR944	TCNTR945	TCNTR946	TCNTR947	TCNTR948	TCNTR949	TCNTR950	TCNTR951	TCNTR952	TCNTR953	TCNTR954	TCNTR955	TCNTR956	TCNTR957	TCNTR958	TCNTR959	TCNTR960	TCNTR961	TCNTR962	TCNTR963	TCNTR964	TCNTR965	TCNTR966	TCNTR967	TCNTR968	TCNTR969	TCNTR970	TCNTR971	TCNTR972	TCNTR973	TCNTR974	TCNTR975	TCNTR976	TCNTR977	TCNTR978	TCNTR979	TCNTR980	TCNTR981	TCNTR982	TCNTR983	TCNTR984	TCNTR985	TCNTR986	TCNTR987	TCNTR988	TCNTR989	TCNTR990	TCNTR991	TCNTR992	TCNTR993	TCNTR994	TCNTR995	TCNTR996	TCNTR997	TCNTR998	TCNTR999	TCNTR1000
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XCAPSM	980	517	619*	632	676	958	986	992											
XCBM	75	177	152	199*	206	958	986	992											
XCBTV	69	221*	231	482	593	805	827												
XCBTV	83	271	287	503	585	547	540	633	671	728	755	768	832	853*					
XCLAS	869																		
XCLAS	1115	1127*																	
XCLOR	1186	1173*																	
XCOMP1	1097	1164*																	
XCOMP2	1096	1165*																	
XCRIF	1114	1180*																	
XD1L	1083	1129*																	
XD1SKG	1087	1172*																	
XDOLPT	857	864*																	
XDOSM	447	444*																	
XDOUY	1082	1128*																	
XDSKON	5313	5325*																	
XERRO	1285	1162*																	
XFIGURE	1093	1154*																	
XFLBUF	1090	1155*																	
XFROCT	1111	1178*																	
XGREG	4455	4469*																	
XHFCM	1088	1153*																	
XHTRK	1182*	1871	1928	1955	1989	2021	2081	2131	2178	2219	2289	2344	2614	2639					
	2711	2737	2897	2922	2923	2961	2977	3278	3333	3531	3532	3940	3951	4110					
XIDNWT	4228	4236																	
XKLBUF	1096	1163*																	
XKLB	1089	1156*																	
XKLB	3684	3752*																	
XLAS	1116	1126*																	
XLDAD	1102	1172*																	
XLDCA	1105	1171*																	
XLDCH	1104	1170*																	
XLDW	1108	1175*																	
XLOAD	1211*	1847	1850	2397	2452	2485	2489	3020	3118	3171	3258	3413	3496						
XLOAD	1211*	1847	1872	1918	1921	1956	1990	2022	2479	2484	2129	2134	2185	2186					
XLOTRK	2246	2247	2286	2293	2341	2386	2618	2622	2637	2642	2719	2724	2783	2787					
	2895	2911	2919	2927	2934	2962	3272	3335	3582	3526	3527	3535	3958	3954					
XMYLAS	5294	5327*																	
XNERO	1094	1161*																	
XONLY	1086	1150*																	
XOVERD	3685	3753*																	
XPRINT	1113	1160*																	
XPRU	1110	1177*																	
XROAD	1101	1168*																	
XROB	1109	1174*																	
XROCH	1100	1167*																	
XROCR	1107	1174*																	
XROST	1299	1164*																	
XREG	4422	4471*																	
XRESTR	1285	1151*																	
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	593	694*																	

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11075	229	320#	373	382	398	392	394	514#	548
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*V5102	951	1010	1000#		
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*V7402	2056	2002#	3090	3757#	
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*V7510	662	605#			
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*V7600	490	503#			
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*V7774	723	900#			